Inefficient Policies in the Green Transition

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Abstract

Countries have employed a wide range of policy instruments to mitigate climate change. These policies share a common pattern: governments initially rely on subsidies, together with command-and-control regulations, and eventually adopt carbon pricing. I develop a dynamic model of climate policymaking that accounts for this pattern. Although the first-best policy is solely a carbon tax, a climate-concerned policymaker uses subsidies to induce investments in emissions-abatement technologies with the goal of building a coalition in support of efficient policies in the future. The model provides additional insights: First, a policy package that satisfies political constraints and passes a cost-benefit analysis only exists if the economic costs of decarbonization are not too large, and the social cost of carbon is intermediate. Second, soft commitments, such as net-zero targets, can have real consequences by shifting expectations, but only if initial political pressure is not too large and policymakers are sufficiently concerned about climate. Finally, a higher risk of electoral turnover that replaces a green proposer with a misaligned proposer can improve prospects for a green transition.

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1. Introduction

Countries representing 81% of global greenhouse gas (GHG) emissions have announced net-zero targets, most of them for 2050 (Climate Watch). According to leading economists, a combination of two policies can achieve these objectives at minimal social cost: carbon pricing and R&D subsidies (e.g., Acemoglu *et al.* 2016; Metcalf 2019; Blanchard *et al.* 2023). In practice, however, governments use a much wider range of policies, which differ substantially in their costs (Mealy *et al.* 2024; Hahn *et al.* 2026). Moreover, policy instruments tend to be sequenced in a predictable pattern. Governments often begin with relatively costly instruments such as subsidies and command-and-control regulations. These policies provide explicit or implicit subsidies for investments in emissions abatement and production of renewable energy. Over time, carbon pricing tends to become the main policy tool (see Linsenmeier *et al.* 2022 for systematic evidence).

The following examples illustrate this pattern. Germany relied on feed-in tariffs (FITs) to support the expansion of renewable energy production beginning in the 1990s (von Hirschhausen *et al.* 2019), later replacing them with auctions in 2014 (Clean Energy Wire). Despite initially being reluctant to participate in the EU Emissions Trading System (ETS) (Ellerman *et al.* 2010), Germany implemented a carbon price for the heating and transport sectors in 2021 (IEA). The EU followed a similar path, first promoting the deployment of renewable energy and only later imposing a meaningful carbon price. In 2001, it enacted the Renewable Energy Directive, which required member states to set national targets for renewable-energy production. In 2009, these targets became legally binding and their scope expanded (Leipprand *et al.* 2020).[1] The ETS was introduced in 2003, and it imposed a low carbon price until its 2018 reform, which took effect in 2021; prices have been above €80 per ton of CO_2 most of the time since 2022 (van den Bergh and Botzen 2024). The Canadian federal government implemented a series of inefficient regulations and subsidies in its unsuccessful attempt to comply with the Kyoto Protocol (Jaccard *et al.* 2006; Samson and Stamler 2009; Harrison 2010b); in 2018, it enacted an ambitious national carbon price

[1] In the top six European countries, "[t]he cost to society implied by the deployment of wind and solar technologies [in 2010] represented €48,300 million" (Dechezleprêtre and Popp 2017).

(Harrison 2023). In all these cases, policies that subsidized emissions reductions came first and meaningful carbon pricing later.

Meckling *et al.* (2015, 2017) and Pahle *et al.* (2018) propose an explanation for the sequencing of policy instruments based on the idea of policy feedback: renewable energy and green investment subsidies ("carrots") can make more stringent policies that impose costs on emissions ("sticks") politically feasible in the future. They do so by creating or expanding low-carbon industries. These industries, in turn, become political constituencies that counterbalance incumbent polluters and build support for more stringent climate policy. This strategy has informed policy recommendations by, for example, World Bank researchers (Hallegatte *et al.* 2024).

The policy feedback argument is compelling but incomplete. It leaves four questions unanswered:

First, why do forward-looking opponents not block carrots if they anticipate that carrots facilitate sticks in the future? The argument starts from the premise that opponents initially have enough political power to block climate policy. Otherwise, a climate-concerned policymaker could implement carbon pricing immediately. For carrots to pass, therefore, at least some opponents must consent to them. But if these opponents are forward-looking and understand that carrots will make sticks politically feasible, why do they consent? Stokes (2020) introduces the concept of "fog of enactment" to answer this question: opponents accept carrots because they fail to anticipate their consequences. This explanation is difficult to sustain when policymakers and policy advisors explicitly advocate carrots as part of a sequencing strategy.

Second, why does the policymaker not make the carrots generous enough to secure support for the optimal carbon price immediately? (By the optimal carbon price, I mean the price that makes polluters pay the social cost of their carbon emissions at the margin.) In other words, why do carrots come *before* sticks? In addition, are the policies required to fracture the opposition necessarily inefficient?

Third, why is policy sequencing not always implemented? Under what conditions is it feasible and, among those, under what conditions do climate-concerned policymakers find it worth

pursuing?

Fourth, how does political uncertainty affect the feasibility of sequencing? The logic of sequencing relies on expectations about future policy, but current policymakers cannot commit their successors to completing the sequence. Political turnover may weaken incentives to make decarbonizing investments, and thereby undermine the strategy.

In this paper, I provide a formal account of the policy feedback mechanism that resolves these puzzles. The main idea in the paper is that inefficient policies are politically effective if (a) politically pivotal polluters directly benefit from them, and (b) they induce these polluters to make investments that reduce their exposure to carbon pricing. These two conditions explain why a coalition of green and polluting industries can support these policies, and why they reduce the opposition to carbon pricing over time.

Subsidies for investments in pollution abatement technologies are the clearest example of such policies. Firms or districts that own carbon-intensive assets are willing to make capital investments that "decarbonize" their assets if the subsidies are sufficiently generous. Once the investments are sunk, these actors are less exposed to future penalties on pollution and have less reason to oppose them. Green investment subsidies can therefore fracture the opposition to climate policy if enough opponents have "decarbonizable" assets (Kupzok and Nahm 2024). This idea resolves the first puzzle: pivotal opponents accept the carrots despite anticipating future sticks because taking the carrots reduces their exposure to those sticks.

I formalize this argument in a two-period dynamic model of climate policymaking. Policies emerge from legislative bargaining between a policy proposer and a legislature as in Romer and Rosenthal (1978). Each legislator represents a district, and districts differ in the productivity of their capital assets after an investment in a low-carbon technology. I assume that districts in which adopting the low-carbon technology is already profitable have already done so; these are the initial green constituencies. Both present climate policy and the expectation of future policy affect economic investments, which, in turn, affect districts' preferences in the future. First-period policies are therefore constrained by the initial legislature, but they also shape the opposition that

the policymaker faces in the second period. The first-period policymaker cares about aggregate economic welfare net of climate damages. With some probability there is political turnover; in that case the second-period policymaker places no weight on climate damages. This captures the possibility that future carbon pricing is not fully credible because a subsequent government may abandon the sequencing strategy.

In the baseline model, I consider two policy instruments: a carbon tax and a green investment subsidy. I assume that the net fiscal result (savings or deficit) at the end is returned to districts through a uniform lump-sum transfer or tax. The first-best policy is a Pigouvian carbon tax that equates the price of carbon emissions to the social cost of carbon. The green investment subsidy is not required in the first-best because the expectation of an optimal carbon price is enough to induce socially optimal investment decisions. The model thus abstracts from other market failures that may justify green subsidies. This allows me to focus on the political rationale for their use.

To focus on the case in which opponents can block climate policy in the first stage of the sequence, I assume that polluting districts are initially a majority in the legislature. The initial status quo is no climate policy. Thus, to implement climate policy, the proposer must form a coalition that includes some polluters. Polluters oppose carbon taxes, but those who own the most "decarbonizable" assets find green subsidies attractive if they are sufficiently generous. These districts are pivotal: the proposer can use subsidies to build a green coalition that includes them. Once these districts take the subsidies and their investments are sunk, their preferences change, since they enjoy the fiscal revenue from carbon taxes but no longer pay the costs. Thus, they support carbon taxes in the second period, which allows the proposer, if still in power, to implement optimal carbon taxes. The model, thus, provides an account of the policy sequencing mechanism.

The second question that the account must answer is why the policymaker does not implement the optimal carbon price in the first stage. The answer is that a higher carbon tax in the first stage of the sequence requires more generous subsidies to compensate pivotal opponents. More generous subsidies, however, create larger welfare losses. As a result, even if the optimal carbon price is politically feasible in the first stage, the policymaker may instead implement a weaker carbon price

or none at all. The model can thus explain why subsidies come first, with little or no use of carbon taxes. Subsidies are necessarily inefficient because of imperfect targeting. The policymaker would like to direct benefits only to pivotal opponents, but benefits must be conditioned on behavior to be effective, not on pivotality.

The answer to the third question is that sequencing is politically feasible exactly when the following two constraints can be satisfied simultaneously. First, subsidies must be generous enough to win the support of the pivotal opponents. Second, they must be acceptable to green districts, who demand the fiscal cost of the subsidies not to exceed the expected revenue from carbon taxes. This is because green districts refuse to pay for subsidies to polluters. Thus, the policy sequence must involve not only a transition from inefficient to efficient policies, but also a transition from fiscally costly policies to policies that raise revenue.

I show that the sequence cannot always be started: green technology must be sufficiently productive, investment costs sufficiently low, and the social cost of carbon cannot be neither too low nor too large. The last condition is perhaps counterintuitive. The reason is that a high social cost of carbon implies a large expected carbon price in the second period. This, in turn, leads pivotal opponents to demand a larger subsidy in the first period. The resulting increase in fiscal cost can make the policy sequence politically infeasible. At the same time, however, for the policy sequence to also be welfare-improving, the social cost of carbon must be sufficiently high. Otherwise the welfare losses generated by compensating pivotal opponents outweigh the environmental benefits. Thus, sequencing requires the social cost of carbon to be neither too low nor too high.

Finally, I argue that political turnover does not always undermine sequencing. Because subsidies cannot be perfectly targeted, the marginal recipient of the subsidy is a non-pivotal opponent. A small increase in the risk of political turnover lowers the expected future carbon tax and thus reduces polluters' incentive to invest in emissions abatement today. The marginal recipient therefore declines the subsidy, reducing the number of non-pivotal opponents who take it. This lowers the fiscal cost of subsidies while keeping them attractive to pivotal opponents, thereby relaxing the political constraints on the initial policy proposal. Political turnover has a second effect that goes

in the opposite direction: a lower expected carbon tax can reduce the revenue available to finance the carrots, which tightens the political constraints. The overall effect is therefore ambiguous. Surprisingly, I show that if initial opposition is sufficiently large, the first effect dominates. In that case, a greater risk of political turnover makes it easier to start the policy sequence.

The dynamic linkage of policies raises a further implication: under some conditions, the model has multiple equilibria.[2] There is always an equilibrium in which opponents expect that failure of climate policy in the present preserves their power to veto carbon pricing in the future. But if the social cost of carbon is sufficiently high, a self-fulfilling prophecy is possible. If districts expect a high carbon tax in the future, enough of them will decide to invest in decarbonization today; thus a green majority will form in the second period, allowing the policymaker to implement the expected carbon tax. In this case, the bargaining power of opponents in the first period goes down significantly, because rejecting carrots today still leads to sticks tomorrow. The policymaker is thus able to enact more ambitious climate policies in the first period. Even in this equilibrium, however, the same sequencing pattern emerges: a green investment subsidy combined with a low carbon price is eventually replaced by a Pigouvian carbon tax.

Literature. The paper contributes most directly to the literature on the domestic political economy of climate policy (e.g., Harrison 2010a; Breetz *et al.* 2018; Dolphin *et al.* 2020; Battaglini and Harstad 2020; Mildenberger 2020; Besley and Persson 2023; Gazmararian and Tingley 2023), and especially to work on policy feedback and policy sequencing (Aklin and Urpelainen 2013; Meckling *et al.* 2015, 2017; Pahle *et al.* 2018; Stokes 2020). My contribution is to develop a formal account of the policy sequencing mechanism that answers the questions I posed above: why forward-looking opponents accept carrots that facilitate future sticks, why the optimal carbon price is delayed, under what conditions policymakers initiate the sequencing strategy, and how political turnover between stages affects the feasibility of the strategy.[3]

[2] Biais and Landier (2022) and Smulders and Zhou (2024) also study models where climate policy creates multiple equilibria.

[3] My account connects to work on the political acceptability of climate policy (Gaikwad *et al.* 2022; Meckling and Nahm 2022; Meckling and Strecker 2023; Bolet *et al.* 2023; Gazmararian and Tingley 2023) and on lobbying and political influence by carbon-intensive interests (Grumbach 2015; Kim *et al.* 2016, 2021; Brulle 2018; Meng and Rode

My account builds on the framework proposed by Gazmararian and Tingley (2026). Gazmararian and Tingley distinguish between *strategic credibility*, which concerns commitment problems and expectations about future policy, and *structural credibility*, which concerns whether clean-energy investments can generate economic benefits for the actors whose support matters. Both determine whether the sequencing strategy works in my model. Strategic credibility determines the political constraints: political actors evaluate policy proposals in light of the policies they expect to follow from their implementation. Structural credibility determines which opponents can be included in the coalition: subsidies are attractive only to those opponents who have access to investments that reduce their carbon emissions without sacrificing their future economic welfare. The contribution of this paper is to propose a concrete mechanism that generates policy sequencing while taking these credibility problems seriously.

I also contribute to the literature on the politics of instrument choice in climate policy (Buchanan and Tullock 1975; Aidt and Dutta 2004; Hughes and Urpelainen 2015; Meckling and Jenner 2016; Cullenward and Victor 2020; Konisky 2024) by showing how several second-best instruments can be used to implement a sequencing strategy. Aidt and Dutta (2004) also propose a model that generates a transition from command-and-control policies to carbon pricing, but in their model the transition is driven by an exogenous tightening of emissions-reduction objectives over time. In my model, the transition emerges endogenously due to policy feedback. The mechanism also provides a concrete political rationale for green industrial policy (Rodrik 2014; Meckling 2021; Allan and Nahm 2024; Juhász and Lane 2024).

In addition, I contribute to a related literature that studies the effects of political turnover on climate policy (Ulph and Ulph 2013; Schmitt 2014; Harstad 2020; Hochman and Zilberman 2021; Behmer 2023). A common result is that the possibility of an "anti-climate" policymaker in the future distorts policy in the present by increasing carbon taxes and clean technology subsidies relative to the first-best. My contribution is to show that turnover has different consequences once we incorporate political constraints. I show that, depending on the initial size of the opposition,

2019; Goldberg *et al.* 2020; Kennard 2020).

turnover can make climate policy politically infeasible, or, instead, make it easier to enact. My model also reconciles the effects of turnover with the empirical observation that green subsidies often coexist with low or zero carbon prices, which this literature cannot explain.

A broader contribution of this paper is to connect the study of climate policy to an older literature on the political economy of economic reform (reviewed by Roland 2002). This literature showed how political constraints can cause reform to proceed gradually rather than all at once, and how the reform path can itself reshape political support for subsequent reforms. Dewatripont and Roland (1992) and Wei (1997), for example, study "divide and rule" tactics that work by gradually fracturing the opposition. Caselli and Gennaioli (2008) provide an example of policy sequencing: financial reform followed by deregulation can be politically more feasible than implementing both reforms simultaneously. Unlike in their model, however, the first-stage instrument in my model is not itself part of the first best; it is used only to make the optimal carbon price politically feasible later. A related insight appears in Acemoglu and Robinson (2001): inefficient policy instruments may be chosen precisely because their economic effects shape future political power. In their model, inefficient redistribution preserves the power of the beneficiary group; in mine, it fractures an entrenched coalition, weakening its political power.

More broadly, the paper contributes to the theoretical literature that studies policies chosen partly for their effects on future political outcomes (Persson and Svensson 1989; Alesina and Tabellini 1990; Besley and Coate 1998; Coate and Morris 1999; Battaglini and Coate 2008; Prato 2017). A particularly close paper is Coate and Morris (1999), which shows how a policy can induce beneficiaries to make economic choices that increase their future willingness to support the policy. My contribution is to show that a policy can instead be chosen because it changes the preferences of its current beneficiaries toward a *different* policy in the future, and that this effect can improve welfare. Baldursson and von der Fehr (2007) also study climate policy taking into account dynamic political effects, but ask a different question: why "brown" governments may favor long-lived emissions allowances over a carbon tax, even though the latter is more efficient. The model is also related to work on dynamic policymaking with an endogenous status quo (Buisseret

and Bernhardt 2017; Dziuda and Loeper 2018; Austen-Smith *et al.* 2019): the fact that a carbon tax enacted today becomes the status quo tomorrow creates an incentive to raise it if there is a risk of political turnover. Austen-Smith *et al.* (2019) use an endogenous status quo to explain the choice of inefficient policies, but their mechanism works differently. In their model, veto players are more willing to accept an inefficient instrument because its inefficiency makes it easier to repeal in the future. In my model, opponents accept an inefficient instrument even though it facilitates a shift in future policy against their current preferences.

My paper also contributes to a large literature that studies policy distortions due to interest-group influence. Two main approaches to modeling influence are reduced-form constraints, and the Grossman and Helpman (1994) common agency framework. An example of the former is Rozenberg *et al.* (2020), who model the political pressure from fossil fuel producers as a "no stranded assets" constraint; an example of the latter is Gerlagh and Liski (2023), who assume that a subset of producers can offer transfers to the policymaker contingent on policies.[4] Existing applications of these approaches to environmental policy treat the political power of special interests as exogenous to policy choices. The contribution of this paper is to show how policy can change that political power over time, and how those dynamic effects shape the policy distortions.

Structure of the Paper. In Section 2 I introduce the model and discuss its assumptions. In Section 3 I determine optimal policy without political constraints, with and without political turnover. In Section 4 I analyze the full model and provide the main results. In Section 5 I show that the main qualitative findings are not affected by considering other commonly used policy instruments nor by changes in several assumptions of the model. Section 6 discusses the possibility that there is political turnover before the investments in clean technology create the political coalition required to lock in the changes in climate policy, and concludes.

[4]Other papers using reduced-form political constraints to study climate policy include Bovenberg *et al.* (2005, 2008); Kalkuhl *et al.* (2013); Kalk and Sorger (2023); Acharya *et al.* (2024). Others using common-agency approaches include Fredriksson (1997); Damania (2001); Fredriksson and Svensson (2003); Damania and Fredriksson (2003); Fredriksson and Sterner (2005); Fredriksson and Wollscheid (2008); Habla and Winkler (2013); Aidt (1998); Aidt and Dutta (2004); Aidt (2010); Lai (2007, 2008); Hanoteau (2014); Grey (2018); Kalkuhl *et al.* (2020); and Winkler (2022).

2. The Model

Model Overview. The model has two periods. The actors are districts and policy proposers G and B. There is a set of districts, which are represented in a legislature and vote over policy proposals under majority rule. Each district is endowed with a unit of capital, which can be either polluting or green. Initially, a majority of districts is endowed with polluting capital. Districts choose how much to produce, and polluting districts can make an investment in the first period that converts their capital into green capital in the second period. Districts care about their income, which is affected by two climate policies: a carbon tax and a green investment subsidy. Policy proposers care about aggregate income; the initial proposer G, in addition, cares about climate damages.

In the first period, G chooses a carbon tax and a green investment subsidy. If a majority of districts prefers the proposal to the status quo, which is no climate policy, the proposal is implemented and it becomes the new status quo. After the policy is chosen, districts decide how much to produce using their initial capital endowment, and polluting districts decide whether to invest in green capital. Their policy preferences and economic decisions depend on their expectations about the future carbon tax and the aggregate level of green investment.

In the second period, a new proposer is drawn. G remains in power with positive probability; otherwise, B takes over. The new proposer chooses a carbon tax, which replaces the status quo if a majority of districts approves it. Districts choose how much to produce, and the game ends. In each period, the legislative pivot is the district whose support is needed to assemble a winning coalition for the proposed policy.

The dynamic link at the center of the model is the following. The first-period policy affects which polluting districts invest; those investments determine whether green districts form a majority in the second period; and the future majority determines the carbon tax that districts anticipate when they invest and vote in the first period.

The Economy. The model has two periods, $t = 1, 2$, and all actors discount second-period

payoffs by $\delta \in (0, 1]$. There is a continuum of districts indexed by $i \in I = [0, 1]$. Each district is endowed with one unit of capital.[5] Capital can be green or polluting. I refer to districts with polluting capital as polluting districts and districts with green capital as green districts. Initially, districts $i \in [0, b_1)$ are polluting, and districts $i \in [b_1, 1]$ are green; I characterize the cutoff b_1 below.

Let $\chi_{it} = 1$ if district i's capital is polluting in period t, and let $\chi_{it} = 0$ if it is green. Polluting districts can invest in green capital in the first period at cost $c > 0$. This investment converts their polluting capital into green capital in the second period. Let $\iota_i = 1$ denote that district i invests in green capital, and let $\iota_i = 0$ otherwise.

The economy has a single good, which is used for production and consumption and serves as the numéraire. A polluting district produces y units of the good at cost $\frac{1}{2}y^2$ and emits y units of carbon. A green district produces y units at cost $\frac{1}{2Ai}y^2$ and emits no carbon; Ai is the productivity of green capital in district i and $A > 1$ is a parameter.[6] This specification captures the assumption that green capital is more productive than polluting capital in some districts but less productive in others.

Policies. There are three policy instruments. The first is a carbon tax $\tau \in [0, 1]$. The second is a green investment subsidy $s \geqslant 0$ paid to any polluting district that invests in green capital. The third is a uniform lump-sum transfer or tax $T \in \mathbb{R}$.

Given carbon tax τ, polluting districts choose production y to maximize profits net of the tax, $(1-\tau)y - \frac{1}{2}y^2$. Thus, $y = 1 - \tau$ and profits are $\frac{1}{2}(1-\tau)^2$. Similarly, green district i chooses $y = Ai$ and earn profits $\frac{1}{2}Ai$.

Let $b_t = \int_0^1 \chi_{it}\, di$ denote the share of polluting districts in period t. I assume that the districts that are initially polluting are precisely those that would not invest absent climate policy, and that these districts form a majority. Thus, the cutoff district b_1 is indifferent between remaining polluting

[5] Capital is a composite of the district's human capital and investments that are location-specific in the short run, such as energy production capacity, infrastructure and firms.

[6] This is equivalent to assuming that firms in district i have production function $f_i(x) = x_B^{1/2}k_B^{1/2} + Aix_G^{1/2}k_G^{1/2}$, where x_B and x_G denote units of the numéraire used in production, k_B denotes units of district i's specific polluting capital, and k_G denotes units of district i's specific green capital.

and investing when carbon taxes and green investment subsidies are zero. If a polluting district i invests in green capital absent climate policy, it pays cost c and earns discounted second-period profits $\delta\frac{1}{2}Ai$. If it does not, it earns discounted second-period profits $\delta\frac{1}{2}$. Therefore b_1 is given by $-c + \frac{\delta A}{2}b_1 = \frac{\delta}{2}$. Formally:

Assumption 1. The initial cutoff b_1 satisfies $-c + \frac{\delta A}{2}b_1 = \frac{\delta}{2}$ and $\frac{1}{2} < b_1 < 1$.[7]

Policy Process. Districts are represented in a legislature. There are two possible policy proposers, G and B, whom I refer to interchangeably as policymakers. Proposer G is green or climate-concerned; proposer B is not. The initial proposer is G. The second-period proposer is G with probability $\rho \in (0, 1]$ and B otherwise. The status quo carbon tax and green investment subsidy are zero. The timing of events is:

First period:

1. G chooses a policy proposal $(\tau_1', s') \in [0, 1] \times \mathbb{R}_{\geqslant 0}$.
2. If a majority of districts prefers the proposal to the status quo, then $\tau_1 = \tau_1'$ and $s = s'$; otherwise $\tau_1 = s = 0$.
3. Districts make production and investment decisions, $y_{i1} \geqslant 0$ and $\iota_i \in \{0, 1\}$.

Second period:

1. The second-period proposer is drawn and proposes $\tau_2' \in [0, 1]$.
2. If a majority of districts prefers τ_2' to the status quo τ_1, then $\tau_2 = \tau_2'$; otherwise $\tau_2 = \tau_1$.
3. Districts make production decisions, $y_{i2} \geqslant 0$.

The uniform transfer is set to satisfy the intertemporal budget constraint:

$$T = \tau_1 \int_0^1 \chi_{i1} y_{i1}\, di - \int_0^1 \iota_i s\, di + \delta\tau_2 \int_0^1 \chi_{i2} y_{i2}\, di.$$

The first and last terms account for revenue from the carbon tax; the second term is the fiscal cost of the subsidy. This formulation assumes that the state can issue public debt to finance a first-period fiscal deficit, but must repay it in the second period.

[7] See Appendix A.8 for a discussion of the robustness of the results to this assumption.

Preferences. District i's payoff is $\pi_{i1} + \delta\pi_{i2} + T$, where

$$\pi_{it} = \begin{cases} (1-\tau_t)y_{it} - \frac{1}{2}y_{it}^2 - \mathbb{1}_{t=1}\iota_i(c-s) & \text{if } i \text{ is polluting in period } t, \\ y_{it} - \frac{1}{2Ai}y_{it}^2 & \text{if } i \text{ is green in period } t, \end{cases}$$

is district i's pre-transfer income in period t, equal to the return on its capital endowment net of carbon taxes and, in the first period, investment costs net of subsidies. The assumption that districts do not internalize climate damages is important for the analysis.[8] Throughout, $\mathbb{1}_X$ equals one if X is true and zero otherwise.

The payoff of proposer $P \in \{G, B\}$ is aggregate district income, including the uniform transfer, net of climate damages:

$$W_P = \sum_{t=1,2} \delta^{t-1}\left[\int_0^1 \pi_{it}\, di - \mathbb{1}_{P=G}\lambda \int_0^1 \chi_{it} y_{it}\, di\right] + T,$$

where $\lambda \in (0, \frac{1}{2})$ measures the social cost of carbon emissions. I assume that only proposer G internalizes this cost, and I impose the following upper bound on λ:

Assumption 2. We have $4\lambda + 2\lambda^2/\delta < 1 + 2c/\delta$.[9]

Equilibrium Concept. Subgame perfect equilibrium.

Comments on the Assumptions. The model of the economy is stylized in order to focus on the political mechanism, but the simplifications are not uncommon in the literature. For example, Acharya *et al.* (2024) also study a one-good economy in which there is no market power, and Coate and Morris (1999) also study a technological change that follows from a binary investment decision. Colmer *et al.* (2024) provide empirical justification for the latter assumption: in their study of the EU ETS the conclude that "[o]ur findings are consistent with firms paying an up-front fixed cost to invest in alternative 'clean' production technologies that reduce marginal variable

[8] In Appendix A.3, I consider an extension in which all districts internalize climate damages. See Section 5 for a discussion.

[9] I only use this assumption to rule out the possibility that the green proposer is willing to use arbitrarily large green subsidies to induce every polluting district to invest, supported by a coalition that excludes initially green districts.

costs." Ramadorai and Zeni (2024) show that firms react to beliefs about future carbon prices by investing in carbon abatement technology, which justifies the assumption that firms anticipate future climate policies. The assumption that the productivity of clean technology is heterogeneous across constituencies can be defended in two ways: the cost-competitiveness of renewable energy depends on location (Davis *et al.* 2023), and the advances in decarbonization technology depend on industry (Victor *et al.* 2019).

The assumption that different proposers may differ in their concerns for climate may be explained by differences in partisanship. Cadoret and Padovano (2016) find that left-wing governments promote the development of renewable energy more than right-wing parties in Europe. Lundquist (2024) finds that the degree of environmentalism expressed in parties' manifestos predicts the level of policy stringency they implement when in power. Knill *et al.* (2010), Jensen and Spoon (2011) and Jahn (2022) obtain similar results. Fankhauser *et al.* (2015) and Dolphin *et al.* (2020), however, do not find an effect of party ideology on climate legislation or the carbon price. In any case, the main results of this paper do not depend on this assumption. What is crucial is that carbon-intensive interests hold political power (because of their representation in the legislature) regardless of which party controls the agenda. Mildenberger (2020)'s logic of the double representation of these interests (with labor being represented via left-wing parties and business via right-wing parties) provides an empirically grounded justification for this assumption.

An important feature of the model is that investments in emissions-substituting capital in the present change the policy preferences of the constituencies where those investments take place. I offer three pieces of evidence to support this assumption. First, Alberdi (2024) finds that subsidized investments in rooftop solar panels increased support for ambitious climate policies and for the Green party in Germany. Second, Urpelainen and Zhang (2022) show that wind turbine installations increased vote shares of climate-concerned Democratic candidates in US House elections, and led to an increase in pro-climate votes in Congress, even though they may have created an electoral backlash among voters located close to the turbines. Third, Vormedal and Meckling (2023) provide evidence that the shale gas revolution led oil and gas industries to sincerely support carbon pricing

(during the Trump administration, Exxon lobbied against withdrawal from the Paris Agreement, and a Republican-backed coalition involving Exxon and other oil companies promoted legislation for a federal carbon tax starting at $40 per ton), and the fuel-efficiency regulations on the car industry imposed by the Obama administration led some car manufacturers to resist Trump's decision to roll back those regulations, due to their investments in clean technologies.

There are many important issues involved in climate policy from which the model abstracts, such as innovation, learning-by-doing and network externalities (Stock 2020; Fischer *et al.* 2021; Bistline *et al.* 2023; Hahn *et al.* 2026), imperfect competition (Kennard 2020), regulation of energy markets (Reguant 2019; Davis *et al.* 2023), land use regulation (Sud *et al.* 2023), other tax distortions (Barrage 2020), international trade (Clausing and Wolfram 2023; Kotchen and Maggi 2024), conservation (Harstad 2023a), international cooperation (Battaglini and Harstad 2016; Harstad 2023b), private politics (Egorov and Harstad 2017), consumer preferences (Besley and Persson 2023), behavioral distortions of energy-efficiency investments (Allcott *et al.* 2014), and the possibility of fiscal illusion (Abbott and Jones 2023).

3. Benchmarks

Optimal Policy. Suppose that the green policymaker can unilaterally choose policy in both periods. What is the optimal policy? To answer this question, I will characterize first how policies affect investment decisions.

If a polluting district i decides to invest in green capital in the first period, they pay the cost of capital c, and receive the investment subsidy s. Their discounted second-period profit is $\delta\frac{1}{2}Ai$. If the district does not invest, its discounted second-period profit is $\delta\frac{1}{2}(1-\tau_2)^2$, taking τ_2 as the expected second-period carbon tax. Therefore, the district invests if and only if $s - c + \frac{\delta Ai}{2} \geqslant \frac{\delta}{2}(1-\tau_2)^2$. The set of polluting districts that invest is $[b_2, b_1)$, where b_2 is given by

$$s - c + \frac{\delta A}{2} b_2 = \frac{\delta}{2}(1-\tau_2)^2. \tag{1}$$

The policy instruments have the following effects on the economy. Carbon taxes reduce

emissions by reducing production in the polluting districts. The second-period carbon tax also induces investment in green capital, since it reduces the expected returns from using polluting capital. The subsidy s increases investment directly.

Proposition 1. *The optimal policy consists of carbon taxes equal to the social cost of carbon in both periods, and no subsidies.*

Proof. All proofs are in Appendix B. ■

There are two economic decisions, production and investment, and each affects emissions. Production creates carbon emissions today, while investment reduces future emissions. The carbon tax induces producers to internalize the first externality; thus the optimal tax is equal to the marginal environmental damage. This is the Pigouvian level. An investment subsidy could address the second externality by encouraging firms to adopt green capital and reduce future emissions. In equilibrium, however, the policymaker does not use the subsidy: once emissions are priced at their marginal environmental cost, firms already face the correct incentive to invest. A subsidy on top of Pigouvian carbon taxes would induce inefficient investment, i.e., investment in technologies whose economic cost exceeds the social benefit from the emissions they avoid.

Political Turnover. Suppose now that there is a probability $1 - \rho$ that the green proposer is replaced with proposer B between periods one and two. Proposer B does not internalize the cost of climate damages, and for that reason chooses zero carbon taxes. Thus, from a first-period perspective, the expected future profit in the polluting sector is greater than if turnover was not possible, because with probability $1 - \rho$ the future carbon tax is zero. As a result, fewer firms decide to transition.

How should the green policymaker respond? As I show in Appendix B.2, the Pigouvian carbon tax is still optimal, but the optimal subsidy is now $s = \delta(1 - \rho)\lambda$. The reason why the subsidy is required in this case is that political turnover reduces investment below the optimal level, and the subsidy is the appropriate instrument to correct this distortion. It is noteworthy and intuitive that the optimal subsidy increases with the probability of turnover and with the social cost of carbon.

The result that the equilibrium carbon tax is not affected by political turnover depends on the assumption that the environmental damage is linear. If the environmental damage is a strictly convex function of the stock of carbon emissions, the effect of an increase in the probability of turnover on the first-period carbon tax is ambiguous. See Appendix B.3 for details.

In sum, government turnover can explain the use of subsidies, but under this mechanism subsidies arise as a complement to carbon pricing, not as a substitute. I now incorporate political constraints into the model, and show that these lead both to positive subsidies and carbon taxes strictly below the optimal level.

4. Political Constraints

I now study the full model. The analysis proceeds in four steps. I first solve the continuation game and characterize the level of investment in the first period. I then derive the two constraints that a first-period proposal must satisfy. The first protects the pivotal polluting district; the second protects the initial green districts from the fiscal cost of the subsidy. I use these constraints to characterize the equilibrium policy, and the conditions under which the policy sequence starts. The final subsection studies when an additional equilibrium exists.

4.1. *The Second-Period Coalition*

I analyze the model by backward induction, looking at the second period first. Suppose that proposer G has implemented a policy (τ_1, s). Suppose that enough polluting districts invested in the first period, and green districts now form a majority. This majority will approve an increase in the carbon tax and reject a decrease: green districts do not pay the tax, but benefit from the resulting fiscal revenue. G's most preferred carbon tax level is λ. Therefore, if G remains the proposer, the second-period tax is $\max\{\tau_1, \lambda\}$. If B becomes the proposer, the inherited tax τ_1 remains in place: proposer B does not wish to raise it, and the green majority blocks any attempt to lower it.

Suppose that polluters are still a majority in the second period. This implies that a majority of polluters did not invest in the first period. Therefore, they did not receive the subsidy, and they must pay the carbon tax in both periods. Some of these polluters must have accepted the first-period

policy proposal. Since they pay the taxes and do not receive the subsidy (but share the fiscal cost), both the tax τ_1 and the subsidy s must be zero. Therefore, G can implement a nonzero policy (τ_1, s) only if enough polluters invest that green districts form a majority in the second period.

Suppose that a green majority is expected in the second period. We can determine investment decisions at the end of the first period in the same way as in the previous section: districts $[b_2, b_1)$ invest, where b_2 is given by

$$s - c + \frac{\delta A}{2} b_2 = \frac{\delta}{2} \mathrm{E}\big[(1 - \tau_2)^2\big]. \tag{2}$$

The difference with (1) is that τ_2 is a random variable that depends on the identity of the second-period proposer.

Notice that first-period policies have an effect on b_2. This is the policy-feedback mechanism in the model. A subsidy induces some polluting districts to invest; investment removes their exposure to the carbon tax; and the resulting change in the majority allows a higher tax to survive in the future. Because districts anticipate this sequence when they vote, G cannot evaluate the first-period package separately from the continuation policy it creates.

The analysis below focuses on the baseline equilibrium selection following rejection of the first-period proposal. Under this selection, rejection preserves a polluting majority and the second-period carbon tax is zero. The beliefs that support a different continuation equilibrium are considered in Subsection 4.5.

4.2. *The Two Political Constraints*

To replace the status quo, G must assemble a coalition containing the initial green districts and enough polluting districts to reach a majority. The pivotal polluting district is $i = \frac{1}{2}$. If this district accepts the package and invests, all initially polluting districts with greater green productivity also invest. The investing districts $[\frac{1}{2}, b_1)$ then join the initial green districts $[b_1, 1]$, producing a second-period green majority.

The two parts of this coalition want different things. The pivotal polluting district bears the first-period carbon tax and must pay the investment cost to receive the subsidy. It therefore requires

enough compensation to prefer the package and its continuation to rejection. The initial green districts receive no subsidy and do not pay the tax. They support the package only if the transfer they expect exceeds the transfer they expect if they reject the proposal.

These requirements depend on what districts expect to happen if they reject the proposal. I will focus on an expectation that is always rational: a rejection is followed by no investment, no green majority in the second period, and therefore no climate policy. Under Assumption 1, this is an equilibrium of the continuation game following a rejection. In Subsection 4.5 I consider the alternative, which is that a green majority forms in the second period after a rejection.

The following proposition states the requirements in the equilibrium in which a rejection is followed by no climate policy in the second period.

Proposition 2. *In an equilibrium in which rejection of the first-period proposal implies zero carbon taxes in the second period, a carbon tax τ_1 and subsidy s are accepted by the legislature if and only if*

$$\frac{1}{2}(1-\tau_1)^2 + s - c + \frac{\delta A}{4} + \mathrm{E}(T) \geqslant \frac{1}{2} + \frac{\delta}{2} \tag{PC$_B$}$$

and

$$\mathrm{E}(T) = b_1\tau_1(1-\tau_1) + \delta b_2 \mathrm{E}[\tau_2(1-\tau_2)] - s(b_1 - b_2) \geqslant 0 \tag{PC$_G$}$$

hold, where b_2 is defined by (2), and

$$\tau_2 = \begin{cases} \max\{\tau_1, \lambda\} & \text{with probability } \rho, \\ \tau_1 & \text{with probability } 1-\rho. \end{cases}$$

Proof. See Subsection B.4. ∎

Constraint (PC$_B$) is the pivotal district's political constraint. The left-hand side is its payoff when it accepts the package and invests: current profits, the subsidy net of the investment cost, future profits from green capital, and the transfer. The right-hand side is its payoff after rejection, in which case it remains polluting and expects no carbon tax.

Constraint (PC_G) is the green districts' fiscal constraint. Because their production profits do not depend on the carbon tax or the investment subsidy, they compare the common transfer under the package with the zero transfer following rejection. The three terms in $E(T)$ are current carbon-tax revenue, expected future carbon-tax revenue, and subsidy spending.

The proposer implements a feasible policy if and only if they prefer it to the status quo. Define the proposer's gain relative to the status quo as

$$\Delta W(\tau_1, s) = W_G(\tau_1, s) - W_G(0, 0). \tag{3}$$

Then G implements a feasible policy (τ_1, s) only if $\Delta W(\tau_1, s) \geqslant 0$.

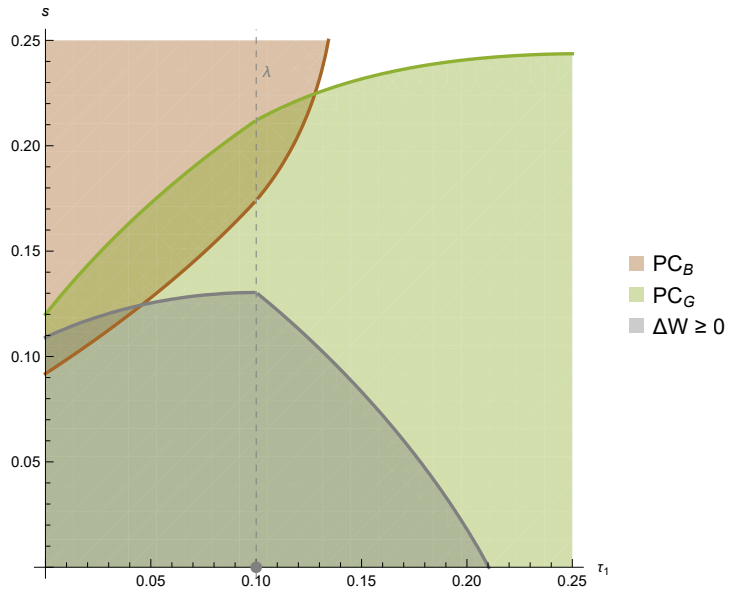


Figure 1: Political constraints when $A = 1.8$, $c = 0.06$, $\delta = 0.9$, $\lambda = 0.1$, and $\rho = 1$. PC_B is the set of policies the pivotal polluting district prefers to rejection. PC_G is the set of policies the initial green districts prefer to rejection. Their intersection is the set of politically feasible policies. The region $\Delta W \geqslant 0$ contains the policies that proposer G prefers to the status quo. The gray dot is G's unconstrained optimum, $(\tau_1, s) = (\lambda, 0)$.

Figure 1 shows the conflict between the two constraints. The pivotal polluting district prefers lower taxes and enough subsidy to cover the cost of investing. The green districts prefer a smaller subsidy and enough carbon-tax revenue to finance it. A package can pass only in the intersection of these regions. The proposer chooses from the part of that intersection that also lies in $\Delta W \geqslant 0$;

if this three-way intersection is empty, the equilibrium policy is the status quo.

4.3. *The Equilibrium Policy*

The first-period carbon tax τ_1 imposes a loss on the pivotal polluting district, which tightens (PC_B). The subsidy relaxes this constraint. Thus, when the constraint binds, a more demanding carbon tax requires a more generous subsidy to compensate the pivotal opponent.

This tradeoff rules out the Pigouvian tax in the first period. The reason is that, at $\tau_1 = \lambda$, a marginal reduction in the tax creates only a second-order loss for the proposer. But it relaxes (PC_B), allowing for a reduction in the subsidy. This leads to a first-order gain for the proposer. Thus, the proposer has an incentive to choose a first-period carbon tax below λ. This argument is not a complete proof that $\tau_1 < \lambda$ in equilibrium, however, because if (PC_G) binds, the proposer cannot reduce τ_1. However, even in this case I can show that there is a reduction in both τ_1 and s that creates a gain for the proposer without violating the political constraints. Thus, we have the following.

Proposition 3. *Under the baseline equilibrium selection, either the first-period carbon tax is below the social cost of carbon and the subsidy is positive, or no climate policy is enacted. That is, either* $\tau_1 < \lambda$ *and* $s > 0$*, or* $\tau_1 = s = 0$*.*

In period 1, G accepts a carbon tax below the Pigouvian level and uses a subsidy to induce the pivotal polluting districts to invest. Those investments create a green majority. If G remains in power, the tax then rises to the Pigouvian level; if B takes power, the new majority protects the inherited tax. The first-period distortion is the price of creating a green coalition in the second period.

This result can explain policy sequences: G initially obtains a partial political victory by enacting a low or zero carbon tax and an inefficiently large subsidy, and this leads over time to an optimal carbon tax. The initial policies are designed to expand the green coalition by inducing pivotal polluting districts to invest.

Germany's experience with the Renewable Energy Sources Act (EEG) of 2000 offers an illus-

tration of the mechanism. The EEG, enacted by the Social Democrat–Green coalition, provided generous feed-in tariffs for renewable energy, which effectively acted as an investment subsidy. While these tariffs were inefficient as policies designed to reduce carbon emissions (Marcantonini and Ellerman 2015), they served to rapidly expand the renewable energy sector. This expansion created a growing constituency of firms, workers, and communities with vested interests in green policies. Over time, as the renewable energy industry matured and costs decreased, Germany was able to gradually reduce the subsidies and implement more market-based mechanisms: the 2017 revision of the EEG introduced competitive auctions for most renewable energy sources. Furthermore, Germany implemented a carbon price for the heating and transport sectors in 2021 (IEA), which complements the EU ETS. This progression exemplifies how an initially inefficient policy paved the way for more comprehensive and efficient climate measures by altering the balance of power between green and polluting economic interests over time.

4.4. *Under What Conditions Does the Policy Sequence Start?*

Two conditions must be met. First, there must be a policy package that is acceptable to both pivotal polluting districts and green districts. Second, there must be a politically feasible policy that the proposer prefers over the status quo.

Political Feasibility. The two constraints respond differently to changes in the economic returns from decarbonization. Better green technology or a lower investment cost makes the subsidy more attractive to the pivotal polluting district, relaxing (PC_B). It also induces more districts to invest. That response raises the fiscal cost of the subsidy and reduces the future carbon-tax base, tightening (PC_G). However, an increase in A or a decrease in c lowers b_1 by Assumption 1, which reduces the number of districts that invest and take the subsidy. These effects go in different directions, but I can show that the first effect dominates, and both increasing A and reducing c expands the set of feasible policies.

Observation 1. *At every politically feasible policy (τ_1, s), an increase in the productivity of green technology A or a decrease in the cost of capital c relaxes the political constraint imposed*

by polluting districts (PC_B) *and tightens the political constraint imposed by green districts* (PC_G). *Moreover, if the constraints are feasible, and A increases or c decreases preserving Assumption 1, then the constraints remain feasible.*

Increasing the expected second-period carbon tax has two effects. It raises expected revenue for a fixed tax base. But it also shrinks the tax base by inducing more investment and thus reducing the share of polluters in the second period. More investment also increases the fiscal cost of the subsidy. Thus, the net effect on the expected fiscal transfer $\mathrm{E}(T)$, and therefore on political feasibility, is ambiguous.

The second effect dominates if A is not too large. The reason is that, by (2), the effect of τ_2 on b_2 depends inversely on A. When A is small, an increase in the expected carbon tax has a large effect on b_2. Thus, it shrinks the tax base and increases the cost of the subsidy significantly, and the second effect dominates.

Therefore, if A is not too large, increasing the social cost of carbon λ tightens the political constraint, because λ determines the second-period tax τ_2 if G remains in power. A larger λ thus makes G more committed to reducing carbon emissions, but precisely because of this, it makes a political bargain harder to implement.

Observation 2. *If green technology is not too productive,* $A < \frac{3\rho}{4(1-b_1)}$, *the political constraints are feasible only if* $\lambda < \overline{\lambda}$, *where* $\overline{\lambda}(A, c, \delta, \rho) < \frac{1}{2}$.

Political turnover acts through the same fiscal tradeoff. A greater probability of turnover lowers the expected second-period carbon tax. This has a revenue effect: a lower future tax raises less revenue. It also has a targeting effect: fewer non-pivotal districts invest, which reduces subsidy spending and preserves the future tax base. When initial opposition is large, the targeting effect can dominate.

Observation 3. *If the political opposition to climate policies is initially high, i.e.,* $b_1 \geqslant \frac{1-\frac{3}{4}\lambda}{1-\frac{1}{2}\lambda}$, *and the political constraints are feasible, then, if the probability that the second-period proposer is polluting increases, the constraints continue to be feasible.*

Thus, if the political opposition to climate policies is initially high, an increase in the probability of turnover can facilitate the policy sequence. In this case the possibility of turnover can paradoxically help a climate-concerned policymaker start a policy sequence that leads to an efficient carbon tax in the future. This is despite the fact that turnover interrupts the sequence, and policy uncertainty reduces green investments.

Proposer's Incentives. A policy that compensates the pivotal polluter may induce so much inefficient investment that G prefers the status quo. Therefore, the marginal damage from carbon emissions, λ, cannot be too low.

Observation 4. *There is a policy that satisfies the political constraints (PC_B and PC_G) and the proposer prefers over the status quo (i.e., $\Delta W \geqslant 0$) only if the social cost of carbon is not too low, i.e., $\lambda \geqslant \underline{\lambda}$, where $\underline{\lambda}(A, c, \delta) > 0$.*

This and Observation 2 place λ between a lower and an upper bound. If λ is too low, the proposer will not pay the economic cost of the political bargain. If λ is too high, the expected future tax can make that bargain politically infeasible. Thus, under the conditions in the two observations, a policy sequence can begin only for intermediate values of the social cost of carbon.

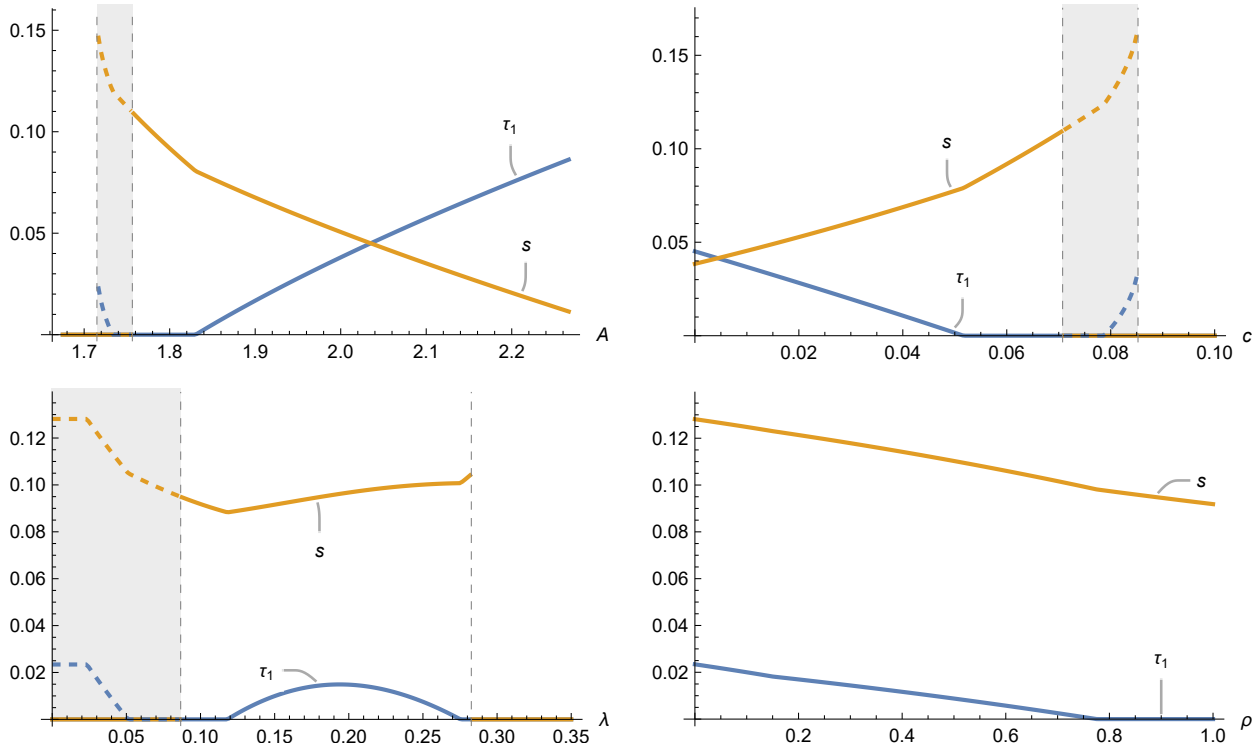


Figure 2: Equilibrium policy when $A = 1.8$, $c = 0.06$, $\delta = 0.9$, $\lambda = 0.1$, and $\rho = 1$, as each parameter changes. The dashed curves show the best politically feasible policy. Gray regions indicate that proposer G prefers the status quo to that policy.

Numerical Example. Figure 2 illustrates how the equilibrium policy depends on parameters. Improvements in technology A or reductions in investment cost c allow G to use a higher carbon tax and a smaller subsidy, except where the fiscal constraint (PC_G) binds. For low enough A, or large enough c, τ_1 is so small and s is so large that the fiscal constraint (PC_G) binds: the carbon tax is not enough to pay for the subsidies. This requires increasing taxes or decreasing spending in the future, which the green districts do not support. To compensate, the policymaker can increase τ_1, which requires an increase in s. This works until the constraints cannot be jointly satisfied. However, as we see in the Figure, the policy does no longer pass a cost-benefit analysis for the policymaker before (PC_G) binds.

In the bottom-left panel, low values of λ leave politically feasible packages that G does not wish to implement, while high values make the two political constraints incompatible. Thus, the sequence starts only at intermediate values of λ.

The bottom-right panel illustrates how the probability of turnover affects equilibrium policies. In this example, more likely turnover (lower ρ) relaxes the political constraints, and leads to an increase in the carbon tax and the subsidy. Consistent with previous literature (Schmitt 2014; Harstad 2020; Behmer 2023), the probability of turnover increases the equilibrium subsidy, but in this model the carbon tax is significantly lower than the social cost of carbon, consistent with the sequencing pattern we observe in the data.

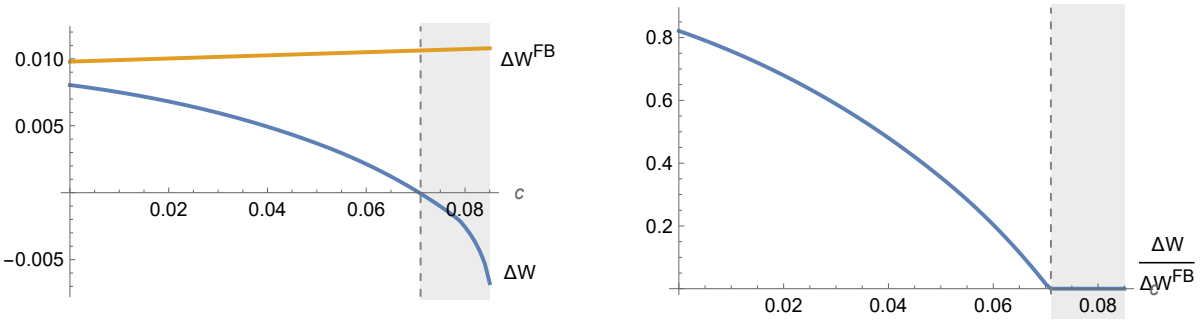


Figure 3: Welfare gains from the equilibrium and first-best policies relative to the status quo when $A = 1.8$, $\delta = 0.9$, $\lambda = 0.1$, and $\rho = 1$, as c changes. Gray regions indicate that the best politically feasible policy is no better than the status quo.

Figure 3 shows the welfare cost of political feasibility. As c grows, the political constraints become harder to satisfy. The improvement of equilibrium climate policy over the status quo decreases, and the distance from the first best policy increases, until feasible policy is no longer better than the status quo. The welfare loss relative to the first best can vary enormously. When $c = 0$, the equilibrium package captures more than 80 percent of the welfare gain from first-best policy in this example. At $c \approx 0.07$, the best politically feasible package is no better than the status quo, and no climate policy is enacted.

4.5. *Green Expectations*

I have considered so far the equilibrium in which districts expect no carbon taxes after a rejection of the first-period policy proposal. Here I consider an alternative equilibrium, the *green expectations* equilibrium, in which districts expect a green majority in the second period following

a rejection of the first-period proposal. This can happen because the expectation of a second-period carbon tax induces some districts to invest even in the absence of an investment subsidy (by Assumption 1 and equation Equation 2). If the number of districts that invest is enough to generate a green majority in the second period, this is an equilibrium.

Under what conditions is there a green expectations equilibrium? Suppose that the status quo policy $\tau_1 = s = 0$ remains, and there is a green majority in the second period. We have $\tau_2 = \lambda$ if G is still the proposer, which happens with probability ρ, and $\tau_2 = 0$ if there is turnover. Let $\tilde{b}_2$ be second-period marginal polluting district. By (2) we have

$$-c + \frac{\delta A}{2}\tilde{b}_2 = \frac{\delta}{2}\big[\rho(1-\lambda)^2 + 1 - \rho\big].$$

Therefore, this is an equilibrium if and only if $\tilde{b}_2 \leqslant \frac{1}{2}$. Using Assumption 1, this is equivalent to

$$b_1 \leqslant \frac{1}{2} + \frac{\rho\lambda(2-\lambda)}{A}. \tag{4}$$

Thus, for the green expectations equilibrium to exist, the initial share of polluting districts cannot be too large. A greater social cost of carbon λ and a smaller probability of turnover $1-\rho$ increase the upper bound, because they lead more districts to transition if they expect a carbon tax in the future. The fact that polluting districts expect a future carbon tax in the future absent any policy in the first period changes the political calculation for the green policymaker, because resisting a climate policy bundle (τ_1, s) is less attractive in this equilibrium than in the equilibrium considered in the previous section. The new political constraints are as follows:

$$\frac{1}{2}(1-\tau_1)^2 + s - c + \frac{\delta A}{4} + T \geqslant \frac{1}{2} + \underbrace{-c + \frac{\delta A}{4}}_{\substack{\text{transition} \\ \text{without subsidy}}} + \tilde{T} \tag{PC$'_B$}$$

and

$$\mathrm{E}(T) = b_1\tau_1(1-\tau_1) + \delta b_2 \mathrm{E}[\tau_2(1-\tau_2)] - s(b_1 - b_2) \geqslant \tilde{T} \equiv \delta \tilde{b}_2 \rho\lambda(1-\lambda), \qquad (\mathrm{PC}'_G)$$

where $\tau_2 = \max\{\tau_1, \lambda\}$ with probability ρ and $\tau_2 = \tau_1$ with probability $1-\rho$.

We have the following.

Proposition 4. *Suppose that $\lambda < \frac{1}{3}$. There is a green expectations equilibrium if and only if $\tilde{b}_2 \leqslant \frac{1}{2}$. In that case, $\tau_1 < \lambda$ and $s > 0$.*

An active climate policy is always feasible if green expectations are an equilibrium of the continuation game as long as λ is not too large, but it is still distorted relative to the first best. The proposer faces the same trade-off: a Pigouvian first-period carbon tax requires a positive subsidy to be feasible. Reducing the tax imposes a second-order loss on the proposer's objective, but makes it possible to reduce the subsidy, which has a first-order positive impact. Thus, green expectations help implement climate policy, but they do not eliminate the policy distortion. See Subsection A.1 for a numerical illustration.

The existence of the green expectations equilibrium suggests that *soft commitments* –non-binding declarations by governments that might be dismissed as mere "cheap talk"– can exert real effects on economic behavior and policy outcomes if they shift expectations of economic actors towards this equilibrium.[10] Examples of such soft commitments include widespread net-zero targets and the Nationally Determined Contributions (NDCs) under the Paris Agreement. Stiglitz (2019) articulates this logic in the context of the latter: "[P]art of the rationale for the Paris strategy [was that] if enough firms believed that there was enough global commitment to climate change that there would be a high carbon price (implicit or explicit) going forward, they would have an incentive to make green investments; and to ensure that they were advantaged over firms that didn't make such investments and to ensure that they obtained the desired returns on those investments,

[10]To be precise, my argument is that the proposer can induce districts to play the green expectations equilibrium by making it focal in the Schelling sense. This is reasonable because, even though the proposer strictly prefers the green expectations equilibrium, their action is not required to implement it. Whether districts play the green expectations equilibrium or not only depends on their expectations about each others' actions (plus subgame perfection).

they would then politically support, in coalition with other like-minded agents, a high carbon price." Ramadorai and Zeni (2024) show that, consistent with this idea, the announcement of the Paris Agreement led to a significant change in beliefs about future carbon taxation in a sample of North American public firms.

The analysis of the model, however, reveals that soft commitments cannot work in all circumstances. Carbon-abating technology needs to be sufficiently advanced, the social cost of carbon perceived to be sufficiently large, and the probability of turnover sufficiently low. Equation (4) makes the conditions precise. The fact that many governments undershot their emissions targets (de Silva and Tenreyro 2021) suggests that policymakers may not always be able to trigger a change in expectations that paves the ground for ambitious policies.

5. Extensions

Targeted Subsidies and Transfers. In Appendix A.2 I consider the implications of allowing the policymaker to target transfers and subsidies, which so far I assumed to be uniform. I first argue that arbitrary targeting is implausible because of an information asymmetry: an optimal targeting strategy would require knowing exactly the identity of the minimal set of polluting districts that are closest to the decarbonization frontier and form a winning coalition. Which constituencies are willing to make investments in the transition given the right incentives is a difficult empirical question. Moreover, to the extent that we observe targeting (for example, the allocation of free allowances in the initial phases of the EU ETS), it does not respond to an optimal coalition-building strategy, but can be explained as a result of lobbying by particularly powerful industries (Winkler 2022).

To take into account the information asymmetry I employ a mechanism design approach. I assume that legislators know the potential productivity of green capital investments in their district, but can choose to withhold that information. The policymaker proposes a menu of targeted taxes, subsidies and transfers. The legislature then votes on the entire menu, after which each district selects its preferred bundle. I find that the proposer can target subsidies and transfers to some

extent, but the incentive compatibility constraints protect the polluting districts excluded from the winning coalition, and prevent the policymaker from restricting the subsidy to only the pivotal districts. Consequently, the same fundamental distortion that arises with uniform subsidies and transfers emerges with this richer set of policy instruments.

Green Preferences. In Appendix A.3 I assume legislators intrinsically care about the environment. In this case climate policy is always implemented, even without a green majority in the second period. Moreover, under some conditions first-best policy can arise in equilibrium. But policy sequences as in Proposition 3 also arise under some conditions. In sum, even though green preferences relax political constraints, the model features the same distortions and dynamics as the baseline.

Production Subsidies, Standards and FITs. I considered so far investment subsidies as the inefficient policy instrument that the policymaker can use to build a winning coalition, but in practice other instruments are used for this purpose. In Appendix A.4 I show how a production subsidy in the green sector can be used with the purpose of inducing polluting districts to transition. In Appendix A.5 I introduce an emissions standard, and show that it is equivalent to a revenue neutral combination of a production subsidy and a carbon tax. I also show that a feed-in tariff (FIT) for goods produced using green capital is equivalent to a tradable emissions standard. I show that the policymaker can (under some conditions) use a standard or a FIT to start a policy sequence, because it implicitly subsidizes production in the green sector.

Imperfect Competition and Pass-through. The main model ignores market imperfections to focus exclusively on the climate externality. As a result, the incidence of the carbon tax falls solely on districts hosting polluting production. In Appendix A.6, I adapt the model from Besley and Persson (2023), which features monopolistic competition and thus allows for pass-through of carbon taxes to consumers. This extension shows that carbon taxes adversely affect all consumers—not only those in polluting districts as in the main model. Nevertheless, green districts still benefit on net after tax rebates, as long as the carbon price is not too large. I show that under certain conditions the same pattern I uncover in the main model emerges: political constraints induce policymakers

to use inefficient subsidies and set first-period carbon taxes below the optimal level.

Endogenous Reelection Probability. In Appendix A.7 I consider the implications of making the probability that the green proposer is reelected endogenous. Because the intensity of policy preferences matters in probabilistic voting models, the probability of reelection depends on the difference between the average expected payoff under each party, where the average is taken over all districts. Thus, even though, once the policy sequence starts, a majority is in favor of increasing the stringency of climate policy in the second period, increasing the salience (i.e., the electoral implications) of climate policy decreases the reelection chances of the green party. This is because voters do not internalize the climate externality in the baseline model, and, therefore, the average voter is against carbon taxes. Counterintuitively, increasing the first-period carbon tax does not lead to a backlash against the green party in the context of a policy sequence, because in that context voters understand that the carbon tax has been locked in, and therefore the anti-climate party cannot credibly promise to eliminate it. In fact, increasing the first-period carbon tax helps the green incumbent get reelected because it reduces the difference in the second-period policies that they and the polluting party implement, since the polluting party is forced to keep the first-period carbon tax in place. Increasing the size of the green subsidy also helps the green policymaker, because it reduces the share of the population affected by the carbon tax in the second period. However, in Appendix A.7 I show that the effect of salience on equilibrium policy is ambiguous. This finding, combined with Observation 3, implies that increasing the salience of climate policy can make it more likely that it is implemented, and can either increase or decrease its ambition, even though it also decreases the electoral prospects of green parties.

6. Conclusion

The model I developed in this paper can explain both the increase in stringency of climate policy over time and the shift from inefficient to efficient instruments. The main idea is that inefficient policies are politically effective if (1) politically pivotal polluters directly benefit from them, and (2) they induce these polluters to make investments that reduce their exposure to carbon pricing.

These two conditions explain why a coalition of green and polluting industries can support these policies, and why they reduce the opposition to carbon pricing over time. Subsidies for investments in pollution abatement technologies are the clearest example of such policies.

Several additional insights emerge from the analysis. First, if clean technologies are costly or immature (c is high or A is low), the economic case for subsidies is strong, because they can stimulate investment and learning-by-doing that markets would underprovide. But, as the analysis showed, these are precisely the conditions under which subsidies are not effective as *political* tools. This finding helps to understand why policy feedback has failed to take hold in several cases. Second, even if there is a policy bundle that is time-consistent and acceptable to a winning coalition of interests, its social costs may be so large that a climate-concerned policymaker will decide not to implement it. This result suggests that a fruitful direction for future research is to estimate empirically the costs that political acceptability adds to climate policies. Recently, researchers have made important methodological advances in the estimation of the costs and benefits of climate policies (Hahn *et al.* 2026), as well as in the design of policies that achieve acceptability by key political actors (Gazmararian and Tingley 2023); future research should combine those efforts in order to determine the cost of political acceptability.

Third, the analysis reveals that managing economic actors' expectations of future policy is important, and can be achieved both by policy design and by soft commitments. The latter can influence beliefs because future policies are affected by investments that economic actors make in the present. This feature creates a coordination game that can have multiple equilibria. Equilibrium policy, in turn, responds to these beliefs—expectations of stringent policy in the future strengthen the bargaining power of policymakers to implement more stringent policies in the present. Thus, voluntary commitments such as the nationally determined contributions made in the context of the Paris Agreement can complement policy enactment domestically.

The analysis of the policy sequencing strategy in this paper shows that it is not always feasible, that it can be interrupted by political turnover, and that it can produce large welfare losses relative to the first best. These conclusions follow even under the favorable assumption that the political effects

of green industrial policies arrive before political turnover. If an election takes place before there are enough investments in decarbonization to tip the balance of power between polluters and green constituencies, and an anti-climate policymaker is elected, we can expect the logic of the sequencing argument to work in reverse: a policymaker opposed to climate policy has dynamic incentives to repeal green subsidies and stall the deployment of green technologies to prevent further growth of the green coalition, even if this is inefficient or entails short-term political costs. The Trump Administration's 2025 rollback of key subsidies in the Inflation Reduction Act of 2022, despite opposition from Republican lawmakers and business lobbies, is consistent with this mechanism.[11] Studying the politics of climate-policy retrenchment is fertile ground for future work.

[11] In March 2025, 21 House Republicans urged Ways & Means Chair Jason Smith to spare the IRA's production and manufacturing credits (Garbarino *et al.* 2025). Their stance dovetailed with lobbying by major clean-energy trade groups and some fossil-fuel and GOP allies (see, e.g., American Clean Power Association 2025; Fuel Cell & Hydrogen Energy Association *et al.* 2025).

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A. Extensions

A.1. *Green Expectations*

Figure 4 shows how green expectations affect equilibrium policy. When they are possible (for large A, small c and large λ), demarcated by the green regions, the first-period carbon tax is increased, and the subsidy decreased, relative to the baseline equilibrium.

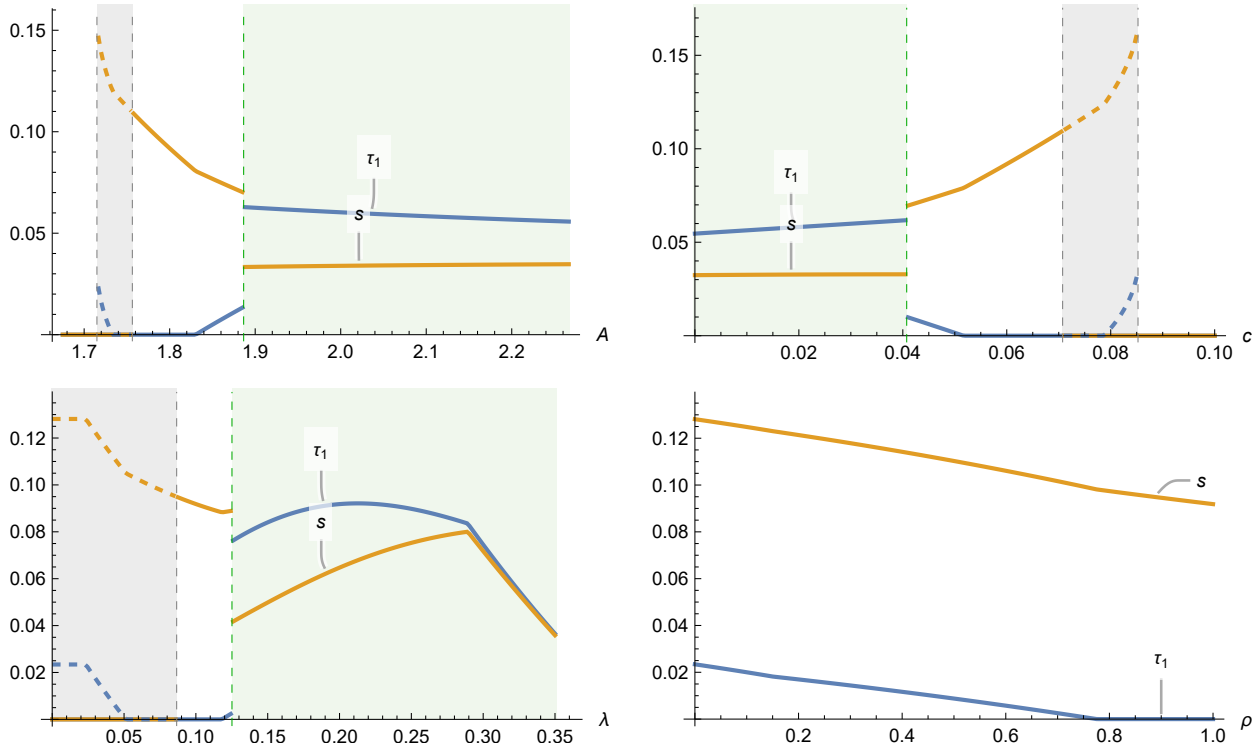


Figure 4: Equilibrium policy when $A = 1.8$, $c = 0.06$, $\delta = 0.9$, $\lambda = 0.1$ and $\rho = 1$, as each parameter changes. In the gray regions the proposer prefers the status quo over the optimal feasible policy (the dashed lines). In the green regions the green equilibrium exists and is displayed.

A.2. *Targeted Transfers and Subsidies*

If policymakers can use arbitrarily targeted lump-sum transfers and have perfect information, optimal climate policy is simple: the government can offer each polluting firm in the economy a transfer contingent on technology adoption that makes the firm indifferent between upgrading or not. This requires no commitment, because lump-sum transfers are costless for proposers. One reason why this is not feasible in reality is that the government lacks information: it does not know what the optimal abatement actions are.

There are two facts that put this argument into question. First, governments do use targeted transfers to obtain political support for policies. This is clear in the case of free allowances in cap-and-trade systems, and is the basis of the idea of "just transition" strategies (Bolet *et al.* 2023), which bundle climate with redistributive policies to support affected communities, and "green bargains" (Meckling and Strecker 2023), which tie regulations to public investments. But, for example, in the first two phases of the EU ETS, even though there is evidence that lobbying by particular firms impacted the allocation of free allowances, policymakers used rigid rules (mainly based on historical emissions) and a large information asymmetry dominated the process (Ellerman *et al.* 2007). The fact that more than 90% of the allowances were allocated for free, plus the fact

that there was substantial overallocation in several industries (i.e., they produced less emissions than the allowances they were given for free; see, e.g., Hanoteau 2014), indicates that the allocation was not targeted in a surgically precise manner to assuage opposition.

If we consider the possibility of quid-pro-quo, which is the implicit assumption of common agency or rent-seeking models, then we should expect a targeted (but distorted) allocation. This targeting, however, responds to the incentives and ability for lobbying of each firm or industry, which can be (and should be expected to be) very different to the targeting that results from a strategy of building a "green coalition"—the optimal green coalition is formed by the set of constituencies that have the greatest expected opportunities in a decarbonized economy. In contrast, the firms or industries most willing to lobby may include declining industries (see Grossman and Helpman 1996 and Baldwin and Robert-Nicoud 2007) and industries with assets that will lose value with decarbonization. This argument helps to justify that a coalition-building strategy cannot plausibly rely on finely targeted transfers, despite the fact that we see quite targeted transfers as part of climate policies in practice.

The second counterargument is that in practice subsidies and regulations tend to be technology- and industry-specific. See, for example, Gawel *et al.* (2017) on the FITs in Germany, and Hahn *et al.* (2026) for a list of specific subsidies in the US. Moreover, Cullenward and Victor (2020) argue that industry-specific regulations are not just prevalent but desirable, since broad policies, despite having the potential for being more efficient, tend to be watered down by the pressure of the most affected industries (the authors do not provide systematic evidence or a sound theoretical argument for this assertion, though; theoretically, targeted interventions may lead to less internalization, and, thus, more lobbying in opposition). However, the fact that in Germany the demand for FITs exploded beyond expectations, and the subsidies in the IRA are uncapped and there is considerable uncertainty about their fiscal cost (Bistline *et al.* 2023), suggest again that subsidies in practice are far from being surgically targeted and policymakers face substantial uncertainty when designing them.

To address these arguments in the context of my model I can relax the assumption that policymakers cannot propose targeted transfers, but also relax the assumption that proposers have complete information. To put structure, suppose that the proposer knows the distribution of productivities A_i (i.e., they know that A_i has a uniform distribution on $[0, A]$), but not the particular value of A_i for any i. In that case, they can design the policies that I analyzed in the main model, but they cannot implement individually targeted rebates or subsidies. The policy choice decision given the information constraint becomes a mechanism design problem. Following Mas-Colell and Vives (1993), an (anonymous direct) *mechanism* is a function $f : [0, A] \to \mathbb{R}^3_+$ that takes the productivity of green technology of a district (which is the relevant characteristic) and assigns a

specific carbon tax, a green subsidy and a tax rebate.[12] It is *strongly incentive compatible* (IC) if each district chooses to reveal their productivity type, i.e., district i communicates A_i and is subject to the policy $f(A_i)$, rather than any other productivity level A_j. The timing of the game is left unchanged: in each period, the agenda-setter proposes an IC mechanism to the legislature (instead of a uniform policy as in the main model); each legislator votes in favor of the proposal if and only if they prefer it over the status quo, and then firms submit their productivity levels to the mechanism, which assigns to them a carbon tax, a green subsidy and a tax rebate, and then decide their production and investment levels.

In the second period, green subsidies are irrelevant, IC mechanisms assign policies τ, r^b for polluters and r^g for green districts, and the IC constraint is that $\frac{1}{2}(1-\tau^b)^2 + r^b \geqslant r^g \geqslant r^b$. If there is a polluting majority, polluters must accept the mechanism for it to be politically implemented. If the alternative is no climate policy then the political constraint is $\frac{1}{2}(1-\tau)^2 + r^b + T \geqslant \frac{1}{2}$, where $T = b\tau(1-\tau) - br^b - (1-b)r^g \leqslant b\tau(1-\tau) - r^b$ (by IC), but $\frac{1}{2}(1-\tau)^2 + r^b + T - \frac{1}{2} \leqslant \frac{1}{2}(1-\tau)^2 + b\tau(1-\tau) - \frac{1}{2} = -[(1-b)\tau + (b-\frac{1}{2})\tau^2] \leqslant 0$, so the political constraint holds only if $\tau = 0$. In other words, if the alternative policy is void, it is impossible to implement a positive carbon tax. If greens are a majority, in contrast, any carbon tax is politically implementable.

Incentive-compatible mechanisms in the first period consist of three policies: a tax and rebate for polluting firms that do not transition, (τ_1^b, r_1^b), a tax, subsidy and rebate for firms that transition, $(\tau_1^{bg}, s, r_1^{bg})$, and a rebate for green firms r_1^g. Incentive compatibility imposes the following conditions: $\frac{1}{2}(1-\tau_1^b)^2+r_1^b \geqslant \frac{1}{2}(1-\tau_1^{bg})^2+r_1^{bg}$, i.e., polluting firms that do not plan to transition do not take the tax and rebate intended for firms that do transition, and $\frac{1}{2}(1-\tau_1^{bg})^2+s+r_1^{bg} \geqslant \frac{1}{2}(1-\tau_1^b)^2+r_1^b$, i.e., polluting firms that plan to transition do not take the tax and rebate intended for firms that do not. In addition, $\frac{1}{2}(1-\tau_1^b)^2 + r_1^b \geqslant r_1^g \geqslant r_1^b$ and $\frac{1}{2}(1-\tau_1^{bg})^2 + s + r_1^{bg} \geqslant r_1^g \geqslant r_1^{bg}$, i.e., polluters and greens do not impersonate each other.

To focus on the information asymmetry, I assume that there is no turnover, i.e., $\rho = 1$. The political constraints demand that polluting firms in $[\frac{1}{2}, b_1]$ and green firms accept the mechanism that implements $(\tau_1^b, r_1^b, \tau_1^{bg}, s, r_1^{bg}, r_1^g)$ in the first period, and green firms accept the mechanism (τ_2, r_2^b, r_2^g) in the second period. The polluting firms that do not transition have no say in the legislature, but they are protected in the first period by the incentive compatibility constraint. The set of firms that transition is given by

$$\tilde{s} - c + \frac{\delta A}{2} b_2 = \frac{\delta}{2}(1-\tau_2)^2,$$

where $\tilde{s} = s - \Delta_1 + \Delta_2 \geqslant 0$, $\Delta_1 = \frac{1}{2}(1-\tau_1^b)^2 + r_1^b - \frac{1}{2}(1-\tau_1^{bg})^2 - r_1^{bg} \geqslant 0$, $\Delta_2 = \delta(r_2^g - r_2^b) \geqslant 0$.

[12] I focus on anonymous mechanisms in order to abstract from information leakage and ratcheting across periods.

The political constraint for polluting districts in the first period can be written as

$$\frac{1}{2}(1-\tau_1^{bg})^2 + s - c + \frac{\delta A}{4} + r_1^{bg} + \delta r_2^g + T \geqslant \frac{1}{2} + \frac{\delta}{2},$$

where

$$\begin{aligned} T = {} & b_2[\tau_1^b(1-\tau_1^b) - r_1^b] + (b_1 - b_2)[\tau_1^{bg}(1-\tau_1^{bg}) - s - r_1^{bg}] - (1-b_1)r_1^g \\ & + \delta b_2[\tau_2^b(1-\tau_2^b) - r_2^b] - \delta(1-b_2)r_2^g. \end{aligned}$$

The political constraint for green districts is

$$r_1^g + \delta r_2^g + T \geqslant 0$$

in the first period and

$$r_2^g + b_2\tau_2^b(1-\tau_2^b) - b_2 r_2^b - (1-b_2)r_2^g \geqslant r_1^g + b_2\tau_1^b(1-\tau_1^b) - b_2 r_1^b - (1-b_2)r_1^g$$

in the second period.

The political constraints are relaxed to some extent relative to uniform rebates and subsidies, but the key inefficiency arises from the fact that the subsidy is used for political acceptability, but it spills over to wasteful investment decisions. The main driver of the results in the baseline model, viz., the fact that the political constraint creates a tradeoff between the first-period carbon tax and the subsidy (a larger carbon tax requires a larger subsidy to compensate the pivotal district, since rebates cannot be perfectly targeted to offset the loss in profits), persists in this variation of the model.

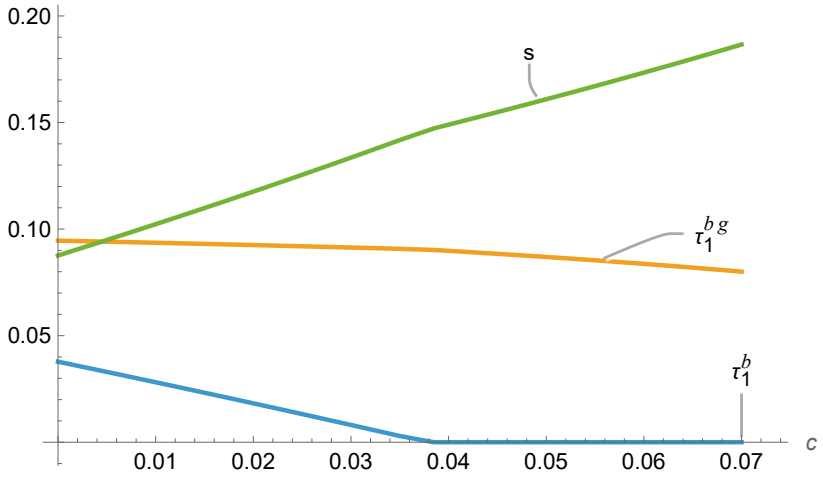


Figure 5: Equilibrium policy when $A = 1.8$, $\delta = 0.9$ and $\lambda = 0.1$, for different values of c.

Figure 5 shows a numerical example. The specific rebates are not used in equilibrium. As in the main model, the carbon tax in the first period is below the Pigouvian level, and the investment subsidy is used. The districts that transition pay a higher carbon tax rate than the ones that do not, even though the latter are left out of the political coalition. The reason is that, without specific rebates, the incentive compatibility constraint $\frac{1}{2}(1-\tau_1^b)^2+r_1^b \geqslant \frac{1}{2}(1-\tau_1^{bg})^2+r_1^{bg}$ implies $\tau_1^b \leqslant \tau_1^{bg}$. If we use the rebate r_1^b to obtain a larger τ_1^b, we must compensate the pivotal district (who bears part of the fiscal cost of the rebate); extracting from the green districts is not feasible because of their own political constraint; therefore increasing r_1^b requires increasing $\frac{1}{2}(1-\tau_1^{bg})^2+s$, which creates a welfare loss. In other words, the flexibility to provide specific policies is useful, but the incentive compatibility condition protects the polluters that are excluded from the political coalition. This, in turn, leads to the same tradeoff as in the main model: an optimal first-period carbon tax requires inefficient compensation to pivotal districts, and is, therefore, not used in equilibrium.

A.3. *Green Preferences*

In this section I assume that districts care about environmental damages as much as the green proposer, so payoffs are the same as before with an added term $-\lambda e_t$, i.e.,

$$\pi_{it} = \begin{cases} (1-\tau)y_{it} - \frac{1}{2}y_{it}^2 - \mathbb{1}_{t=1}\iota_i(c-s) - \lambda e_t & \text{if } i \text{ is polluting,} \\ y_{it} - \frac{1}{2Ai}y_i^2 - \lambda e_t & \text{if } i \text{ is green,} \end{cases}$$

where $e_t = \int_0^1 \chi_{it} y_{it}\, di$ are aggregate emissions at time t. For simplicity I assume that the first-period proposer stays in the second period, i.e., $\rho = 1$. In a one-period interaction, if the polluting districts are $[0, b)$ and the carbon tax is τ, the payoff of a polluting district is $\frac{1}{2}(1-\tau)^2+b\tau(1-\tau)-\lambda b(1-\tau)$. If $(1+\lambda)b > 1$, which requires $b > \frac{1}{2}$, then their ideal tax is $\tau = \frac{(1+\lambda)b-1}{2b-1} < \lambda$. Otherwise, their ideal tax is $\tau = 0$.

Assumption A.1. $(1+\lambda)b_1 < 1$.

Under Assumption A.1, polluting districts always oppose a carbon tax.

As in the baseline model, the green districts prefer a high carbon tax. But, given that they care about carbon emissions, they are now willing to pay some taxes themselves in order to pay for investment subsidies if that is required to obtain a reduction in GHGs emissions.

Polluting districts impose a political constraint. The proposer needs to induce $b_2 \leqslant \frac{1}{2}$ and obtain the approval of the median district, i.e.,

$$\underbrace{\frac{1}{2}(1-\tau_1)^2 + s - c + \frac{\delta A}{4} + T}_{\text{economic payoff}} - \underbrace{\lambda b_1(1-\tau_1) + \delta\lambda b_2(1-\tau_2)}_{\text{environmental damage}} \geqslant \underbrace{(1+\delta)\left(\frac{1}{2} - \lambda b_1\right)}_{\text{status quo payoff}}. \qquad (\text{PC}_B)$$

Define the gain of an initially green district relative to the status quo as $G \equiv T + \lambda b_1 \tau_1 + \delta\lambda[b_1 - b_2(1-\tau_2)]$.

Let P^N denote the gain of a polluting district that does not invest. Since such a district bears the carbon tax in both periods, while an initially green district does not, we have $P^N \equiv G - (\tau_1 - \tau_1^2/2) - \delta(\tau_2 - \tau_2^2/2)$.

Finally, whenever $b_2 > 0$, let P^I denote the gain of the median polluting district when it invests. Using the investment cutoff, $s - c + \delta A b_2/2 = \delta(1-\tau_2)^2/2$, we can write $P^I \equiv P^N + \delta A(1/2 - b_2)/2$.

Thus $G \geqslant 0$, $P^N \geqslant 0$, and $P^I \geqslant 0$ are respectively the political constraints of the initially green districts, a polluting district that does not invest, and the median polluting district when it invests. We have the following.

Proposition A.1. *Suppose that $\rho = 1$ and Assumptions 1, 2, and A.1 hold. The status quo is never an equilibrium outcome. Every equilibrium is of one of the following three types.*

(i) Partial transition. *Conditional on a polluting-majority continuation, (τ_1, s) maximizes W subject to $P^N \geqslant 0$. We have $b_2 > \frac{1}{2}$, $\tau_1 = \tau_2 < \lambda$, and $s > 0$.*

(ii) Policy sequence. *Conditional on a green-majority continuation, (τ_1, s) maximizes W subject to $P^I \geqslant 0$ and $G \geqslant 0$. We have $0 < b_2 < \frac{1}{2}$, $\tau_1 < \lambda$, $\tau_2 = \lambda$, and $s > 0$.*

(iii) First best. *We have $\tau_1 = \tau_2 = \lambda$ and $s = 0$.*

The main difference with the baseline model is that climate policy is always implemented, even without a green majority in the second period. Moreover, under some conditions first-best policy is feasible. However, the same distortions in the baseline model are present, and under some conditions the equilibrium has the same qualitative pattern as the baseline. Thus, this extension can also explain policy sequences.

A.4. *Production Subsidies*

Consider a production subsidy for the green sector $\sigma \geqslant 0$. If a firm in district i produces y_i, the government pays them σy_i. Given σ, the firm chooses y_i to maximize profits, $\pi_i = (1+\sigma)y_i - \frac{1}{2Ai}y_i^2$, so $y_i = (1+\sigma)Ai$. Thus, equilibrium profits are $\pi_i = (1+\sigma)^2 \frac{1}{2}Ai$, and the fiscal cost is $\sigma(1+\sigma)Ai$, so the contribution to aggregate welfare is $(1+\sigma)^2\frac{1}{2}Ai - \sigma(1+\sigma)Ai = \frac{1}{2}(1-\sigma^2)Ai$. This is clearly decreasing in σ, which means that green production subsidies are nothing more than an inefficient form of redistribution in this model: they transfer income to green producers without reducing carbon emissions. (This result is, of course, not robust: green production subsidies do reduce carbon emissions in more general models; my model captures in an extreme way the fact that production subsidies are an inefficient policy instrument for reducing carbon emissions.)

To affect investment, the policymaker could use a production subsidy that starts acting in the second period. The political constraint for polluting districts $[\frac{1}{2}, b_1)$ is

$$\frac{1}{2}(1-\tau_1)^2 + s - c + \underbrace{(1+\sigma)^2\frac{\delta A}{4}}_{\substack{\text{period-2 profit} \\ \text{with production subsidy}}} + T \geqslant (1+\delta)\frac{1}{2}, \tag{A1}$$

and the political constraint for the green districts $[b_1, 1]$ is

$$\underbrace{(1+\sigma)^2\frac{\delta A}{2}b_1}_{\substack{\text{period-2 profit} \\ \text{with production subsidy}}} + T \geqslant \frac{\delta A}{2}b_1, \tag{A2}$$

where

$$T = \underbrace{b_1\tau_1(1-\tau_1) + \delta b_2\tau_2(1-\tau_2)}_{\text{carbon tax revenue}} - \underbrace{s(b_1-b_2)}_{\substack{\text{fiscal cost of the} \\ \text{investment subsidy}}} - \underbrace{\delta\int_{b_2}^{1}\sigma(1+\sigma)Ai\,di}_{\substack{\text{fiscal cost of the} \\ \text{production subsidy}}}.$$

The objective of the proposer is

$$W = \frac{b_1}{2}(1-\tau_1^2) + \frac{A}{4}(1-b_1^2) - c(b_1-b_2) - \lambda b_1(1-\tau_1) + \delta\left[\frac{b_2}{2}(1-\tau_2^2) + \underbrace{(1-\sigma^2)}_{\substack{\text{distortion due to the} \\ \text{production subsidy}}}\frac{A}{4}(1-b_2^2) - \lambda b_2(1-\tau_2)\right],$$

where b_2 is now given by $s - c + (1+\sigma)^2\frac{\delta A}{2}b_2 = \frac{\delta}{2}(1-\tau_2)^2$, and $\tau_2 = \max\{\tau_1, \lambda\}$.

If both a production and an investment subsidy are available, the proposer does not use the production subsidy.

Proposition A.2. *If s and σ are available, in every equilibrium $\sigma = 0$.*

The investment subsidy is a better instrument in this model. This justifies the focus on an investment subsidy in the baseline model. If the investment subsidy is not available, the production subsidy can be used to start a policy sequence.

There are theoretical reasons for subsidizing output rather than inputs (capital), but in general it depends. If the objective is to increase output, then subsidizing inputs will distort productive efficiency, so subsidizing output is prima facie better. However, even taking into account this distortion, it may be cheaper to subsidize one input if it's a lot more elastic than another one. See Parish and McLaren (1982) for an exposition. Now, even though this is theoretically ambiguous,

Aldy *et al.* (2023) show that, empirically, production subsidies for wind energy are more cost effective than investment subsidies. However, in settings such as wind generation, environmental benefits arise at the production margin because additional renewable output displaces polluting energy generation. In my model, by contrast, firms decarbonize existing production through technology adoption, so the relevant environmental margin is investment rather than output. Investment subsidies are therefore better targeted in my model.

I show next that production subsidies can be useful in equilibrium to unblock political constraints:

Proposition A.3. *Suppose that s is not available. If the status quo is replaced, then $\tau_1 < \lambda$, $\tau_2 = \lambda$, and $\sigma > 0$ iff $Ab_1 > 6\lambda - 5\lambda^2$.*

The only difference with Proposition 3 is that under some conditions the proposer implements climate policy without using the subsidy. The intuition for $Ab_1 > 6\lambda - 5\lambda^2$ is that the production subsidy is a good tool to relax the political constraint when the extra rents to the pivotal polluter exceed the fiscal cost. This cost is small when the number of green districts is small (so b_1 is large) and the reduction in the revenue from the carbon tax due to fewer polluters in the second period is small (λ small).

A.5. *Standards and Feed-in Tariffs*

Consider a clean production standard $\mu \in [0, 1]$, such as a RPS (see Helfand 1991; Holland *et al.* 2009; Holland 2012; Schmalensee 2012). It forces firms to emit no more than μ units of GHGs per unit of the good produced. Firms that produce more emissions than allowed can buy permits from firms that produce less emissions than allowed for a market price p. In the model, firms that use green capital do not emit GHGs, so if they produce y, they are allowed to emit μy, which they can sell, earning $p\mu y$. Firms with polluting capital emit one unit of GHGs per unit produced, so they need to buy $(1-\mu)y$ allowances, paying a cost $p(1-\mu)y$. Profit maximization implies that $y_i = (1 + p\mu)Ai$ in green districts i, and $y_i = 1 - (1-\mu)p$ in polluting districts. The price of allowances p is set by the market clearing condition:

$$\underbrace{\int_0^b (1-\mu) y_i \, di}_{\substack{\text{permits bought} \\ \text{by polluting firms}}} = \underbrace{\int_b^1 \mu y_i \, di}_{\substack{\text{permits sold} \\ \text{by green firms}}},$$

which implies $\int_0^b y_i\,di = \mu\int_0^1 y_i\,di$, so, as intended, emissions are capped at a fraction μ of aggregate production. The equilibrium price of the permits is

$$p = \frac{(1-\mu)b - \mu\frac{A}{2}(1-b^2)}{(1-\mu)^2 b + \mu^2\frac{A}{2}(1-b^2)}$$

if $b \geqslant \mu\Big(b + \frac{A}{2}(1-b^2)\Big)$, i.e., if the standard is binding: emissions per unit produced exceed μ in the absence of climate policy. Otherwise, the price is 0. The clean production standard is thus equivalent to a carbon tax $\tau = p(1-\mu)$ combined with a green production subsidy $p\mu$.

Let $\tau = p(1-\mu)$ and $\sigma = p\mu$ be the equivalent carbon tax and green production subsidy given a standard μ. The market clearing condition can be written in terms of τ and σ as

$$b\tau(1-\tau) = \frac{A}{2}(1-b^2)\sigma(1+\sigma).$$

Notice that the LHS is the fiscal revenue from the equivalent carbon tax τ, and the RHS is the fiscal cost of the equivalent production subsidy σ, so the market clearing condition is effectively a joint revenue neutrality condition for the tax and the subsidy. Thus, choosing a standard is equivalent to choosing a carbon tax τ and using the revenue to finance a green production subsidy σ. The greater the tax τ, the greater the subsidy σ, as long as $\tau \leqslant \frac{1}{2}$.

We can introduce a feed-in tariff (FIT) into the model as follows. A retailer is forced to buy the numéraire good at price $p_G \geqslant 1$ from green producers, and sets the price $p_B \geqslant 0$ it buys from polluting producers to maximize profits from selling the good, whose price is fixed at 1. Profits are given by

$$\begin{aligned}
\pi &= (1-p_B)\int_0^b y_i\,di - (p_G-1)\int_b^1 y_i\,di \\
&= (1-p_B)\int_0^b p_B\,di - (p_G-1)\int_b^1 Aip_G\,di \\
&= -\frac{A}{2}(1-b^2)p_G(p_G-1) + bp_B(1-p_B).
\end{aligned}$$

Perfect competition or free entry in the retail market brings profits to zero, so the price paid to polluting producers is given implicitly by $bp_B(1-p_B) = \frac{A}{2}(1-b^2)p_G(p_G-1)$. A solution exists only if $\frac{A}{2}(1-b^2)p_G(p_G-1) \leqslant b/4$. On this domain, I select the solution $p_B \geqslant \frac{1}{2}$, which is continuous from $p_B = 1$ at $p_G = 1$. This is again equivalent to a revenue-neutral combination of a carbon tax $\tau = 1 - p_B$ and an output subsidy for green producers $\sigma = p_G - 1$. Thus, in this simple model a standard and a FIT are equivalent. (The instruments differ in reality if, for example, there is uncertainty about demand or productivity. A standard fixes the quantity of green production,

creating price risk for clean producers; a FIT fixes the price of renewable energy, which reduces risk for producers and can stimulate investment, but increases risk for consumers. See Schmalensee 2012.)

If a carbon tax and a production subsidy are available, a green proposer would not use the standard or FIT, because its effects can be replicated, and in general improved, by using the other two instruments. However, if the carbon tax and the subsidy are not available, the standard can be used in equilibrium. Notice that polluting districts will oppose any binding standard in a static environment. However, if the median district transitions, they expect to receive a subsidy in the second period, which acts as an investment incentive, and, thus, as a carrot for accepting a standard today. The policymaker cannot commit in the first period to a generous standard in the future, though, except by setting the standard high in the first period and relying on the fact that, once a majority of districts is green, they will not accept a reduction in the standard, because they profit from the allowances. However, increasing the standard in the first period creates a cost for the median district that exceeds the benefit it provides by increasing the implicit subsidy in the future.

The green policymaker chooses the second period standard (τ_2, σ_2) to maximize W given b_2. Anticipating this, the share of polluting districts that decide to transition in the first period is given by

$$-c + (1+\sigma_2)^2 \frac{\delta A}{2} b_2 = \frac{\delta}{2}(1-\tau_2)^2.$$

The political constraint in the first period is

$$\frac{1}{2}(1-\tau_1)^2 - c + (1+\sigma_2)^2 \frac{\delta A}{4} - (1+\delta)\frac{1}{2} \geqslant 0. \qquad (\mathrm{PC}_G)$$

Notice that in the first period the policymaker cannot affect σ_2 in order for PC_G to hold, except by making $\sigma_1 > \sigma_2$, but, as mentioned before, this is counterproductive as it increases τ_1.[13] Therefore, the policymaker can only implement a standard in the first period if PC_G holds for $\tau_1 = 0$. In that case, they can increase τ_1 until PC_G binds or the efficiency cost of the subsidy σ_1 is greater than the benefit of the tax. In the first case, the first-period standard is too low relative to the optimum choice.

Thus, climate policy in the first period relies on the second-period standard being large enough that it provides an incentive for the median district to invest. This, in turn, requires that the social cost of carbon λ is large enough to justify a stringent standard. In fact, for large λ the standard may be feasible when a carbon tax plus an investment subsidy is not, because the subsidy has to be paid by green districts as well as polluting districts, which makes it less appealing to the pivotal

[13] Suppose $\sigma_2 = \sigma_1$. Differentiating the fiscal neutrality condition yields $b_1(1-2\tau_1)\frac{d\tau_1}{d\sigma_1} = \frac{A}{2}(1-b_1^2)(1+2\sigma_1)$. Differentiating the LHS of PC_G we obtain $-(1-\tau_1)\frac{d\tau_1}{d\sigma_1} + \frac{\delta A}{2}(1+\sigma_2) = \frac{A}{2}\left[-\frac{1-b_1^2}{b_1}\frac{1-\tau_1}{1-2\tau_1}(1+2\sigma_1) + \delta(1+\sigma_2)\right] \leqslant 0$ if $1-b_1^2 \geqslant b_1$, which is true if $b_1 \leqslant 0.618$. In that case, as claimed, increasing σ_1 beyond σ_2 is counterproductive.

polluting district, to the point where the political constraints become infeasible. This seems to contradict the assertion I made before that the standard can be improved upon by a combination of a tax and a subsidy. What's happening is that if the other instruments are available the policymaker will use them instead of the standard, but, given the time-inconsistency problem, the fact that the policymaker has that power may make it harder to satisfy the political constraints.

A.6. *Imperfect Competition and Pass-through*

In this section I adapt the model in Besley and Persson (2023) to incorporate an investment subsidy and legislative bargaining. As in the main model, there is a set of districts indexed by $i \in I = [0, 1]$. Each district produces a specific variety with a clean or dirty production technology, using labor as an input. Producing x_i units of variety i using the dirty technology costs x_i units of labor, and releases x_i units of carbon. Producing x_i units using the clean technology costs $c(i)x_i$ units of labor, with c decreasing, and has a fixed cost f. There is a numéraire good produced competitively using labor with unit cost. The wage is therefore equal to 1. Each district is endowed with L units of labor.

Consumer preferences in each period are given by

$$U = \frac{1}{1-\sigma}\int_0^1 x_i^{1-\sigma}\,di + n,$$

where x_i is the quantity of variety i, n is the quantity of the numéraire consumed, and $\sigma \in (0, 1)$ is an inverse measure of the elasticity of substitution between varieties. Let p_i be the price of variety i. The representative consumer in district i chooses $x_i^{-\sigma} = p_i$, assuming that L is large enough so that the solution is interior.

Producers are monopolistic and maximize profits $\pi_i = (p_i - \tau_i - c_i)x_i - f_i$, where τ_i is a production tax, c_i is the marginal labor cost (i.e., $c_i = 1$ if i is polluting, $c_i = c(i)$ if i is green, employing a slight abuse of notation), and f_i is the fixed cost of production (i.e., $f_i = 0$ if i is polluting, $f_i = f$ if i is green). Therefore, they choose prices $p_i = \frac{1}{1-\sigma}(c_i + \tau_i)$. Profits are thus $\pi_i = \sigma p_i x_i - f_i$.

Given that districts are ordered by the productivity of the green technology, varieties $i \in [0, b)$ are produced with the dirty technology, and $i \in (b, 1]$ are produced with the clean technology, for some $b \in [0, 1]$.

The planner's problem is to choose $(x_i)_{i\in I}$ and b to maximize

$$\frac{1}{1-\sigma}\int_0^1 x_i^{1-\sigma}\,di - \int_0^1 (c_i x_i + f_i)\,di + L - \lambda\int_0^b x_i\,di,$$

where the last term measures climate damages. The solution is given by $x_i^{-\sigma} = c_i + \mathbb{1}_{i\leqslant b}\lambda$ and

$\frac{\sigma}{1-\sigma}((1+\lambda)^{1-\frac{1}{\sigma}} - c(b)^{1-\frac{1}{\sigma}}) + f = 0$. We can implement the social optimum using production taxes τ_i and a subsidy for the fixed cost of green varieties s. To find the optimal taxes we use that in the decentralized equilibrium $x_i^{-\sigma} = p_i = \frac{1}{1-\sigma}(c_i + \tau_i)$, and therefore we can implement the optimal quantities if $\frac{1}{1-\sigma}(c_i + \tau_i) = c_i + \mathbb{1}_{i \leqslant b}\lambda$, i.e.,

$$\tau_i = \mathbb{1}_{i \leqslant b}(1-\sigma)\lambda - \sigma c_i.$$

The first term is the usual carbon tax, which is equal to the social cost of carbon λ multiplied by $1-\sigma$ in order to correct for the market imperfection. The second term corrects the underproduction due to market power. The marginal green producer is indifferent between profits using each technology, taking into account the fixed cost and the green subsidy, and thus b is given in the decentralized equilibrium by $\sigma c(b)^{1-\frac{1}{\sigma}} - f + s = \sigma(1+\lambda)^{1-\frac{1}{\sigma}}$. Therefore, to implement the optimal b we need $s = \sigma f$, which is due to the fact that profits do not fully cover fixed costs.

Putting everything together, to implement the optimal allocation we need (1) production subsidies $\sigma(c_i x_i + f_i)$ given variable cost c_i, production quantity x_i and fixed cost f_i, and (2) a carbon tax $(1-\sigma)\lambda x_i$ given a quantity x_i produced with dirty technology. The first instrument corrects the distortion due to the market imperfection, and the second instrument corrects the distortion due to the climate externality. In order to focus on the climate issue I will assume that the instrument (1) is already in place, and, as always, the tax revenue is rebated as a uniform lump-sum to every district. For notational convenience I will write the carbon tax as $(1-\sigma)\tau$, noting that the optimal tax is $\tau = \lambda$, and the fixed cost as σf.

Thus, as in the main model, there are two climate policies: a carbon tax $\tau \geqslant 0$ that collects $(1-\sigma)\tau x_i$ from polluting districts, and a green subsidy $s \geqslant 0$ that pays $\delta\sigma s$ to polluting districts that switch to green technology ($\delta\sigma$ is a normalizing factor). The policy process is the same as in the main model.

Given these policy instruments, statically we have $p_i = x_i^{-\sigma} = c_i + \mathbb{1}_{i \leqslant b}\tau$ and the variable component of profits (taking production subsidies and carbon taxes into account) is $\pi_i = \sigma p_i x_i$. Utility from consumption of varieties is $\frac{1}{1-\sigma}\int_0^1 x_i^{1-\sigma}\,di - \int_0^1 p_i x_i\,di = \frac{\sigma}{1-\sigma}\int_0^1 x_i^{1-\sigma}\,di$. The tax rebate, ignoring the green subsidy, is $T = -\sigma\int_0^1 c_i x_i\,di + (1-\sigma)\int_0^b \tau x_i\,di = -\sigma\int_0^1 p_i x_i\,di + \tau\int_0^b x_i\,di$. Putting everything together, we define

$$\begin{aligned} V(\tau, b) &\equiv \frac{1}{1-\sigma}\int_0^1 x_i^{1-\sigma}\,di - \int_0^1 p_i x_i\,di + T \\ &= \frac{\sigma^2}{1-\sigma}\int_0^1 x_i^{1-\sigma}\,di + \tau\int_0^b x_i\,di \\ &= \frac{\sigma^2}{1-\sigma}\left(b(1+\tau)^{1-\frac{1}{\sigma}} + \int_b^1 c(i)^{1-\frac{1}{\sigma}}\,di\right) + b\tau(1+\tau)^{-\frac{1}{\sigma}}. \end{aligned}$$

We can write the payoff of district i as

$$U_i = V(\tau_1, b_1) + \delta V(\tau_2, b_2) + \pi_{i1} + \delta\pi_{i2} - \delta\sigma((1-\sigma)f - s)\iota_i - \delta\sigma(s + \sigma f)(b_1 - b_2),$$

with $\pi_{it} = \sigma[\mathbb{1}_{i\leqslant b_t}(1+\tau_t)^{1-\frac{1}{\sigma}} + \mathbb{1}_{i>b_t}c(i)^{1-\frac{1}{\sigma}}]$, and $\iota_i = \mathbb{1}_{i\in(b_2,b_1)}$, i.e., an indicator of technology switching.

I assume as in the main model that the initial marginal polluting district equalizes profits using each technology assuming no climate policies, and so it's given by $c(b_1)^{1-\frac{1}{\sigma}} - (1-\sigma)f = 1$. The share of second-period polluting districts is determined endogenously in the model, and is given by the equation

$$c(b_2)^{1-\frac{1}{\sigma}} - (1-\sigma)f + s = (1+\tau_2)^{1-\frac{1}{\sigma}}.$$

As before, b_2 decreases if the subsidy increases or if the expectation of the second-period carbon tax increases.

The objective of the policymaker is

$$\begin{aligned} W &= \int_0^1 U_i\, di - \lambda\left(\int_0^{b_1} x_{i1}\, di + \delta\int_0^{b_2} x_{i2}\, di\right) \\ &= \frac{\sigma}{1-\sigma}\left(b_1(1+\tau_1)^{1-\frac{1}{\sigma}} + \int_{b_1}^1 c(i)^{1-\frac{1}{\sigma}}\, di\right) + (\tau_1 - \lambda)b_1(1+\tau_1)^{-\frac{1}{\sigma}} - \delta\sigma f(b_1 - b_2) \\ &\quad + \delta\left[\frac{\sigma}{1-\sigma}\left(b_2(1+\tau_2)^{1-\frac{1}{\sigma}} + \int_{b_2}^1 c(i)^{1-\frac{1}{\sigma}}\, di\right) + (\tau_2 - \lambda)b_2(1+\tau_2)^{-\frac{1}{\sigma}}\right]. \end{aligned}$$

As in the main model, this is maximized with $\tau_1 = \tau_2 = \lambda$ and $s = 0$.

Notice that

$$\frac{dV}{d\tau} = b\frac{(1+\tau)^{-1-\frac{1}{\sigma}}}{\sigma}(\sigma(1-\sigma)(1+\tau) - \tau),$$

so $\frac{dV}{d\tau} > 0$ iff $\tau < \overline{\tau} \equiv \frac{\sigma(1-\sigma)}{1-\sigma(1-\sigma)}$. This means that after tax rebates every consumer benefits from a carbon tax as long as it is not too large. Of course, polluting districts dislike carbon taxes because the negative effect on their profits dominates. But carbon taxes only affect green districts as consumers (i.e., through V), and therefore those districts support increases in the carbon tax (after rebates). Therefore, the distributive tension is exactly as in the main model. In the second period, if a majority of districts is green then a green proposer can implement an optimal carbon tax (as long as λ is not too large), but if a majority is still polluting then polluters block any proposal that increases the carbon taxes.

To implement climate policy, the policymaker needs to build a winning coalition in the first period, which requires approval by green districts and by the median (polluting) district. The

political constraints are thus

$$V(\tau_1, b_1) + \delta V(\tau_2, b_2) - \delta\sigma(s + \sigma f)(b_1 - b_2) \geqslant (1 + \delta)V(0, b_1) \tag{PC$_G$}$$

for green districts, and

$$\begin{aligned} &V(\tau_1, b_1) + \delta V(\tau_2, b_2) + \sigma(1 + \tau_1)^{1-\frac{1}{\sigma}} \\ &\quad + \delta\sigma\left[-(1-\sigma)f + s + c\left(\frac{1}{2}\right)^{1-\frac{1}{\sigma}} - (s + \sigma f)(b_1 - b_2)\right] \geqslant (1 + \delta)(V(0, b_1) + \sigma) \end{aligned} \tag{PC$_P$}$$

for polluting districts. As in the main model, the optimal policy is not feasible. Polluters demand a subsidy (up to some level) and resist first-period carbon taxes unless they are compensated. Therefore, accommodating the pivotal polluters creates an incentive to raise s and reduce τ_1 from the Pigouvian level, as in Proposition 3.

To make the argument work, we need a few assumptions. Suppose that $c(i) = 1/(Ai + B)$ with $A, B > 0$. Let $H(i) \equiv c(i)^{1-\frac{1}{\sigma}}$. The definition of b_1 yields $H(b_1) = 1 + (1 - \sigma)f$. Let $m \equiv \inf_{i\in[0,b_1]} H'(i)$ and $M \equiv \sup_{i\in[0,b_1]} H'(i)$. I will assume that:

H1 $M \leqslant 2m$.

H2 $m > \max\left\{\frac{2(1-\sigma)(1+\delta)\lambda}{\delta\sigma b_1}, \frac{\lambda(1+\lambda)^{-\frac{1}{\sigma}}}{\sigma\min\{b_1-\frac{1}{2}, 1-b_1\}}, \frac{1+\delta}{2\delta(b_1-\frac{1}{2})^2}\left[1 - (1+\overline{\tau})^{-\frac{1-\sigma}{\sigma}}\right]\right\}$.

H1 requires the slope of H not to vary too much across districts. I use this to show that a pure polluting coalition $[b_1 - \frac{1}{2}, b_1)$ cannot support policy in equilibrium. H1 ensures that the gain of district $b_1 - \frac{1}{2}$ from becoming green cannot be disproportionately large relative to the fiscal cost of inducing the entire block of districts between b_2 and b_1 to invest.

If $H'(b_2)$ is small, a small increase in s induces a lot of polluters to invest. H2 ensures that this does not happen. It ensures that reducing b_2 is sufficiently costly to rule out a complete transition, first-best policy without compensation, and a carbon tax higher than λ. It also ensures that increasing s doesn't create a huge fiscal cost, which guarantees that a small subsidy relaxes the pivotal polluter's constraint when $s = 0$.

Under H1–2 we obtain the same conclusion as in Proposition 3:

Proposition A.4. *Suppose that $\frac{1}{2} < b_1 < 1$, $\lambda < \overline{\tau}$, and H1–2. If the proposer replaces the status quo in equilibrium, then $0 < b_2 < \frac{1}{2}$, $\tau_2 = \max\{\tau_1, \lambda\}$, and τ_1, s maximize W subject to (PC$_G$) and (PC$_P$). Moreover, $\tau_1 < \lambda$ and $s > 0$.*

A.7. *Endogenous Reelection Probability*

I now allow the reelection probability to depend on the anticipated second-period policies. Following the probabilistic voting approach in Persson and Tabellini (2000, Ch. 3), voter j in district i supports G if $V_i^G + \sigma_{ij} + \nu \geqslant V_i^B$, where V_i^P is the expected payoff if P has agenda-setting power in the second period, and σ_{ij}, ν are individual- and population-level preference biases (capturing differences on dimensions orthogonal to climate policy). The individual bias is uniform on $\left[-\frac{1}{2\psi}, \frac{1}{2\psi}\right]$, and the aggregate bias is uniform on $\left[-\frac{\rho_0}{\phi}, \frac{1-\rho_0}{\phi}\right]$ with $\rho_0 \in (0,1)$. Assume $\psi, \phi > 0$ are small enough so that all probabilities are interior. The aggregate vote share, conditional on ν, is $\int_I \Pr(\sigma_{ij} \leqslant V_i^G - V_i^B - \nu)\, di = \int_I \left(\frac{1}{2} + \psi(V_i^G - V_i^B - \nu)\right) di = \frac{1}{2} + \psi(V^G - V^B - \nu)$, where $V^P = \int_I V_i^P\, di$ for $P = G, B$. It exceeds one half iff $V^G - V^B \geqslant \nu$. This is the case with probability $\phi(V^G - V^B + \frac{\rho_0}{\phi}) = \rho_0 + \phi(V^G - V^B)$. In sum, the probability that G is reelected, when $b_2 < \frac{1}{2}$ and $\tau_2 = \max\{\tau_1, \lambda\}$, is

$$\rho = \rho_0 + \phi(V^G - V^B) = \rho_0 - \phi\frac{\delta b_2}{2}\max\{\lambda^2 - \tau_1^2, 0\}.$$

In the main model I studied the case $\phi = 0$. Here I am interested in determining the effect of increasing ϕ, which can be understood as the salience of the climate issue, by a small amount. Notice that, since voters are assumed to not internalize the climate externality, the average voter is against climate policy, and, therefore, increasing the salience of climate policy reduces the chances of victory of the green party as long as $\tau_1 < \lambda$, i.e., if there is a difference in second-period policy between the parties. The green proposer can increase their chances of reelection by reducing the share of polluting districts in the second period (b_2), which can be achieved by providing a more generous subsidy s, or by increasing the first-period carbon tax τ_1, thus reducing the difference in expected policies between the parties (due to the fact that in the second period a majority is pro-climate and can impose the costs of climate policy on the polluting minority).

Notice that a large first-period carbon tax would intuitively lead to backlash against the green party. However, a carbon tax in the first period is only possible in the context of a policy sequence. And, in that context, voters realize that the carbon tax has been locked in, and therefore the anti-climate party cannot credibly promise to eliminate it. I show below that, in fact, increasing τ_1 helps the green proposer get reelected, despite the fact that the average voter dislikes climate policy.

If we ignore political constraints, how does ϕ (the electoral consequences of climate policy) affect the optimal τ_1 and s? G chooses $\tau_2 = \lambda$, while B chooses $\tau_2 = 0$. Thus,

$$\rho = \rho_0 + \phi(V^G - V^B) = \rho_0 - \phi\frac{\delta b_2 \lambda^2}{2}. \tag{A3}$$

Greater electoral salience therefore reduces G's reelection probability. The first-period carbon

tax has no effect on either b_2 or ρ in this benchmark. Thus $\tau_1 = \lambda$. At $\phi = 0$, the optimal subsidy is $s = \delta(1-\rho_0)\lambda$ by Observation A.1. At this policy, after substituting for b_2 using $s - c + \frac{\delta A}{2} b_2 = \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$,

$$\frac{\partial^2 \Delta W}{\partial s^2} = -\frac{2}{\delta A}, \qquad \frac{\partial^2 \Delta W}{\partial s \partial \rho} = -\frac{2\lambda}{A}, \qquad \frac{\partial \Delta W}{\partial \rho} = \frac{\delta b_2 \lambda^2}{2}.$$

Moreover, (A3) implies that, at $\phi = 0$,

$$\frac{d\rho}{d\phi} = -\frac{\delta b_2 \lambda^2}{2}, \qquad \frac{\partial^2 \rho}{\partial s \partial \phi} = \frac{\lambda^2}{A}.$$

Differentiating the first-order condition for s gives

$$\left.\frac{ds}{d\phi}\right|_{\phi=0} = \frac{\delta^2 b_2 \lambda^3 (2+\lambda)}{4} > 0.$$

Thus, when political constraints are ignored, electoral salience has no effect on the optimal first-period carbon tax, but increases the optimal green subsidy. The reason is that a higher ϕ makes turnover more likely. This reduces the expected second-period carbon tax and therefore weakens private incentives to invest in green technology. A larger subsidy partly offsets this distortion. In addition, the subsidy now has an electoral benefit: by inducing more districts to become green, it reduces the number of voters who bear the cost of G's second-period carbon tax.

With political constraints, if climate policy occurs in equilibrium we know from Proposition 3 that $\tau_1 < \lambda$, and therefore the parties present a meaningful difference to voters if $\phi > 0$. In fact, $\rho = \rho_0 - \phi \frac{\delta b_2}{2}(\lambda^2 - \tau_1^2)$. Differentiating, we get

$$\begin{aligned}
\left.\frac{\partial \rho}{\partial \phi}\right|_{\phi=0} &= -\frac{\delta b_2}{2}(\lambda^2 - \tau_1^2) < 0, \\
\left.\frac{\partial^2 \rho}{\partial \tau_1 \partial \phi}\right|_{\phi=0} &= \delta b_2 \tau_1 + \frac{\delta(1-\rho)(1-\tau_1)}{A}(\lambda^2 - \tau_1^2) > 0, \\
\left.\frac{\partial^2 \rho}{\partial s \partial \phi}\right|_{\phi=0} &= \frac{\lambda^2 - \tau_1^2}{A} > 0.
\end{aligned}$$

Thus, once climate policy becomes electorally salient, both a larger carbon tax and a larger subsidy increase G's reelection probability at the margin.

This suggests that an increase in salience should increase τ_1 and s. In fact, Figure 6 shows a numerical example in which this is the case: we see that ϕ increases both the first-period carbon tax τ_1 and the green subsidy s.

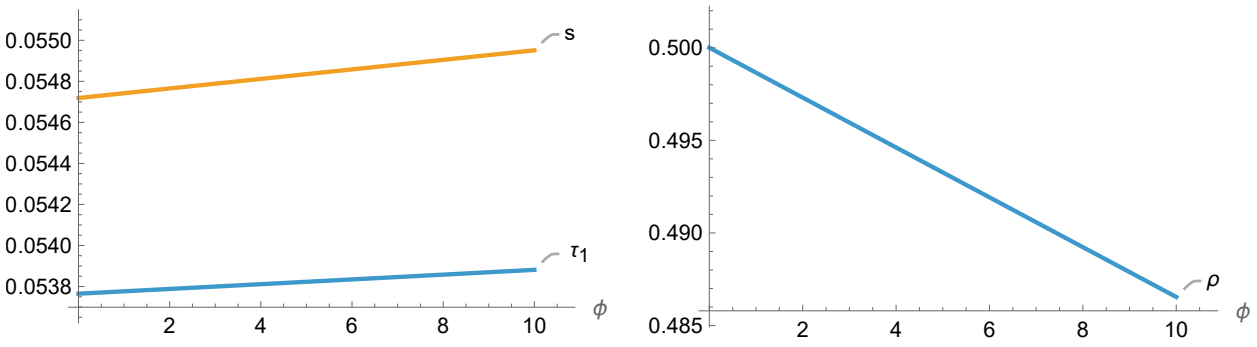


Figure 6: Equilibrium policy when $A = 1.8$, $c = 0.06$, $\delta = 0.9$, $\lambda = 0.1$ and $\rho_0 = \frac{1}{2}$, as ϕ goes from 0 to 10.

However, Figure 7 shows a numerical example in which an increase in ϕ *decreases* both the first-period carbon tax τ_1 and the green subsidy s.

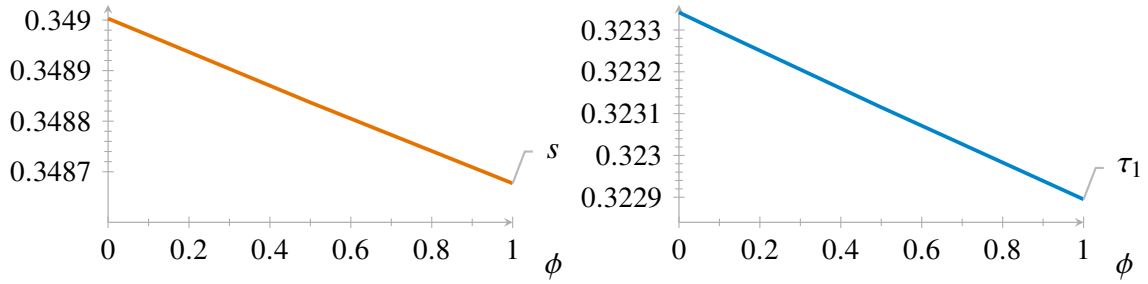


Figure 7: Equilibrium policy when $A = 8$, $b_1 = 0.56$, $c = 1.653$, $\delta = 0.95$, $\lambda = 0.44$, and $\rho_0 = 0.92$, as ϕ goes from 0 to 1. In this example, increasing the electoral salience of climate policy reduces both the first-period carbon tax τ_1 and the green subsidy s.

The intuition is that greater salience lowers ρ. In this parameter region, lower ρ relaxes the pivotal polluter's constraint: fewer districts invest because the future carbon tax is less credible, so the government wastes less money subsidizing inframarginal transitions. The proposer uses part of that political slack to reduce the costly subsidy. Since the subsidy and the first-period carbon tax are linked through the pivotal district's compensation requirement, the carbon tax falls as well. The direct electoral incentives to raise s and τ_1 are present, but here they are dominated by this turnover effect.

A.8. *Robustness of Assumption 1*

I am going to discuss to what extent Assumption 1 is necessary for the results first, and, second, a different way of making b_1 endogenous.

Necessity of Assumption 1. The main result does not hold for *any* exogenous b_1. The reason is that if b_1 is large, it can happen that a majority of districts prefers to invest in green capital even without climate policy. In other words, it can happen that there is already a majority of districts that want to be green, even though initially they happen to be polluting. In that case optimal climate policy ($\tau_1 = \lambda$ and $s = 0$) can be implemented if λ is not too large and $\rho < 1$. This is because climate policy in the first period is costly, but it creates fiscal revenue in both periods even if there is political turnover. Formally, without subsidies the expected transfer is positive, so green districts approve any carbon tax. For polluters, the median accepts iff

$$\frac{1}{2}(1-\lambda)^2 - c + \frac{\delta A}{4} + (b_1 + \delta b_2)\lambda(1-\lambda) \geqslant \frac{1}{2} - c + \frac{\delta A}{4} + \rho b_2 \lambda(1-\lambda),$$

i.e., iff

$$(b_1 + \delta(1-\rho)b_2 - 1)(1-\lambda) \geqslant \frac{1}{2}\lambda,$$

which is true if b_1 is close to 1, $\rho < 1$, $b_2 > 0$, and λ is close to 0. This is not the logic that the paper intends to study, namely, how climate policy is distorted when polluters resist optimal policy. Therefore, I need to make *some* assumption in order to focus on the interesting case.

It is reasonable to assume that firms that would profit from green capital even without climate policy will invest. Formally:

Assumption 1'. Let b_1^* such that $-c + \frac{\delta A}{2} b_1^* = \frac{\delta}{2}$. We have $\frac{1}{2} < b_1 \leqslant b_1^* < 1$.

The only reason not to invest in that case would be to wait for a subsidy. I consider this possibility when I discuss how to make b_1 endogenous below. If we assume that firms invest in green capital if it is profitable even without climate policies, optimal climate policy cannot be implemented in equilibrium. The reason is that in the political constraints b_1 only enters through $\mathrm{E}(T)$, and it does as a term $b_1\tau_1(1-\tau_1)$. Therefore $P \leqslant P|_{b_1=b_1^*}$, and we know from Proposition 2 and Proposition 3 that $P|_{b_1=b_1^*} < 0$ when $\tau_1 = \lambda$, $s = 0$. So, indeed, optimal climate policy is not feasible under this assumption. In sum, Assumption 1', though weaker than Assumption 1, is enough to focus the analysis on the interesting case. The main result, namely, that in equilibrium $\tau_1 < \lambda$ and $s > 0$, still holds under this weaker assumption. The proof in Proposition 3 can be easily adapted.

Endogenous b_1. Under Assumption 1, b_1 is endogenous. The implicit assumption is that before the game starts climate policy was not available to policymakers, or the green proposer was not aware of climate damages. Under that assumption the only reason to invest in green capital is to exploit its productivity advantages, which happens exactly for the districts i such that $-c + \frac{\delta A}{2} i \geqslant \frac{\delta}{2}$; this is what Assumption 1 asserts.

Now, districts could have anticipated in period $t = 0$ that the green proposer was going to be

in power in period $t = 1$. In this case, even if climate policy was not available in period $t = 0$, b_1 should take into account the expectation of climate policies in periods $t = 1, 2$.

There is a problem: with three periods, the value b_1 that obtains if no climate policy is expected should be given by

$$-c + \frac{\delta A}{2} b_1 + \frac{\delta^2 A}{2} b_1 = \frac{\delta}{2} - \delta c + \frac{\delta^2 A}{2} b_1,$$

i.e., $-(1-\delta)c + \frac{\delta A}{2} b_1 = \frac{\delta}{2}$, which determines a different (smaller) value to the value of b_1 that results from Assumption 1. (That value solves $-c + \frac{\delta A}{2} b_1 = \frac{\delta}{2}$.) This inconsistency is an artifact of the two-period model; it would not arise in an infinite period version. To fix it we can make the cost of green capital in period $t = 0$ equal to $(1+\delta)c$.

With that fix, the endogenous b_1 is given by

$$-(1+\delta)c + \frac{\delta A}{2} b_1 + \frac{\delta^2 A}{2} b_1 = \frac{\delta}{2}(1-\tau_1)^2 - \delta(c-s) + \frac{\delta^2 A}{2} b_1,$$

i.e.,

$$-c + \frac{\delta A}{2} b_1 = \frac{\delta}{2}(1-\tau_1)^2 + \delta s. \tag{A4}$$

The expectation of a carbon tax in period 1 creates an incentive to invest preemptively in period 0, but the expectation of an investment subsidy creates an incentive to wait in order to obtain it. Both implications are intuitive. b_2 is still given by $s - c + \frac{\delta A}{2} b_2 = \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$. Let $b_1^* = b_1^*(\tau_1, s)$ be the value of b_1 that solves (A4).

This change in b_1 doesn't change optimal policies, because they are $\tau_1 = \tau_2 = \lambda$ and $s = 0$, which don't depend on b_1. This also doesn't change Proposition 2, because agents take b_1 as given in period 1 by subgame perfection.

Now, suppose that when $\tau_1 = \lambda$ and $s = 0$ we have that $b_1^* \leqslant \frac{1}{2}$. Then there is an equilibrium in which there is (endogenously) a green majority in the first period, and the green proposer therefore implements optimal policy, which districts expect in period zero. This is not an interesting case for the question I study in the paper, which is how equilibrium climate policy is distorted by political opposition from polluting constituencies (because in this case polluters cannot block policies). Therefore, it is reasonable to assume that $b_1^* > \frac{1}{2}$ when $\tau_1 = \lambda$ and $s = 0$ (and, moreover, that $b_1^* < 1$, so that the formula is valid). Under this assumption it must be the case that $b_1 > \frac{1}{2}$ in equilibrium, because if $b_1 \leqslant \frac{1}{2}$ then the proposer chooses $\tau_1 = \lambda$ and $s = 0$, which contradicts that $b_1^* > \frac{1}{2}$ for these policies. In sum, we can replace Assumption 1 for this:

Assumption A.2. Let b_1^* such that $-c + \frac{\delta A}{2} b_1^* = \frac{\delta}{2}(1-\lambda)^2$. We have $\frac{1}{2} < b_1^* < 1$.

Given $b_1 > \frac{1}{2}$, let $\tau_1^*(b_1)$ and $s^*(b_1)$ be the equilibrium policies in the subgame that starts in period $t = 1$. The equilibrium b_1 is such that b_1 satisfies (A4) for $\tau_1 = \tau_1^*(b_1)$ and $s = s^*(b_1)$.

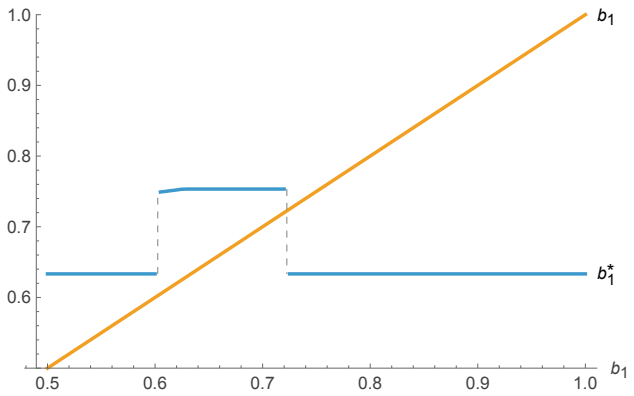


Figure 8: Illustration of the possibility of equilibrium nonexistence when b_1 is endogenous. b_1^* is the value of b_1 given by $(*)$. There is an equilibrium iff $b_1 = b_1^*$. Parameters: $A = 2$, $c = 0.12$, $\delta = 0.9$, $\lambda = 0.1$, $p = 0.6$.

In other words, $b_1 = b_1^*(\tau_1^*(b_1), s^*(b_1))$. A problem that arises is that there may not be any b_1 that satisfies this condition. This is because we are looking for a fixed point of the function $f(b_1) = b_1^*(\tau_1^*(b_1), s^*(b_1))$, which is not necessarily continuous. The reason is that there may be feasible (and better than status quo) policies τ_1, s for some values of b_1 but not others. When we go from feasible to not feasible, τ_1^*, s^* drop to 0 discontinuously, which creates a discontinuity for $b_1^*(\tau_1^*, s^*)$.

Figure 8 illustrates the problem. Let b_1^0 be given by $-c + \frac{\delta A}{2} b_1^0 = \frac{\delta}{2}$. This is the value of b_1 that obtains if districts expect no climate policy. In the example we have $b_1^0 \approx 0.63$. Now, if $b_1 = b_1^0$, the policymaker chooses policies $\tau_1 \approx 0.004$ and $s \approx 0.12$, and they are implemented. Therefore, $b_1^* \approx 0.75 > b_1$, so $b_1 = b_1^0$ is not an equilibrium. Now, if $b_1 = 0.75$, say, there are no feasible policies τ_1, s, so $\tau_1^* = 0$ and $s^* = 0$, and $b_1^* = b_1^0 < b_1$. So $b_1 = 0.75$ is not an equilibrium either. The jump from $b_1^* > b_1$ to $b_1^* = b_1^0 < b_1$ happens at $b_1 = \hat{b}_1 \approx 0.72$.

Now, there is an argument to accept $b_1 = \hat{b}_1$ and $(\tau_1, s) = 0$ as an equilibrium in this case, even though it's not. The reason is that if we see the game as a limit of a game with finitely many districts, this is the limit of the equilibrium in that game as the number of districts grows to infinity. Suppose that we have districts $i_1 < \ldots < i_n$ with, say, $i_k = \frac{k}{n}$, so that the distribution with CDF $F_n(x) = \frac{1}{n} \sum_{i=1}^n 1_{x \leqslant i_k}$ converges to a uniform distribution on $[0, 1]$ (in the weak topology) as $n \to \infty$. Notice that if a district i_k decides to invest in green capital with positive probability at $t = 0$, every district $i_{k'}$ with $k' > k$ will invest for sure, because the benefit of doing so is $\frac{\delta A}{2} i_{k'} + \frac{\delta^2 A}{2} i_{k'}$ and the cost is the same for both districts. For the same reason, if i_k does not invest with positive probability, then every district $k' < k$ does not invest for sure. Assume that there is one i_k such that if exactly the districts $k' \geqslant k$ invest then $b_1 < \hat{b}_1$, but if exactly the districts $k' > k$ invest (i.e., if i_k doesn't invest) then $b_1 > \hat{b}_1$. In other words, suppose that i_k is pivotal in the relevant sense. (We can clearly construct the $\{i_k\}_{k=1,\ldots,n}$ to force this.) Suppose that exactly the

districts $k' > k$ invest, so $b_1 > \hat{b}_1$. If k decides to invest then $b_1 < \hat{b}_1$.

We have $b_1 > \hat{b}_1$ and $i_k, b_1 \to \hat{b}_1$ as $n \to +\infty$, so $\tau_1^* = s^* = 0$. Therefore, every $k' > k$ prefers to invest, because if they don't invest then they increase b_1, which keeps $\tau_1^* = s^* = 0$, and under the expectation of $\tau_1 = s = 0$ every $i \geqslant b_1^0$ is happy to invest, so in particular $i_{k'} > b_1 > \hat{b}_1 > b_1^0$ strictly prefers to invest. Also, every k prefers not to invest, because if they do invest, then $b_1 < \hat{b}_1$, so b_1^* jumps strictly above $\hat{b}_1$, which means that $i = \hat{b}_1$ strictly prefers not to invest; since $i_k \to \hat{b}_1$, for n sufficiently large i_k prefers not to invest. In sum, as I claimed, if n is large enough, investment by districts $i_k > \hat{b}_1$ followed by policies converging to $\tau_1^* = s^* = 0$ are an equilibrium. The intuition is that with a finite number of firms, firms internalize their effect on the policy choice around a discontinuity of the policy function. As a result, a number of polluting firms a bit to the right of the discontinuity constitutes an equilibrium. This provides an argument to take $b_1 = \hat{b}_1$ and $\tau_1 = s = 0$ as an equilibrium in the model with infinite districts.

In sum, if we take this argument as valid, in general there is an equilibrium. In cases where a "proper" equilibrium exists we will have $-c + \frac{\delta A}{2} b_1 = \frac{\delta}{2}(1 - \tau_1^*)^2 + \delta s^*$, where τ_1^* and s^* are the equilibrium policies, instead of $-c + \frac{\delta A}{2} b_1 = \frac{\delta}{2}$ as I assume in the paper. The question is whether the main result, namely, that either $\tau_1^* = s^* = 0$ or $\tau_1^* < \lambda$ and $s^* > 0$, continues to hold.

The answer is surprisingly no. Figure 9 shows a counterexample. We see that there is an equilibrium ($b_1 = b_1^*$), but in that equilibrium $\tau_1^* > \lambda$. The green proposer chooses a carbon tax that is *larger* than the social cost of carbon in the first period (in addition to subsidies, $s^* > 0$). What happens is that a few districts choose to remain polluting in the first period because they expect to receive subsidies. The constraint imposed by green districts binds, so having more polluters increases the fiscal cost of the subsidies, which increases the carbon tax required to satisfy green districts. This pushes the optimal carbon tax above the social cost of carbon, which contradicts the main result. Now, this counterexample is extreme: green technology is 120 times more productive than polluting technology in some districts, and the social cost of carbon is extremely large.

B. Proofs

B.1. *Proof of Proposition 1*

Suppose that the policymaker can make production and investment decisions directly. Then they would choose $\{y_{it}\}_{i \in I, t=1,2}$ and $b_2 \in [0, b_1]$ to maximize aggregate welfare ignoring policies, which is

$$\begin{aligned} W &= \int_0^1 \pi_{i1}\, di - \lambda \int_0^{b_1} y_{i1}\, di + \delta \left[\int_0^1 \pi_{i2}\, di - \lambda \int_0^{b_2} y_{i2}\, di \right] \\ &= \int_0^{b_2} \left(y_{i1} - \frac{1}{2} y_{i1}^2 \right) di + \int_{b_2}^{b_1} \left(y_{i1} - \frac{1}{2} y_{i1}^2 - c \right) di + \int_{b_1}^1 \left(y_{i1} - \frac{1}{2Ai} y_{i1}^2 \right) di - \lambda \int_0^{b_1} y_{i1}\, di \end{aligned}$$

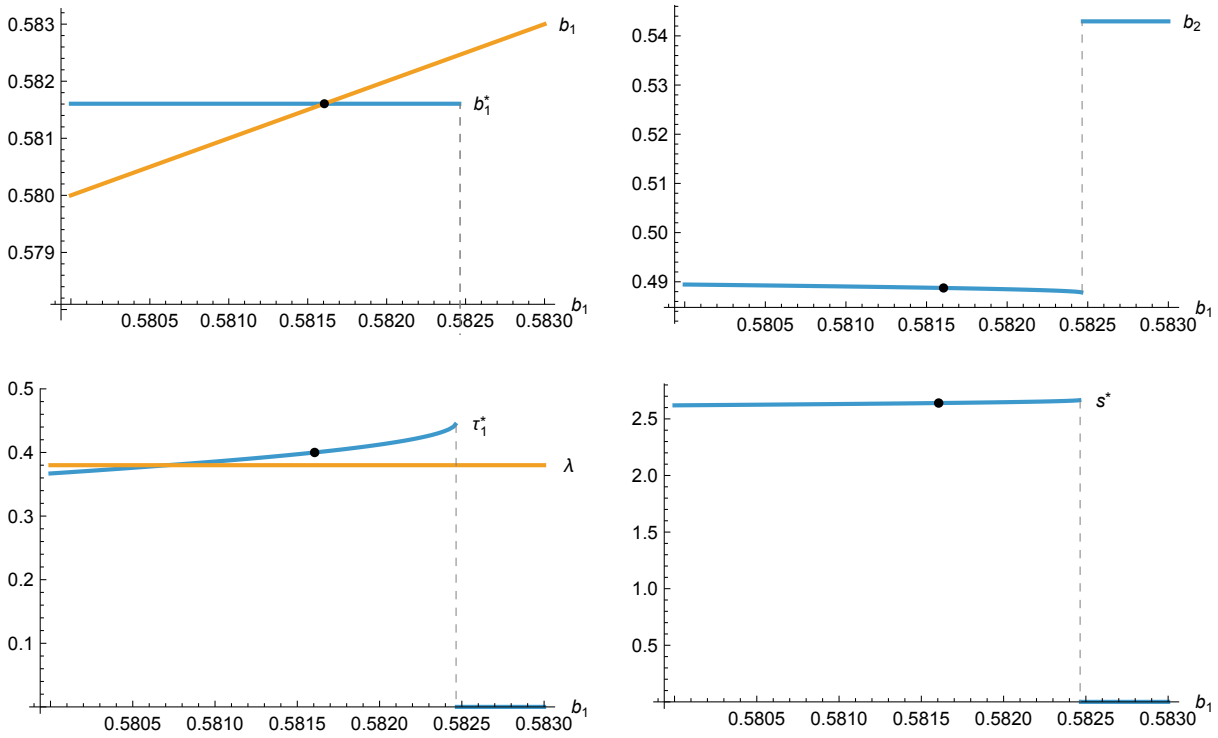


Figure 9: Counterexample of the main result with endogenous b_1. b_1^* is the value of b_1 given by $(*)$. Black dots are the equilibrium values. The main result is violated because $\tau_1^* > \lambda$. Parameters: $A = 120.036$, $c = 28.878$, $\delta = 0.9$, $\lambda = 0.38$, $\rho = 0.1$.

$$
\begin{aligned}
&+ \delta\left[\int_0^{b_2}\left(y_{i2} - \frac{1}{2}y_{i2}^2\right)di + \int_{b_2}^{1}\left(y_{i2} - \frac{1}{2Ai}y_{i2}^2\right)di - \lambda\int_0^{b_2} y_{i2}\,di\right] \\
= &\int_0^{b_1}\left((1-\lambda)y_{i1} - \frac{1}{2}y_{i1}^2\right)di + \int_{b_1}^{1}\left(y_{i1} - \frac{1}{2Ai}y_{i1}^2\right)di - c(b_1 - b_2) \\
&+ \delta\left[\int_0^{b_2}\left((1-\lambda)y_{i2} - \frac{1}{2}y_{i2}^2\right)di + \int_{b_2}^{1}\left(y_{i2} - \frac{1}{2Ai}y_{i2}^2\right)di\right].
\end{aligned}
$$

Pointwise maximization of the integrals requires $y_{it} = 1 - \lambda$ almost everywhere for polluting i and $y_{it} = Ai$ almost everywhere for green i. In that case,

$$
W = \frac{b_1}{2}(1-\lambda)^2 + \frac{A}{4}(1 - b_1^2) - c(b_1 - b_2) + \delta\left[\frac{b_2}{2}(1-\lambda)^2 + \frac{A}{4}(1 - b_2^2)\right].
$$

We have $\frac{dW}{db_2} = c + \frac{\delta}{2}(1-\lambda)^2 - \frac{\delta A}{2}b_2$. Therefore, the unconstrained optimal b_2 is given by $-c + \frac{\delta A}{2}b_2 = \frac{\delta}{2}(1-\lambda)^2$. By Assumption 1, $-c + \frac{\delta A}{2}b_1 = \frac{\delta}{2}$, so $b_2 = b_1 - \frac{1}{A}(2-\lambda)\lambda < b_1$, and $b_2 = \frac{2}{\delta A}\left[\frac{\delta}{2}(1-\lambda)^2 + c\right] > 0$, so this satisfies the constraint $b_2 \in [0, b_1]$. To summarize, the necessary and sufficient conditions for the allocation $\{y_{it}\}_{i\in I, t=1,2}, b_2$ to maximize welfare W are

$$
\begin{cases}
y_{it} = \begin{cases} 1-\lambda & \text{almost everywhere for polluting } i, \\ Ai & \text{almost everywhere for green } i, \text{ and} \end{cases} \\
-c + \frac{\delta A}{2}b_2 = \frac{\delta}{2}(1-\lambda)^2.
\end{cases} \tag{$*$}
$$

Now, the policymaker chooses carbon taxes $\tau_1, \tau_2 \geqslant 0$ and an investment subsidy $s \geqslant 0$. Polluting firms choose $y_{it} = 1 - \tau_t$, and green firms choose $y_{it} = Ai$. Equilibrium investment decisions are implicitly determined by $s - c + \frac{\delta A}{2}b_2 = \frac{\delta}{2}(1-\tau_2)^2$. Therefore, in equilibrium we obtain

$$
\begin{cases}
y_{it} = \begin{cases} 1-\tau_t & \text{for polluting } i, \\ Ai & \text{for green } i, \text{ and} \end{cases} \\
s - c + \frac{\delta A}{2}b_2 = \frac{\delta}{2}(1-\tau_2)^2.
\end{cases}
$$

By direct comparison we see that the policymaker implements the allocation defined by $(*)$ if and only if they choose the policies $\tau_1 = \tau_2 = \lambda$ (matching the conditions for y_{it}) and, therefore, $s = 0$ (matching the conditions for b_2 using $\tau_2 = \lambda$). This shows that $\tau_1 = \tau_2 = \lambda$ and $s = 0$ uniquely implement the optimal allocation. Given that, by assumption, raising taxes and providing transfers can be done via lump-sum transfers (i.e., they have no shadow costs), these must be the optimal policy choices, as desired.

B.2. *Proof of Observation A.1*

Observation A.1. *With political turnover the optimal carbon taxes are Pigouvian, and the investment subsidy is* $s = \delta(1-\rho)\lambda$.

Proof. The objective of the green proposer in period 2, if they keep agenda power, is to maximize

$$\begin{aligned}
&\delta\left[\int_0^1 \pi_{i2}\,di - \lambda\int_0^{b_2} y_{i2}\,di\right] + T \\
&= \delta\left[\int_0^{b_2}\frac{1}{2}(1-\tau_2)^2\,di + \int_{b_2}^1 \frac{1}{2}Ai\,di - \lambda\int_0^{b_2}(1-\tau_2)\,di + \tau_2\int_0^{b_2}(1-\tau_2)\,di\right] + T_1 \\
&= \delta\left[\frac{b_2}{2}(1-\tau_2)^2 + \frac{A}{4}(1-b_2^2) - b_2\lambda(1-\tau_2) + b_2\tau_2(1-\tau_2)\right] + T_1
\end{aligned}$$

where T_1 is the period-1 component of the transfer. I used that profit maximization implies $y_{it} = 1-\tau_t$ for polluting i and $y_{it} = Ai$ for green i. The derivative of this with respect to τ_2 is $\delta b_2(\lambda - \tau_2)$, so the optimal τ_2 is λ. Therefore, if the proposer in period 2 is G, they choose $\tau_2 = \lambda$. Similarly, if the proposer is B, they choose $\tau_2 = 0$.

The objective of the green proposer in period 1 is to maximize

$$\begin{aligned}
W_G = &\int_0^1 \pi_{i1}\,di - \lambda\int_0^{b_1} y_{i1}\,di + \delta\mathrm{E}\left[\int_0^1 \pi_{i2}\,di - \lambda\int_0^{b_2} y_{i2}\,di\right] + \mathrm{E}(T) \\
= &\int_0^{b_1}\frac{1}{2}(1-\tau_1)^2\,di + \int_{b_1}^1\frac{1}{2}Ai\,di - (c-s)(b_1-b_2) - \lambda\int_0^{b_1}(1-\tau_1)\,di \\
&+ \delta\left[\int_0^{b_2}\frac{1}{2}[\rho(1-\lambda)^2 + (1-\rho)]\,di + \int_{b_2}^1\frac{1}{2}Ai\,di - \lambda\int_0^{b_2}[\rho(1-\lambda)+(1-\rho)]\,di\right] \\
&+ \int_0^{b_1}\tau_1(1-\tau_1)\,di + \delta\rho\int_0^{b_2}\lambda(1-\lambda)\,di - s(b_1-b_2) \\
= &\frac{b_1}{2}(1-\tau_1^2) + \frac{A}{4}(1-b_1^2) - c(b_1-b_2) - b_1\lambda(1-\tau_1) \\
&+ \delta\left[\frac{b_2}{2}(1-\rho\lambda^2) + \frac{A}{4}(1-b_2^2) - b_2\lambda(1-\rho\lambda)\right].
\end{aligned}$$

We have, using $-c + s + \frac{\delta A}{2}b_2 = \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$, that

$$\begin{aligned}
\frac{\partial W_G}{\partial b_2} &= c + \frac{\delta}{2}(1-\rho\lambda^2) - \frac{\delta A}{2}b_2 - \delta\lambda(1-\rho\lambda) \\
&= c + \frac{\delta}{2}(1-\rho\lambda^2) - \left(c - s + \frac{\delta}{2}[\rho(1-\lambda)^2 + (1-\rho)]\right) - \delta\lambda(1-\rho\lambda) \\
&= s - (1-\rho)\delta\lambda.
\end{aligned}$$

Therefore,

$$\frac{\partial W_G}{\partial s} = \frac{\partial W_G}{\partial b_2}\frac{\partial b_2}{\partial s} = \frac{2}{\delta A}((1-\rho)\delta\lambda - s),$$
$$\frac{\partial W_G}{\partial \tau_1} = (\lambda - \tau_1)b_1 + \frac{\partial W_G}{\partial b_2}\frac{\partial b_2}{\partial \tau_1}.$$

The first equation implies that the optimal s is $(1-\rho)\delta\lambda$. This implies $\frac{\partial W_G}{\partial b_2} = 0$, so the optimal τ_1 is λ, as desired. ■

B.3. *Turnover and Convex Damages*

In order to relax the assumption that environmental damages are linear, let $D_t(E)$ be the damage function at time $t = 1, 2$, which takes the stock of carbon pollution added by the country since period 1 as its argument. We have $E_1 = b_1(1-\tau_1)$ and $E_2 = E_1 + b_2(1-\tau_2)$. For simplicity I will assume that $D_t''(E) = \kappa \geqslant 0$ for all E and $t = 1, 2$. Let E_2^P be the stock of emissions if $P = G, B$ is the proposer in period two, and let τ_2^P be the choice of period-two carbon tax if $P = G, B$ is the second period proposer.

Using a similar computation as in Subsection B.2, we obtain that the second period objective of proposer P is to maximize

$$W_{P2} = \frac{b_2}{2}(1-\tau_2)^2 + \frac{A}{4}(1-b_2^2) - \mathbb{1}_{P=G}D_2(E_1 + b_2(1-\tau_2)) + b_2\tau_2(1-\tau_2).$$

We have $\frac{dW_{P2}}{d\tau_2} = b_2(\mathbb{1}_{P=G}D_2'(E_2^P) - \tau_2)$, so $\tau_2^G = D_2'(E_2^G)$ and $\tau_2^B = 0$. Notice that $D_2'(E_2^B) = D_2''(\xi)(E_2^B - E_2^G) + D_2'(E_2^G) = \kappa(E_2^B - E_2^G) + D_2'(E_2^G) = \tau_2^G(1+\kappa b_2)$, where ξ is between E_2^G and E_2^B.

Now, again using a similar computation as in Subsection B.2, we obtain

$$W_G = \frac{b_1}{2}(1-\tau_1^2) + \frac{A}{4}(1-b_1^2) - c(b_1 - b_2) - D_1(E_1)$$
$$+ \delta\left[\frac{b_2}{2}(1-\rho(\tau_2^G)^2) + \frac{A}{4}(1-b_2^2) - \rho D_2(E_2^G) - (1-\rho)D_2(E_2^B)\right].$$

Using that $s - c + \frac{\delta A}{2}b_2 = \frac{\delta}{2}[\rho(1-\tau_2^G)^2 + (1-\rho)]$, and the envelope theorem for τ_2^G (because τ_2^G is chosen to maximize W_G taking τ_1, s as given), we have

$$\begin{aligned}\frac{\partial W_G}{\partial b_2} &= c + \frac{\delta}{2}\left[\rho(1-\tau_2^G)^2 + (1-\rho)\right] - \frac{\delta A}{2}b_2 - \delta(1-\rho)D_2'(E_2^B)\\ &= s - \delta(1-\rho)D_2'(E_2^B)\\ &= s - (1-\rho)\delta\tau_2^G(1+\kappa b_2).\end{aligned}$$

Therefore, using $\tau_2^G = D_2'(E_2^G)$,

$$\frac{\partial W_G}{\partial s} = \frac{\partial W_G}{\partial b_2}\frac{\partial b_2}{\partial s} = \left[s - (1-\rho)\delta\tau_2^G(1+\kappa b_2)\right]\frac{\partial b_2}{\partial s},$$
$$\frac{\partial W_G}{\partial \tau_1} = b_1[D_1'(E_1) + \delta\tau_2^G(1+(1-\rho)\kappa b_2) - \tau_1] + \frac{\partial W_G}{\partial b_2}\frac{\partial b_2}{\partial \tau_1}.$$

Since the investment condition implies $\frac{\partial b_2}{\partial s} < 0$, the first-order conditions imply $s = (1-\rho)\delta\tau_2^G(1+\kappa b_2)$ and $\tau_1 = D_1'(E_1) + \delta\tau_2^G(1+(1-\rho)\kappa b_2)$. Differentiating, we obtain

$$\frac{\partial \tau_2^G}{\partial \rho} = D_2''(E_2^G)\frac{\partial E_2^G}{\partial \rho} = \kappa\left[-b_1\frac{\partial \tau_1}{\partial \rho} + \frac{\partial b_2}{\partial \rho}(1-\tau_2^G) - b_2\frac{\partial \tau_2^G}{\partial \rho}\right],$$
$$\frac{\partial s}{\partial \rho} = -\delta\tau_2^G(1+\kappa b_2) + (1-\rho)\delta\left[\frac{\partial \tau_2^G}{\partial \rho}(1+\kappa b_2) + \tau_2^G\kappa\frac{\partial b_2}{\partial \rho}\right],$$
$$\frac{\partial b_2}{\partial \rho} = -\frac{2}{\delta A}\left[\frac{\partial s}{\partial \rho} + \delta\tau_2^G\left(1-\frac{\tau_2^G}{2}\right) + \delta\rho(1-\tau_2^G)\frac{\partial \tau_2^G}{\partial \rho}\right],$$
$$\frac{\partial \tau_1}{\partial \rho} = -\kappa b_1\frac{\partial \tau_1}{\partial \rho} + \delta\frac{\partial \tau_2^G}{\partial \rho}(1+(1-\rho)\kappa b_2) - \delta\tau_2^G\kappa b_2 + \delta\tau_2^G(1-\rho)\kappa\frac{\partial b_2}{\partial \rho}.$$

Taking $\rho = 1$, we obtain $\frac{\partial s}{\partial \rho} = -\delta\tau_2^G(1+\kappa b_2)$, so

$$\frac{\partial b_2}{\partial \rho} = \frac{2}{A}\left[\tau_2^G(1+\kappa b_2) - \tau_2^G\left(1-\frac{\tau_2^G}{2}\right) - (1-\tau_2^G)\frac{\partial \tau_2^G}{\partial \rho}\right]$$
$$= \frac{2}{A}\left[\tau_2^G\left(\kappa b_2 + \frac{\tau_2^G}{2}\right) - (1-\tau_2^G)\frac{\partial \tau_2^G}{\partial \rho}\right].$$

Using this,

$$\frac{\partial \tau_2^G}{\partial \rho} = \kappa\left\{-b_1\frac{\partial \tau_1}{\partial \rho} + \frac{2}{A}\left[\tau_2^G\left(\kappa b_2 + \frac{\tau_2^G}{2}\right) - (1-\tau_2^G)\frac{\partial \tau_2^G}{\partial \rho}\right](1-\tau_2^G) - b_2\frac{\partial \tau_2^G}{\partial \rho}\right\}$$
$$= \frac{\kappa}{1+\kappa b_2 + \kappa\frac{2}{A}(1-\tau_2^G)^2}\left[-b_1\frac{\partial \tau_1}{\partial \rho} + \frac{2}{A}\tau_2^G(1-\tau_2^G)\left(\kappa b_2 + \frac{\tau_2^G}{2}\right)\right].$$

Therefore, $\frac{\partial \tau_1}{\partial \rho} = -\kappa b_1\frac{\partial \tau_1}{\partial \rho} + \delta\frac{\partial \tau_2^G}{\partial \rho} - \delta\tau_2^G\kappa b_2$, so

$$\left[1+\kappa b_1 + \frac{\delta\kappa b_1}{1+\kappa b_2 + \kappa\frac{2}{A}(1-\tau_2^G)^2}\right]\frac{\partial \tau_1}{\partial \rho} = -\delta\tau_2^G\kappa\left[b_2 - \frac{2}{A}(1-\tau_2^G)\left(\kappa b_2 + \frac{\tau_2^G}{2}\right)\frac{1}{1+\kappa b_2 + \kappa\frac{2}{A}(1-\tau_2^G)^2}\right].$$

If b_2 is small enough, the RHS is positive, so $\frac{\partial \tau_1}{\partial \rho} > 0$, which implies that an increase in the probability of turnover, i.e., a drop in ρ, leads to a reduction in the first-period carbon tax. However, if τ_2^G is small, the RHS is negative, so $\frac{\partial \tau_1}{\partial \rho} < 0$ and we obtain the opposite conclusion. Thus, the effect of turnover on the first-period carbon tax is ambiguous, as claimed in the text.

Here are two numerical examples. First, let $b_1 = .6$, $A = 1$, $\delta = .9$, $\kappa = 3$, $c = .0405$, and $D_1(E) = D_2(E) = .012E + 1.5E^2$. At $\rho = 1$, the first-order conditions and the investment condition are solved by $\tau_1 = .84$, $\tau_2^G = .6$, $b_2 = .25$, and $s = 0$. At this point, $\frac{\partial \tau_1}{\partial \rho} \approx .0286 > 0$. Thus, an increase in turnover reduces the first-period carbon tax. For example, solving the same system at $\rho = .99$ gives $\tau_1 \approx .83971$, $\tau_2^G \approx .59477$, $b_2 \approx .24204$, and $s \approx .00924$.

For the opposite case, let $b_1 = .6$, $A = 3$, $\delta = .9$, $\kappa = 2$, $c = .468$, and $D_1(E) = D_2(E) = .064E + E^2$. At $\rho = 1$, the solution is $\tau_1 = .82$, $\tau_2^G = .6$, $b_2 = .4$, and $s = 0$. Now, $\frac{\partial \tau_1}{\partial \rho} \approx -.1004 < 0$, so an increase in turnover raises the first-period carbon tax. Indeed, at $\rho = .99$ the solution is $\tau_1 \approx .82099$, $\tau_2^G \approx .59768$, $b_2 \approx .39627$, and $s \approx .00964$.

B.4. *Proof of Proposition 2*

First, I show that in the second period we have $\tau_2 \leqslant \frac{1}{2}$ or $b_2 = 0$.

Fix the investment decisions made after the first-period policy, and hence fix b_2. The components of the uniform transfer generated in each period are

$$T_1 = b_1\tau_1(1-\tau_1) - s(b_1 - b_2) \quad \text{and} \quad T_2 = b_2\tau_2(1-\tau_2).$$

Consequently, when comparing a second-period proposal τ_2 with the status quo τ_1, a green district compares $\frac{1}{2}Ai + b_2\tau_2(1-\tau_2)$ and $\frac{1}{2}Ai + b_2\tau_1(1-\tau_1)$, whereas a polluting district compares $\frac{1}{2}(1-\tau_2)^2 + b_2\tau_2(1-\tau_2)$ with $\frac{1}{2}(1-\tau_1)^2 + b_2\tau_1(1-\tau_1)$.

Suppose that $b_2 > 0$ and that in equilibrium we have $\tau_2 > \frac{1}{2}$. Let $\hat{\tau}_2 = 1 - \tau_2 < \frac{1}{2}$. The two tax rates generate the same revenue because $\hat{\tau}_2(1-\hat{\tau}_2) = \tau_2(1-\tau_2)$. Green districts are therefore indifferent between them, while polluting districts strictly prefer $\hat{\tau}_2$ because $\hat{\tau}_2 < \tau_2$. Let $\lambda_P = \lambda$ when the second-period proposer is G, and let $\lambda_P = 0$ when it is B. The second-period proposer's payoff is

$$\begin{aligned} W_P &= \int_0^1 \pi_{i1}\,di - \lambda_P \int_0^{b_1} y_{i1}\,di + \delta \mathrm{E}\left[\int_0^1 \pi_{i2}\,di - \lambda_P \int_0^{b_2} y_{i2}\,di\right] + T \\ &= \int_0^{b_1} \frac{1}{2}(1-\tau_1)^2\,di + \int_{b_1}^1 \frac{1}{2}Ai\,di - (c-s)(b_1-b_2) - \lambda_P \int_0^{b_1} (1-\tau_1)\,di \\ &\quad + \delta\left[\int_0^{b_2} \frac{1}{2}(1-\tau_2)^2\,di + \int_{b_2}^1 \frac{1}{2}Ai\,di - \lambda_P \int_0^{b_2} (1-\tau_2)\,di\right] \end{aligned}$$

$$
\begin{aligned}
&+ \int_0^{b_1} \tau_1(1-\tau_1)\,di + \delta \int_0^{b_2} \tau_2(1-\tau_2)\,di - s(b_1 - b_2) \\
&= \frac{b_1}{2}(1-\tau_1^2) + \frac{A}{4}(1-b_1^2) - c(b_1-b_2) - b_1\lambda_P(1-\tau_1) \\
&\quad + \delta\left[\frac{b_2}{2}(1-\tau_2^2) + \frac{A}{4}(1-b_2^2) - b_2\lambda_P(1-\tau_2)\right]. \qquad \text{(A5)}
\end{aligned}
$$

It follows that

$$
\begin{aligned}
W_P|_{\tau_2=\hat{\tau}_2} - W_P &= \delta b_2\left[\frac{1}{2}(1-\hat{\tau}_2^2) - \lambda_P(1-\hat{\tau}_2)\right] - \delta b_2\left[\frac{1}{2}(1-\tau_2^2) - \lambda_P(1-\tau_2)\right] \\
&= \delta b_2(2\tau_2 - 1)\left(\frac{1}{2} - \lambda_P\right) > 0,
\end{aligned}
$$

since $\lambda < \frac{1}{2}$. Therefore the proposer strictly prefers to implement $\hat{\tau}_2$ instead of τ_2, which is politically feasible. This contradicts that $\tau_2 > \frac{1}{2}$, as desired.

Second, I show that if the status quo is replaced in the first period, then there is a green majority in the second period.

Suppose first that the first-period proposal fails, so $\tau_1 = s = 0$. If polluting districts remain a majority, they reject every positive second-period tax and $\tau_2 = 0$. Given this continuation, a polluting district i invests if and only if

$$
-c + \frac{\delta A}{2} i \geqslant \frac{\delta}{2}.
$$

By Assumption 1, this condition is equivalent to $i \geqslant b_1$. Thus no initially polluting district invests, $b_2 = b_1 > \frac{1}{2}$, and $\tau_2 = 0$. This is the continuation after a failure of a first-period proposal in the equilibria we are focusing on.

A polluting district i accepts a first-period policy (τ_1, s) if they prefer it to no climate policy in either period, i.e., iff

$$
\underbrace{\frac{1}{2}(1-\tau_1)^2}_{\text{period-1 profit}} + \max\Bigg\{\underbrace{s - c + \frac{\delta}{2}Ai}_{\substack{\text{period-2 profit in} \\ \text{the green sector plus} \\ \text{net cost of transition}}},\ \underbrace{\frac{\delta}{2}\mathrm{E}\left[(1-\tau_2)^2\right]}_{\substack{\text{expected} \\ \text{period-2 profit in} \\ \text{the polluting sector}}}\Bigg\} + \underbrace{\mathrm{E}(T)}_{\substack{\text{expected} \\ \text{transfers}}} \geqslant \underbrace{\frac{1}{2} + \frac{\delta}{2}}_{\substack{\text{status quo} \\ \text{profits}}}. \qquad \text{(A6)}
$$

Notice that the LHS is weakly increasing in i, so if i does not accept, every polluting district $j \leqslant i$ does not accept. Therefore, the median district $i = \frac{1}{2}$, which is polluting because $b_1 > \frac{1}{2}$ by Assumption 1, must approve the proposal.

Suppose that the median district approves but does not invest, so $b_2 \geqslant \frac{1}{2}$. They accept (τ_1, s) iff

$$\frac{1}{2}(1-\tau_1)^2 + \frac{\delta}{2}\mathrm{E}\left[(1-\tau_2)^2\right] + \mathrm{E}(T) \geqslant \frac{1}{2} + \frac{\delta}{2},$$

where

$$\mathrm{E}(T) = b_1\tau_1(1-\tau_1) + \delta b_2 \mathrm{E}[\tau_2(1-\tau_2)] - s(b_1 - b_2).$$

Equivalently,

$$-\left(1 - b_1 + \left(b_1 - \frac{1}{2}\right)\tau_1\right)\tau_1 - \delta\mathrm{E}\left[\left(1 - b_2 + \left(b_2 - \frac{1}{2}\right)\tau_2\right)\tau_2\right] - s(b_1 - b_2) \geqslant 0.$$

Because $1 > b_1 \geqslant b_2 \geqslant \frac{1}{2}$, every term on the left is non-positive. The inequality can therefore hold only if all three terms vanish. Looking at the first and second terms, this requires $\tau_1 = 0$ and $\tau_2 = 0$. If $s > 0$, the subsidy induces some initially polluting districts to invest, so $b_2 < b_1$ and the final term is strictly negative. Hence $s = 0$ as well, and the status quo is unchanged. Therefore, every equilibrium in which the status quo is replaced requires a green second-period majority $b_2 < \frac{1}{2}$, as desired.

Third, I show that in equilibrium $\tau_1 \leqslant \frac{1}{2}$.

Suppose that an equilibrium has $\tau_1 > \frac{1}{2}$. The preceding argument implies $b_2 < \frac{1}{2}$ and green districts form a second-period majority. Consider the first-period proposal $(\hat{\tau}_1, s) = (1 - \tau_1, s)$. Given that the second-period majority is green, a tax $\tilde{\tau}_2$ defeats a status quo τ_1 if and only if it doesn't reduce revenue, i.e., iff $b_2\tilde{\tau}_2(1-\tilde{\tau}_2) \geqslant b_2\tau_1(1-\tau_1)$. Because $\hat{\tau}_1(1-\hat{\tau}_1) = \tau_1(1-\tau_1)$, the two first-period proposals induce the same set of feasible second-period policies. Therefore, they induce the same continuation policy, investment decisions, b_2, subsidy expenditure, and transfers.

Green districts are indifferent between the two first-period proposals, because they produce the same tax revenue. Every initially polluting district strictly prefers $(\hat{\tau}_1, s)$ because $\hat{\tau}_1 < \tau_1$. Hence every coalition that approves (τ_1, s) also approves $(\hat{\tau}_1, s)$. Using (A5), the deviation raises G's payoff by

$$\begin{aligned} W_G|_{\tau_1=\hat{\tau}_1} - W_G &= b_1\left[\frac{1}{2}(1-\hat{\tau}_1^2) - \lambda(1-\hat{\tau}_1)\right] - b_1\left[\frac{1}{2}(1-\tau_1^2) - \lambda(1-\tau_1)\right] \\ &= b_1(2\tau_1 - 1)\left(\frac{1}{2} - \lambda\right) > 0, \end{aligned}$$

a contradiction. Therefore, $\tau_1 \leqslant \frac{1}{2}$, as desired.

Fourth, I show that $b_2 > 0$ *in every equilibrium, and thus* $s - c + \frac{\delta}{2}Ab_2 = \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$.

This is the case under the status quo, since then $b_2 = b_1 > \frac{1}{2}$. Suppose instead that $(\tau_1, s) \neq$

$(0,0)$ and $b_2 = 0$. The proposer has the option to keep the status quo in place, so we must have $W_G \geqslant W_G|_{\text{SQ}}$. But

$$\begin{aligned}
W_G - W_G|_{\text{SQ}} &\leqslant W_G|_{\tau_1=\lambda} - W_G|_{\text{SQ}} \\
&= \left\{\frac{b_1}{2}(1-\lambda^2) + \frac{A}{4}(1-b_1^2) - cb_1 - b_1\lambda(1-\lambda) + \delta\frac{A}{4}\right\} \\
&\quad - \left\{\frac{b_1}{2} + \frac{A}{4}(1-b_1^2) - b_1\lambda + \delta\left[\frac{b_1}{2} + \frac{A}{4}(1-b_1^2) - b_1\lambda\right]\right\} \\
&= b_1\frac{\lambda^2}{2} - cb_1 + \delta\left[\frac{A}{4}b_1^2 - b_1\left(\frac{1}{2} - \lambda\right)\right] \\
&= \frac{\delta b_1}{4}\left(\frac{2\lambda^2}{\delta} + 4\lambda - 1 - \frac{2c}{\delta}\right) < 0.
\end{aligned}$$

The last equality uses $Ab_1 = 1 + 2c/\delta$ from Assumption 1, and the last inequality is Assumption 2. Thus G prefers the status quo to any proposal that produces $b_2 = 0$, so such a proposal cannot occur in equilibrium. Therefore, $b_2 > 0$.

A polluting district i invests iff $s - c + \frac{\delta}{2}Ai \geqslant \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$. Therefore, because $b_2 > 0$, district $i = b_2$ must be indifferent, and b_2 is given by $s - c + \frac{\delta}{2}Ab_2 = \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$.

Fifth, I derive the equilibrium second-period policy.

If no policy is implemented in the first period, $b_2 = b_1 > \frac{1}{2}$, and polluters force $\tau_2 = 0$. If the status quo is changed in the first period, we saw that $0 < b_2 < \frac{1}{2}$. Therefore τ_2 is feasible iff $b_2\tau_2(1-\tau_2) \geqslant b_2\tau_1(1-\tau_1)$, which is the case iff $(\tau_2 - \tau_1)(1 - \tau_1 - \tau_2) \geqslant 0$. This is equivalent to $\tau_2 \geqslant \tau_1$ because $\tau_1, \tau_2 \leqslant \frac{1}{2}$ as we proved. In other words, τ_2 can be implemented iff it's in $[\tau_1, \frac{1}{2}]$.

Proposer G's ideal tax is λ, while proposer B's ideal tax is zero. Therefore, G implements $\max\{\tau_1, \lambda\}$ and B leaves the status quo τ_1 in place. Since the proposer in the second period is G with probability ρ, we obtain

$$\tau_2 = \begin{cases} \max\{\tau_1, \lambda\}, & \text{with probability } \rho, \\ \tau_1, & \text{with probability } 1-\rho. \end{cases}$$

Sixth, I prove that in any equilibrium in which the status quo is replaced, green districts approve the first-period proposal.

If green districts don't approve the proposal, a majority composed entirely of polluters must approve it. Now, polluting district i approves iff (A6) holds, and if i approves then all polluting $j > i$ do so too. Therefore, a majority of polluters approves iff district $i = b_1 - \frac{1}{2}$ approves. As we

saw earlier, this implies that i transitions, i.e., that $s - c + \frac{\delta}{2}Ai \geqslant \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$. Therefore (A6) is

$$\frac{1}{2}(1-\tau_1)^2 + s - c + \frac{\delta}{2}A\left(b_1 - \frac{1}{2}\right) + \mathrm{E}(T) \geqslant \frac{1}{2} + \frac{\delta}{2}.$$

Using Assumption 1, which says $-c + \frac{\delta}{2}Ab_1 = \frac{\delta}{2}$, and $\mathrm{E}(T) = b_1\tau_1(1-\tau_1) + \delta b_2 \mathrm{E}[\tau_2(1-\tau_2)] - s(b_1 - b_2)$, this becomes

$$-\frac{1}{2}(2\tau_1 - \tau_1^2) + s - \frac{\delta}{4}A + b_1\tau_1(1-\tau_1) + \delta b_2\mathrm{E}[\tau_2(1-\tau_2)] - s(b_1-b_2) \geqslant 0.$$

Now, $-c + \frac{\delta}{2}Ab_1 = \frac{\delta}{2}$ and $s - c + \frac{\delta}{2}Ab_2 = \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$ (by step 4) imply

$$\begin{aligned} s &= \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2] - \frac{\delta}{2}Ab_2 + c \\ &= \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2] - \frac{\delta}{2}Ab_2 + \frac{\delta}{2}Ab_1 - \frac{\delta}{2} \\ &= \frac{\delta}{2}A(b_1-b_2) - \frac{\delta}{2}\mathrm{E}(2\tau_2 - \tau_2^2). \end{aligned}$$

Replacing, i approves iff

$$\begin{aligned} 0 \leqslant\ & -\frac{1}{2}(2\tau_1 - \tau_1^2) + \frac{\delta}{2}\left[A(b_1-b_2) - \mathrm{E}(2\tau_2 - \tau_2^2)\right](1 - b_1 + b_2) - \frac{\delta}{4}A \\ & + b_1\tau_1(1-\tau_1) + \delta b_2\mathrm{E}[\tau_2(1-\tau_2)] \\ =\ & -\tau_1 \underbrace{\left[1 - b_1 + \left(b_1 - \frac{1}{2}\right)\tau_1\right]}_{>0} + \frac{\delta}{2}A \underbrace{\left[(b_1-b_2)(1-b_1+b_2) - \frac{1}{2}\right]}_{<0 \text{ because } x(1-x)\leqslant 1/4 \text{ for all } x} \\ & - \frac{\delta}{2}\mathrm{E}(2\tau_2 - \tau_2^2)\underbrace{(1 - b_1 + b_2)}_{>b_2} + \delta b_2\mathrm{E}[\tau_2(1-\tau_2)] \\ <\ & -\frac{\delta}{2}\mathrm{E}(2\tau_2 - \tau_2^2)b_2 + \delta b_2\mathrm{E}[\tau_2(1-\tau_2)] = -\frac{\delta}{2}b_2\mathrm{E}(\tau_2^2) \leqslant 0, \end{aligned}$$

i.e., $0 < 0$, which is impossible. Therefore, a coalition only consisting of polluters will block any policy that changes the status quo, as desired.

Seventh, the first-period policy preserves the status quo or maximizes W_G subject to (PC_B) and (PC_G).

We proved that in any equilibrium in which the status quo is replaced, green districts approve the proposal. This is (PC_G). Since they are a minority, the polluter $i = \frac{1}{2}$ must approve, and, in fact, if the median district approves and (PC_G), then the policy is implemented. Notice that this

condition is exactly (PC_B). Therefore, in equilibrium the proposer chooses the policy (τ_1, s) that maximizes their payoff W_G subject to (PC_B) and (PC_G), as desired. This finishes the proof.

B.5. *Proof of Proposition 3*

Suppose that the status quo is replaced by (τ_1, s) in the first period. By Proposition 2 and its proof, (τ_1, s) maximizes $\Delta W(\tau_1, s) = W_G(\tau_1, s) - W_G(0, 0)$ subject to the two political constraints, and $\Delta W(\tau_1, s) \geqslant 0$. We have to prove that $\tau_1 < \lambda$ and $s > 0$. I assume that $\tau_1 \geqslant \lambda$ or $s = 0$. The basic idea is to find local profitable deviations in each case, which produces a contradiction.

The argument has three steps. First, I show that if $\tau_1 \geqslant \lambda$ then both political constraints must bind. The reason is that, once the first-period tax reaches the optimal tax λ, the proposer does not want a higher tax, and wants a zero subsidy. If (PC_B) is slack, the proposer can reduce the subsidy locally, making green districts better off. If (PC_G) is slack, the proposer can reduce τ_1 if $\tau_1 > \lambda$, or else reduce both τ_1 and s by a small amount to keep (PC_B). Both cases produce a profitable deviation. Therefore, both constraints must bind.

Second, assuming that $\tau_1 \geqslant \lambda$ and both constraints bind, I show that either (a) it is possible to reduce τ_1 and increase s by a small amount in a way that relaxes both constraints, and raises the proposer's payoff, or (b) the proposer prefers the status quo. This is the hard part of the proof. With this, we obtain that $\tau_1 < \lambda$.

Third, I show that $\tau_1 < \lambda$ requires $s > 0$. At $s = 0$, raising τ_1 strictly raises the proposer's payoff, while raising s creates at most a second order loss. Moreover, the green political constraint is slack, and a marginal increase in the subsidy relaxes the polluter's political constraint. Hence the proposer can increase τ_1, raising s if necessary to compensate the pivotal polluter. This gives a profitable feasible deviation, so $s = 0$ cannot be optimal. Putting all together, we obtain $\tau_1 < \lambda$ and $s > 0$, as desired.

Preliminary calculations.

Let $x \equiv b_1 - b_2$. By Proposition 2, $0 < b_2 < \frac{1}{2} < b_1 < 1$, so $0 < x < b_1$. Abusing notation slightly, write the expected transfer as T and the pivotal polluting district's gain as P:

$$T \equiv b_1\tau_1(1 - \tau_1) + \delta b_2 \mathrm{E}[\tau_2(1 - \tau_2)] - sx, \tag{A7}$$

$$P \equiv \frac{1}{2}(1 - \tau_1)^2 + s - c + \frac{\delta A}{4} + T - \frac{1 + \delta}{2}. \tag{A8}$$

Thus (PC_B) and (PC_G) are $P \geqslant 0$ and $T \geqslant 0$. The continuation tax is $\tau_2 = \max\{\tau_1, \lambda\}$ with probability ρ and $\tau_2 = \tau_1$ otherwise.

Recall that the cutoff conditions in Assumption 1 and (2) are

$$-c + \frac{\delta A}{2} b_1 = \frac{\delta}{2}, \tag{A9}$$

$$s - c + \frac{\delta A}{2} b_2 = \frac{\delta}{2} \mathrm{E}\big[(1-\tau_2)^2\big]. \tag{A10}$$

Subtracting (A9) from (A10), and (A7) from (A8), and substituting (A9), gives

$$s = \frac{\delta}{2}\big[Ax - \mathrm{E}(2\tau_2 - \tau_2^2)\big], \tag{A11}$$

$$P - T = s - \tau_1 + \frac{\tau_1^2}{2} - \frac{\delta A}{2}\left(b_1 - \frac{1}{2}\right). \tag{A12}$$

Subtracting $W_G(0,0)$ from (A5) gives

$$\begin{aligned}
\Delta W = {} & \frac{b_1}{2}(1-\tau_1^2) + \frac{A}{4}(1-b_1^2) - c(b_1 - b_2) - b_1\lambda(1-\tau_1) \\
& + \delta \mathrm{E}\left[\frac{b_2}{2}(1-\tau_2^2) + \frac{A}{4}(1-b_2^2) - b_2\lambda(1-\tau_2)\right] \\
& - (1+\delta)\left[\frac{b_1}{2} + \frac{A}{4}(1-b_1^2) - b_1\lambda\right] \\
= {} & b_1\tau_1\left(\lambda - \frac{\tau_1}{2}\right) + \delta b_2 \mathrm{E}\left[\tau_2\left(\lambda - \frac{\tau_2}{2}\right)\right] + \delta\lambda(b_1 - b_2) \\
& - \left(c + \frac{\delta}{2}\right)(b_1 - b_2) + \frac{\delta A}{4}(b_1^2 - b_2^2).
\end{aligned}$$

By (A9), $c + \frac{\delta}{2} = \frac{\delta A}{2} b_1$. Using $x = b_1 - b_2$, the last two terms are

$$-\left(c + \frac{\delta}{2}\right)x + \frac{\delta A}{4}x(b_1 + b_2) = -\frac{\delta A}{2}b_1 x + \frac{\delta A}{4}x(b_1 + b_2) = -\frac{\delta A}{4}x^2.$$

Therefore,

$$\Delta W = b_1\tau_1\left(\lambda - \frac{\tau_1}{2}\right) + \delta b_2 \mathrm{E}\left[\tau_2\left(\lambda - \frac{\tau_2}{2}\right)\right] + \delta\lambda x - \frac{\delta A}{4}x^2. \tag{A13}$$

By (A11), for fixed τ_1 the subsidy is strictly increasing in x, so we can equivalently work with (τ_1, x) whenever convenient.

When $\tau_1 \geqslant \lambda$, we have $\tau_2 = \tau_1$ with probability one. Therefore

$$s = \frac{\delta}{2}\big[Ax - (2\tau_1 - \tau_1^2)\big], \tag{A14}$$

$$T = (1+\delta)b_1\tau_1(1-\tau_1) + \frac{\delta}{2}x\tau_1^2 - \frac{\delta A}{2}x^2, \tag{A15}$$

$$\Delta W = (1+\delta)b_1\tau_1\left(\lambda - \frac{\tau_1}{2}\right) + \delta x\left\{\lambda - \tau_1\left(\lambda - \frac{\tau_1}{2}\right)\right\} - \frac{\delta A}{4}x^2. \tag{A16}$$

First, I show that $\tau_1 \geqslant \lambda$ implies $P = T = 0$.

I first rule out the corner $\tau_1 = \lambda$, $s = 0$. By (A14), $x = (2\lambda - \lambda^2)/A$. Since $b_2 = b_1 - x < \frac{1}{2}$,

we have $\frac{1}{2} < b_1 < \frac{1}{2} + x$. Substituting into P gives an affine function of b_1. At the two endpoints,

$$P\big|_{b_1=1/2} = -\frac{\lambda}{2A}[A(1-\delta(1-\lambda)) + 2\delta\lambda(1-\lambda)(2-\lambda)] < 0,$$
$$P\big|_{b_1=1/2+x} = -\frac{\lambda}{2A}[A(1+\delta) - 2\lambda(1-\lambda)(2-\lambda)] < 0.$$

The first inequality uses $1-\delta(1-\lambda) > 0$; the second uses $A > 1$ and $2\lambda(1-\lambda)(2-\lambda) < 1$. Hence $P < 0$ throughout the interval, a contradiction. Thus $\tau_1 = \lambda$ requires $s > 0$.

Now suppose $\tau_1 \geqslant \lambda$. Define

$$\kappa \equiv \begin{cases} 1+\delta, & \tau_1 > \lambda, \\ 1+\delta(1-\rho), & \tau_1 = \lambda, \end{cases}$$

where at $\tau_1 = \lambda$ derivatives with respect to τ_1 are taken from the left. Holding x fixed,

$$\frac{\partial P}{\partial \tau_1} = \kappa[b_1(1-2\tau_1) - (1-\tau_1)] + (\kappa-1)x\tau_1 < -\tau_1 < 0,$$

because $b_1 < 1$ and $x < 1$. If $\tau_1 > \lambda$, holding x fixed also gives $\frac{\partial \Delta W}{\partial \tau_1} = (\lambda - \tau_1)(b_1 + \delta b_2) < 0$; at $\tau_1 = \lambda$ the corresponding left derivative is zero. Therefore T cannot be slack. If it were, lowering τ_1 slightly would relax P and, when $\tau_1 > \lambda$, strictly raise welfare. At $\tau_1 = \lambda$, also lower x by a sufficiently small amount proportional to the tax reduction. This strictly raises welfare because $s > 0$ and $\frac{\partial \Delta W}{\partial x} = -s < 0$ at $\tau_1 = \lambda$, while the tax reduction creates enough slack in P and T remains feasible by continuity. Hence $T = 0$.

Since $T = 0$ and $P \geqslant 0$, (A12) implies $s > 0$. Holding τ_1 fixed, (A16) and (A15) give

$$\frac{\partial \Delta W}{\partial x} = \delta(1-\tau_1)(\lambda - \tau_1) - s < 0, \qquad \frac{\partial T}{\partial x} = \delta\left(\frac{\tau_1^2}{2} - Ax\right) < 0,$$

where the second inequality follows from $s \geqslant 0$ in (A14). Thus P cannot be slack either: if $P > 0$, lowering x slightly raises both welfare and T, while P and s remain feasible. Therefore any candidate with $\tau_1 \geqslant \lambda$ must satisfy $P = T = 0$.

Second, I rule out that $\tau_1 \geqslant \lambda$.

I now show that $\tau_1 \geqslant \lambda$ cannot be optimal. Along the curve $P = T$, equations (A11) and (A12) imply

$$\frac{dx}{d\tau_1} = \frac{2\kappa(1-\tau_1)}{\delta A} > 0. \tag{A17}$$

Differentiating the common political constraint along this curve at the candidate point gives

$$\frac{dP}{d\tau_1} = \frac{dT}{d\tau_1} = \kappa b_1(1-2\tau_1) + (\kappa-1)x\tau_1 - 2\kappa x(1-\tau_1) + \frac{\kappa(1-\tau_1)\tau_1^2}{A}. \tag{A18}$$

Because ΔW is increasing in λ when $\tau_1 \geqslant \lambda$, $\Delta W \geqslant 0$ implies $\Delta W|_{\lambda=\tau_1} \geqslant 0$. Using $T = 0$ to eliminate the quadratic term in (A16),

$$0 \leqslant \Delta W\big|_{\lambda=\tau_1} = \frac{\tau_1}{2}\left[-(1+\delta)b_1(1-2\tau_1) + \delta x\left(2 - \frac{3}{2}\tau_1\right)\right],$$

so

$$(1+\delta)b_1(1-2\tau_1) \leqslant \delta x\left(2 - \frac{3}{2}\tau_1\right). \tag{A19}$$

Moreover, $P = T$ at the candidate point implies, using (A14) and (A12),

$$x = b_1 - \frac{1}{2} + \frac{1+\delta}{\delta A}(2\tau_1 - \tau_1^2) > \frac{1+\delta}{\delta A}\tau_1(2-\tau_1). \tag{A20}$$

Since $1 \leqslant \kappa \leqslant 1+\delta$, we have $\kappa - 1 \leqslant \delta\kappa/(1+\delta)$. Applying this, (A19), and (A20) to (A18) gives

$$\begin{aligned}\frac{dP}{d\tau_1} &< \frac{\kappa x}{1+\delta}\left[-2 + 2\tau_1 + \frac{3}{2}\delta\tau_1 + \frac{\delta\tau_1(1-\tau_1)}{2-\tau_1}\right] \\ &< \frac{\kappa x}{1+\delta}[-2 + 2\tau_1 + 2\delta\tau_1] \leqslant \frac{\kappa x}{1+\delta}(-2+4\tau_1) \leqslant 0,\end{aligned}$$

where the second line uses $(1-\tau_1)/(2-\tau_1) < 1/2$, $\delta \leqslant 1$, and $\tau_1 \leqslant 1/2$. Thus lowering τ_1 slightly along $P = T$ makes both political constraints strictly slack and, by (A17), also lowers x. If $\tau_1 > \lambda$, both changes strictly raise welfare. If $\tau_1 = \lambda$, the direct left derivative with respect to τ_1 is zero while $\frac{\partial \Delta W}{\partial x} = -s < 0$, so the reduction in x strictly raises welfare. Either way we contradict optimality. Therefore $\tau_1 < \lambda$.

Third, I show that $\tau_1 < \lambda$ implies $s > 0$.

Suppose instead that $\tau_1 < \lambda$ and $s = 0$. Then $\tau_2 = \lambda$ with probability ρ and $\tau_2 = \tau_1$ otherwise. Differentiating (A13) and using (A11) gives

$$\begin{aligned}\frac{\partial \Delta W}{\partial s} &= \frac{2(1-\rho)}{A}(\lambda - \tau_1)(1-\tau_1) \geqslant 0, \\ \frac{\partial \Delta W}{\partial \tau_1} &= (\lambda - \tau_1)\left[b_1 + \delta(1-\rho)b_2 + \frac{2\delta(1-\rho)^2(1-\tau_1)^2}{A}\right] > 0.\end{aligned}$$

Thus raising τ_1 strictly raises welfare, while raising s weakly raises it.

At $s = 0$, differentiating P while taking the induced change in b_2 into account gives

$$\frac{\partial P}{\partial s} = 1 - x - \frac{2}{A}\mathrm{E}[\tau_2(1-\tau_2)].$$

Equation (A11) gives $Ax = \mathrm{E}(2\tau_2 - \tau_2^2)$. Since $\rho > 0$ and $\lambda > 0$, $\frac{2}{A}\mathrm{E}[\tau_2(1-\tau_2)] < x$, so it is enough to show that $x < \frac{1}{2}$. If instead $x \geqslant \frac{1}{2}$, then $b_1 - \frac{1}{2} \geqslant b_2$, and (A7)–(A12) give

$$P = b_1\tau_1(1-\tau_1) - \left(\tau_1 - \frac{\tau_1^2}{2}\right) + \delta\left\{b_2\mathrm{E}[\tau_2(1-\tau_2)] - \frac{A}{2}\left(b_1 - \frac{1}{2}\right)\right\} < 0.$$

Indeed, the first two terms are nonpositive because $b_1 < 1$, while the expression in braces is at most $b_2(\frac{1}{4} - \frac{A}{2}) < 0$. This contradicts feasibility. Hence $x < \frac{1}{2}$, and therefore $\frac{\partial P}{\partial s} > 1 - 2x > 0$.

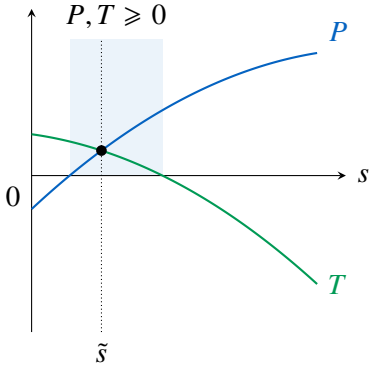
Finally, at $s = 0$, (A12) gives

$$P - T = -\tau_1 + \frac{\tau_1^2}{2} - \frac{\delta A}{2}\left(b_1 - \frac{1}{2}\right) < 0.$$

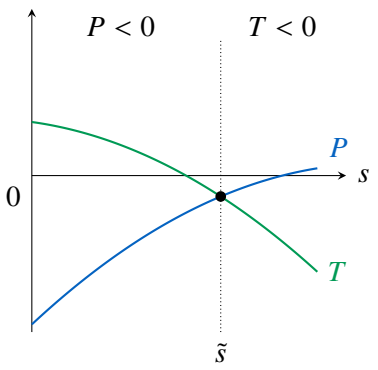
Thus T is strictly slack whenever $P \geqslant 0$. If $P > 0$, a small increase in τ_1 with $s = 0$ is feasible and strictly raises welfare. If $P = 0$, increase τ_1 slightly and raise s just enough to keep $P \geqslant 0$; this is possible because $\frac{\partial P}{\partial s} > 0$, while T remains feasible because it is initially slack. Since $\frac{\partial \Delta W}{\partial \tau_1} > 0$ and $\frac{\partial \Delta W}{\partial s} \geqslant 0$, this deviation also strictly raises welfare. Therefore $s = 0$ is impossible, as desired.

B.6. *A Useful Lemma*

Using (A7) and (A8), the political constraints are $P, T \geqslant 0$. Fix τ_1. The constraints are feasible given τ_1 iff there is $s \geqslant 0$ such that $P(\tau_1, s) \geqslant 0$ and $T(\tau_1, s) \geqslant 0$. The functions $P(\tau_1, \cdot)$ and $T(\tau_1, \cdot)$ are concave, so the set of s that satisfy both constraints is an interval. As the following figure illustrates, if the interval is nonempty then the graphs of the two functions intersect in the interval. The graphs intersect exactly at one point, $s = \tilde{s}(\tau_1)$. Therefore, the political constraints are feasible iff $\tilde{P}(\tau_1) \equiv P(\tau_1, \tilde{s}(\tau_1)) \geqslant 0$ for some τ_1. This is very helpful, because it reduces testing political feasibility to a condition in terms of only one parameter.



Feasible



$P|_{s=\tilde{s}} < 0$

Let $\tilde{s}$ be given by $P|_{s=\tilde{s}} = T|_{s=\tilde{s}}$, i.e., $\tilde{s} \equiv \tau_1 - \frac{1}{2}\tau_1^2 + \frac{\delta A}{2}(b_1 - \frac{1}{2}) > 0$.

Lemma A.1. *If $P, T \geqslant 0$ at (τ_1, s) then $P|_{s=\tilde{s}} \geqslant 0$.*

Proof. We have $\tilde{s} > 0$, $\frac{\partial^2 T}{\partial^2 s} = \frac{\partial^2 P}{\partial^2 s} = -\frac{4}{\delta A} < 0$ and $\frac{\partial T}{\partial s}\big|_{s=0} = -\frac{1}{A}\mathrm{E}[(4 - 3\tau_2)\tau_2] \leqslant 0$, so $\frac{\partial T}{\partial s}\big|_{s=\tilde{s}} < 0$. If $\frac{\partial P}{\partial s}\big|_{s=\tilde{s}} \geqslant 0$ we are done, because if $P|_{s=\tilde{s}} < 0$ then $P < 0$ for $s \leqslant \tilde{s}$, and $T < 0$ for $s > \tilde{s}$, so $P \geqslant 0$ and $T \geqslant 0$ cannot happen simultaneously, a contradiction. So it's enough to show that $\frac{\partial P}{\partial s}\big|_{s=\tilde{s}} \geqslant 0$. Notice, first, that $\frac{\partial P}{\partial s}\big|_{s=\tilde{s}}$ is linear in ρ, so it's enough to prove $\frac{\partial P}{\partial s}\big|_{s=\tilde{s}} \geqslant 0$ for $\tau_2 = \max\{\lambda, \tau_1\}$ and $\tau_2 = \tau_1$.

We have $\frac{\partial P}{\partial s}\big|_{s=\tilde{s}} = 2(1 - b_1) - \frac{1}{\delta A}[2(2 - \tau_1)\tau_1 + \delta(4 - 3\tau_2)\tau_2]$. If $\tau_2 = 0$ then $\tau_1 = 0$, and $\frac{\partial P}{\partial s}\big|_{s=\tilde{s}} \geqslant 0$. So we can assume $\tau_2 > 0$. Suppose that $\frac{\partial P}{\partial s}\big|_{s=\tilde{s}} < 0$. We must have

$$A < \tilde{A} = \frac{2(2 - \tau_1)\tau_1 + \delta(4 - 3\tau_2)\tau_2}{2\delta(1 - b_1)}.$$

Let $\overline{P} = \max_{s' \in \mathbb{R}} P|_{s=s'}$. I will show that if $1 < A < \tilde{A}$ then $\overline{P} < 0$, which contradicts that $P \geqslant 0$ for some s.

We have

$$\overline{P} = -(\tau_1 + \delta\tau_2)(1 - b_1) - \left(b_1 - \frac{1}{2}\right)\tau_1^2 + \frac{\delta}{8}(A + 2\tau_2^2)(3 - 4b_1) + \frac{\delta\tau_2^4}{8A}. \qquad (*)$$

Notice that if $b_1 < \frac{3}{4}$ then $\overline{P}$ is strictly convex in A, in which case it's maximized at either $A = 1$ or $A = \tilde{A}$, and if $b_1 \geqslant \frac{3}{4}$ then it's decreasing, in which case it's maximized at $A = 1$. So we only have to verify that $\overline{P}|_{A=1} \geqslant 0$ and $\overline{P}|_{A=\tilde{A}} \geqslant 0$.

Let $\tilde{P} = \overline{P}|_{A=\tilde{A}}$. By inspecting $(*)$ it's clear that $\tilde{P}$ is concave in b_1, since the only nonlinear term in b_1 is $\propto \frac{3-4b_1}{1-b_1}$, whose second derivative is $\frac{-2}{(1-b_1)^3} < 0$. Now,

$$\frac{d\tilde{P}}{db_1} = \tau_1 + \delta\tau_2 - \tau_1^2 - \frac{\delta}{2}(\tilde{A} + 2\tau_2^2) + \frac{\delta}{8}\frac{\tilde{A}}{1 - b_1}(3 - 4b_1) - \frac{\delta\tau_2^4}{8\tilde{A}(1 - b_1)},$$

so

$$\left.\frac{d\tilde{P}}{db_1}\right|_{b_1=\frac{1}{2}} = -\frac{1}{2}\tau_1^2 - \frac{\delta}{4}\tau_2^2 - \frac{\delta^2\tau_2^4}{4(2(2-\tau_1)\tau_1 + \delta(4-3\tau_2)\tau_2)} < 0.$$

This shows that $\tilde{P}$ is decreasing in b_1. Now,

$$\tilde{P}|_{b_1=\frac{1}{2}} = -\frac{1}{4}\tau_1^2 - \frac{\delta}{8}\tau_2^2 + \frac{\delta^2\tau_2^4}{8(2(2-\tau_1)\tau_1 + \delta(4-3\tau_2)\tau_2)} < 0,$$

as we can easily verify. This proves that $\tilde{P} < 0$, as desired.

We have to prove that $\overline{P}|_{A=1} < 0$. Now, $\overline{P}$ is linear in δ, so it's enough to check this for $\delta = 0$, which is clear, and $\delta = 1$, so assume $\delta = 1$. Since $\tilde{A} > 1$, we have that $b_1 > \underline{b}_1$, where $\underline{b}_1$ is such that $\tilde{A}|_{b=\underline{b}_1} = 1$. Since $\overline{P}|_{A=1}$ is linear in b_1, it's enough to check the inequality for $b_1 = \underline{b}_1$ and $b_1 = 1$. In the first case, $\overline{P}|_{A=1,b_1=\underline{b}_1} = \overline{P}|_{A=\tilde{A},b_1=\underline{b}_1} = \tilde{P}|_{b_1=\underline{b}_1} < 0$ by the above. In the second case, $\overline{P}|_{A=1,b_1=1} = -\frac{1}{2}\tau_1^2 - \frac{\delta}{8}(1 + 2\tau_2^2 - \tau_2^4) < 0$, as desired. ■

B.7. *Proof of Observation 1*

Following Subsection B.5, let P and T denote the quantities that must be nonnegative for (PC_B) and (PC_G) to hold. Following Subsection B.6, let $\tilde{s} = \tau_1 - \frac{1}{2}\tau_1^2 + \frac{\delta A}{2}(b_1 - \frac{1}{2})$ be the unique subsidy for which $P = T$. Suppose that the political constraints hold at (τ_1, s).

First, I establish the bound

$$\tau_1(1-\tau_1) + \delta \mathrm{E}[\tau_2(1-\tau_2)] < \frac{\delta A}{4}. \tag{A21}$$

By Lemma A.1, $(\tau_1, \tilde{s})$ is politically feasible. Let $\tilde{b}_2$ be the corresponding second-period cutoff. Using (A9) and (A10),

$$\tilde{b}_2 = \frac{1}{2} - \frac{2}{\delta A}\left[\tau_1 - \frac{1}{2}\tau_1^2 + \delta \mathrm{E}\left(\tau_2 - \frac{1}{2}\tau_2^2\right)\right]. \tag{A22}$$

We must have $\tilde{b}_2 > 0$. Indeed, if $\tilde{b}_2 \leqslant 0$, then

$$\begin{aligned} T|_{s=\tilde{s}} &= b_1\tau_1(1-\tau_1) + \delta\tilde{b}_2\mathrm{E}[\tau_2(1-\tau_2)] - \tilde{s}(b_1 - \tilde{b}_2) \\ &= b_1[\tau_1(1-\tau_1) - \tilde{s}] + \tilde{b}_2[\delta\mathrm{E}[\tau_2(1-\tau_2)] + \tilde{s}] < 0, \end{aligned}$$

because $\tilde{s} > \tau_1(1-\tau_1)$. This contradicts Lemma A.1. Hence

$$\tau_1 - \frac{1}{2}\tau_1^2 + \delta\mathrm{E}\left(\tau_2 - \frac{1}{2}\tau_2^2\right) < \frac{\delta A}{4},$$

and the desired bound (A21) follows.

Second, I show that an increase in A or a decrease in c relax (PC_B) and tighten (PC_G).

From (A9) and (A10), $\frac{db_1}{dA} = -\frac{b_1}{A}$ and $\frac{db_2}{dA} = -\frac{b_2}{A}$. Therefore $\frac{dT}{dA} = -\frac{T}{A}$ and $\frac{dP}{dA} = \frac{\delta}{4} - \frac{T}{A}$. At a politically feasible policy, $T \geqslant 0$, so $\frac{dT}{dA} \leqslant 0$. Moreover, using the bound (A21),

$$T \leqslant b_1\tau_1(1-\tau_1) + \delta b_2 \mathrm{E}[\tau_2(1-\tau_2)] < \tau_1(1-\tau_1) + \delta \mathrm{E}[\tau_2(1-\tau_2)] < \frac{\delta A}{4}.$$

Hence $\frac{dP}{dA} > 0$. Thus an increase in A relaxes (PC_B) and tightens (PC_G).

Similarly, $\frac{db_1}{dc} = \frac{db_2}{dc} = \frac{2}{\delta A}$, so

$$0 < \frac{dT}{dc} = \frac{2}{\delta A}[\tau_1(1-\tau_1) + \delta \mathrm{E}[\tau_2(1-\tau_2)]] < \frac{1}{2}$$

by (A21), while $\frac{dP}{dc} = -1 + \frac{dT}{dc} < 0$. Thus a decrease in c also relaxes (PC_B) and tightens (PC_G).

Third, I show that increasing A or decreasing c preserves feasibility.

By Lemma A.1, feasibility is equivalent to $\tilde{P}(\tau_1) \equiv P(\tau_1, \tilde{s}) \geqslant 0$ for some τ_1. We have $0 < \tilde{b}_2 < \frac{1}{2}$ by the argument above and (A22). Differentiating $\tilde{P}$ with respect to A gives

$$\begin{aligned} A\frac{\partial \tilde{P}}{\partial A} &= (b_1 - \tilde{b}_2)\left(\frac{\delta A}{4} + \tilde{s}\right) + \delta\left(\frac{1}{2} - \tilde{b}_2\right)\mathrm{E}[\tau_2(1-\tau_2)] + \frac{\tilde{s}}{2} - b_1\tau_1(1-\tau_1) \\ &> \left(b_1 - \frac{1}{2}\right)\frac{\delta A}{4} + b_1[\tilde{s} - \tau_1(1-\tau_1)] > 0. \end{aligned}$$

Thus increasing A raises $\tilde{P}$ whenever the constraints are feasible.

Finally,

$$\frac{\partial \tilde{P}}{\partial c} = \frac{\delta(A-2) - 4c - 2\tau_1 - \delta \mathrm{E}[(2-\tau_2)\tau_2]}{\delta A} < 0.$$

Indeed, Assumption 1 and $b_1 > \frac{1}{2}$ imply $4c > \delta(A-2)$. Hence decreasing c also raises $\tilde{P}$. Therefore an increase in A or a decrease in c preserves political feasibility, provided Assumption 1 continues to hold.

B.8. *Proof of Observation 2*

By Lemma A.1 and Proposition 2, the political constraints are feasible iff $\tilde{P}(\tau_1) \geqslant 0$ for some $\tau_1 \in [0, \frac{1}{2}]$. We have

$$\tilde{P}'(0) = -\frac{1+\delta(1-\rho)}{4A}\left[5\rho - 4A(1-b_1)\right] < 0,$$

where the inequality follows from $A < \frac{3\rho}{4(1-b_1)}$. Moreover,

$$\tilde{P}'\left(\frac{1}{2}\right) = -\frac{6 + 5\delta + 3\delta(1-\rho) + 2\delta^2(1-\rho) + 4A\delta(2b_1 - 1) + 2A\delta^2(1-\rho)(2b_1 - 1)}{8A\delta} < 0,$$

and

$$\tilde{P}'''(\tau_1) = \frac{6[1 + \delta(1-\rho)]}{\delta A}[2(1-\tau_1) + \delta(1-\rho)(3 - 4\tau_1)] > 0.$$

Hence $\tilde{P}'$ is strictly convex on $[0, \frac{1}{2}]$. Since it is negative at both endpoints, it is negative throughout the interval. Therefore $\tilde{P}$ is maximized at $\tau_1 = 0$.

At $\tau_1 = 0$,

$$\tilde{P}(0) = -\frac{\delta}{16A}\left[2A^2(2b_1 - 1)^2 - A(5 - 6b_1)\rho + 3\rho^2\right].$$

The term in brackets is positive. If $b_1 \geqslant \frac{5}{6}$, this is immediate. If $b_1 < \frac{5}{6}$, then

$$\begin{aligned} 2A^2(2b_1 - 1)^2 - A(5 - 6b_1)\rho + 3\rho^2 &> 3\rho^2 - A(5 - 6b_1)\rho \\ &> 3\rho^2 - \frac{3\rho^2(5 - 6b_1)}{4(1 - b_1)} \\ &= \frac{3\rho^2(2b_1 - 1)}{4(1 - b_1)} > 0. \end{aligned}$$

Thus $\tilde{P}(\tau_1) < 0$ for every $\tau_1 \in [0, \frac{1}{2}]$ when $\lambda = \frac{1}{2}$.

Finally, $\tilde{P}(\tau_1, \lambda)$ is continuous in (τ_1, λ). Hence $\max_{\tau_1 \in [0, \frac{1}{2}]} \tilde{P}(\tau_1, \lambda)$ is continuous in λ. Since this maximum is strictly negative at $\lambda = \frac{1}{2}$, there is some $\overline{\lambda} < \frac{1}{2}$ such that it remains negative for every $\lambda \in [\overline{\lambda}, \frac{1}{2}]$. Thus feasibility requires $\lambda < \overline{\lambda}$, as desired.

B.9. *Proof of Observation 3*

Using Lemma A.1 it's enough to show that if $\tilde{P} \geqslant 0$ for some τ_1 then, if we decrease ρ, $\tilde{P} \geqslant 0$ still holds. It is therefore enough to show that $\tilde{P}$ weakly decreases in ρ.

If $\tau_1 \geqslant \lambda$, the second-period tax does not depend on ρ, so neither does $\tilde{P}$. Suppose instead that $\tau_1 < \lambda$. At $s = \tilde{s}$, using (A22), the investment cutoff satisfies $b_2 \leqslant \frac{1}{2}$. Moreover, $\tilde{s} \geqslant \frac{\delta A}{2}(b_1 - \frac{1}{2})$. Differentiating $\tilde{P}$ with respect to ρ therefore gives

$$\begin{aligned} \frac{d\tilde{P}}{d\rho} &= (\lambda - \tau_1)\left\{\delta b_2(1 - \lambda - \tau_1) - \frac{2}{A}(\tilde{s} + \delta \mathrm{E}[\tau_2(1 - \tau_2)])\left(1 - \frac{\lambda + \tau_1}{2}\right)\right\} \\ &\leqslant \delta(\lambda - \tau_1)\left\{\frac{1}{2}(1 - \lambda - \tau_1) - \left(b_1 - \frac{1}{2}\right)\left(1 - \frac{\lambda + \tau_1}{2}\right)\right\}. \end{aligned}$$

The expression in curly brackets is strictly decreasing in τ_1, since its derivative is $-\frac{1}{4}(3 - 2b_1) < 0$.

It is therefore bounded above by its value at $\tau_1 = 0$, which is

$$\frac{1}{2}(1-\lambda) - \left(b_1 - \frac{1}{2}\right)\left(1 - \frac{\lambda}{2}\right) \leqslant 0$$

if and only if $b_1 \geqslant \frac{1-\frac{3}{4}\lambda}{1-\frac{1}{2}\lambda}$. Hence $\frac{d\tilde{P}}{d\rho} \leqslant 0$. This proves the result.

B.10. *Proof of Observation 4*

By the proof of Proposition 2, any policy that satisfies the political constraints induces $0 < b_2 < \frac{1}{2}$. Using (A13) and $t(\lambda - \frac{1}{2}t) \leqslant \frac{1}{2}\lambda^2$ for every $t \geqslant 0$, it follows that

$$\begin{aligned}\Delta W &= b_1 \underbrace{\tau_1\left(\lambda - \frac{\tau_1}{2}\right)}_{\leqslant \frac{1}{2}\lambda^2} + \delta \underbrace{b_2}_{<\frac{1}{2}} \underbrace{\mathrm{E}\left[\tau_2\left(\lambda - \frac{\tau_2}{2}\right)\right]}_{\leqslant \frac{1}{2}\lambda^2} + \delta\lambda \underbrace{(b_1 - b_2)}_{<b_1} - \frac{\delta A}{4}\underbrace{(b_1 - b_2)^2}_{>(b_1-\frac{1}{2})^2} \\ &< \frac{1}{2}\left(b_1 + \frac{\delta}{2}\right)\lambda^2 + \delta b_1 \lambda - \frac{\delta A}{4}\left(b_1 - \frac{1}{2}\right)^2.\end{aligned}$$

The final expression is continuous in λ and is strictly negative at $\lambda = 0$. Therefore, there is $\underline{\lambda}(A, c, \delta) > 0$ such that $\Delta W < 0$ for every policy satisfying the political constraints whenever $\lambda < \underline{\lambda}$. Thus a politically feasible policy with $\Delta W \geqslant 0$ can exist only if $\lambda \geqslant \underline{\lambda}$, as desired.

B.11. *Proof of Proposition 4*

Suppose that $\lambda < \frac{1}{3}$ and $\tilde{b}_2 \leqslant \frac{1}{2}$. We want to show that there is a green expectations equilibrium with first-period policy $(\tau_1, s) \neq (0, 0)$, and that in every such equilibrium we have $\tau_1 < \lambda$ and $s > 0$.

The proof has five steps. I first define $x \equiv b_1 - b_2$. I then rule out $\tau_1 > \lambda$. In that case, a small reduction in τ_1, possibly accompanied by an adjustment in the subsidy, improves welfare without violating the political constraints. I next rule out $\tau_1 = \lambda$ by showing that there is a small reduction of τ_1 and s that improves the proposer's objective without violating the political constraints. Hence any equilibrium policy must have $\tau_1 < \lambda$.

Suppose that $s = 0$. When $\tau_1 < \lambda$, a small subsidy is valuable because it changes b_2 while leaving the second-period tax equal to λ if policy changes. Under $\lambda < \frac{1}{3}$, this marginal change relaxes the relevant political constraint and raises welfare, so $s = 0$ cannot be optimal. Finally, suppose that $\tau_1 = s = 0$. The political constraints hold with equality. A small joint increase in the tax and the subsidy raises welfare and relaxes both political constraints. Thus, the solution to the proposer's problem, which exists because of continuity and compactness, is an equilibrium with $(\tau_1, s) \neq (0, 0)$, as desired. This completes the proof.

Step 1. Preliminary calculations.

Let (τ_1, s) be a solution to the proposer's problem: maximize $W_G(\tau_1, s)$ subject to (PC'_B) and (PC'_G). Abusing notation slightly, define

$$T \equiv b_1\tau_1(1-\tau_1) + \delta b_2 \mathrm{E}[\tau_2(1-\tau_2)] - s(b_1 - b_2) - \delta\rho\tilde{b}_2\lambda(1-\lambda),$$
$$P \equiv -\left(\tau_1 - \frac{\tau_1^2}{2}\right) + s + T.$$

The constraints are $P, T \geqslant 0$. Recall that $\tilde{b}_2 = b_1 - \frac{\rho}{A}\lambda(2-\lambda)$, while $\tau_2 = \max\{\tau_1, \lambda\}$ with probability ρ and $\tau_2 = \tau_1$ otherwise. The equations for b_1 and b_2 are $-c + \frac{\delta A}{2}b_1 = \frac{\delta}{2}$ and $s - c + \frac{\delta A}{2}b_2 = \frac{\delta}{2}\mathrm{E}[(1-\tau_2)^2]$.

Let $x \equiv b_1 - b_2$. Subtracting the equations for b_1, b_2 gives $s = \frac{\delta}{2}\left[Ax - \mathrm{E}(2\tau_2 - \tau_2^2)\right]$, so for fixed τ_1 the subsidy is strictly increasing in x.

Suppose first that $\tau_1 \geqslant \lambda$. Then $\tau_2 = \tau_1$ with probability one, so $s = \frac{\delta}{2}\left[Ax - (2\tau_1 - \tau_1^2)\right]$. Substituting this into T gives

$$T = (1+\delta)b_1\tau_1(1-\tau_1) + \frac{\delta}{2}x\tau_1^2 - \frac{\delta A}{2}x^2 - \delta\rho\tilde{b}_2\lambda(1-\lambda). \tag{A23}$$

Holding x fixed, $\frac{\partial W}{\partial \tau_1} = (b_1 + \delta b_2)(\lambda - \tau_1)$. Holding τ_1 fixed instead, $\frac{\partial W}{\partial x} = -s - \delta(\tau_1 - \lambda)(1-\tau_1)$. Finally, $\frac{\partial T}{\partial x} = \delta\left(\frac{\tau_1^2}{2} - Ax\right) < 0$, because $s \geqslant 0$ implies $Ax \geqslant 2\tau_1 - \tau_1^2 > \tau_1^2/2$, and $\frac{\partial P}{\partial \tau_1} = (1+\delta)[b_1(1-2\tau_1) - (1-\tau_1)] + \delta x\tau_1$.

When $\tau_1 < \lambda$, we instead have $s = \frac{\delta}{2}\left[Ax - (1-\rho)(2\tau_1 - \tau_1^2) - \rho\lambda(2-\lambda)\right]$. At $\tau_1 = \lambda$, holding x fixed and taking left derivatives, $\frac{\partial W}{\partial \tau_1}\big|_{\tau_1 = \lambda^-} = 0$ and

$$\left.\frac{\partial P}{\partial \tau_1}\right|_{\tau_1 = \lambda^-} = [1 + \delta(1-\rho)][b_1(1-2\lambda) - (1-\lambda)] + \delta(1-\rho)x\lambda < -\lambda < 0,$$

where the inequality uses $b_1 < 1$ and $x < 1$.

Step 2. I rule out $\tau_1 > \lambda$.

Suppose that $\tau_1 > \lambda$. The derivatives above show that both lowering τ_1 and lowering x raise welfare. I first show that both political constraints must bind. Using $x < b_1 < 1$ and $\tau_1 \leqslant \frac{1}{2}$,

$$\frac{\partial P}{\partial \tau_1} < b_1[1 + \delta - (2+\delta)\tau_1] - (1+\delta)(1-\tau_1) < -\tau_1 < 0.$$

Therefore T cannot be slack: if $T > 0$, we can lower τ_1 slightly, holding x fixed. This raises both P and W, while T remains positive by continuity. Hence $T = 0$.

Given $T = 0$, we have $P = s - \tau_1 + \tau_1^2/2$, so $P \geqslant 0$ implies $s \geqslant \tau_1 - \tau_1^2/2 > 0$. If $P > 0$, lower x slightly, holding τ_1 fixed. Since $\frac{\partial T}{\partial x} < 0$ and $\frac{\partial W}{\partial x} < 0$, this raises both T and welfare, while for a sufficiently small change P and s remain positive. Hence $P = 0$ as well.

Thus $P = T = 0$, so $s = \tau_1 - \tau_1^2/2$. Combining this with $s = \frac{\delta}{2}\left[Ax - (2\tau_1 - \tau_1^2)\right]$ gives $x(\tau_1) = \frac{1+\delta}{\delta A}\tau_1(2 - \tau_1)$ and $\frac{dx}{d\tau_1} = \frac{2(1+\delta)(1-\tau_1)}{\delta A} > 0$.

Let $\tilde{P}(\tau_1)$ denote the common value of P and T with $x = x(\tau_1)$. At the candidate optimum, $\tilde{P} = 0$. Differentiating (A23) gives

$$\tilde{P}'(\tau_1) = (1 + \delta)b_1(1 - 2\tau_1) + \delta x\tau_1 + \delta\left(\frac{\tau_1^2}{2} - Ax\right)\frac{dx}{d\tau_1}.$$

I claim that this derivative is strictly negative.

To see this, use $T = 0$ in (A23). Since $\tau_1 > \lambda$ and $\tau_1 \leqslant \frac{1}{2}$, we have $\tau_1(1 - \tau_1) > \lambda(1 - \lambda)$. Hence

$$[(1 + \delta)\tau_1(1 - \tau_1) - \delta\rho\lambda(1 - \lambda)]b_1 < \frac{\delta A}{2}x^2,$$

while the coefficient on b_1 is larger than $\tau_1(1 - \tau_1)$. Therefore $b_1 < \frac{\delta A x^2}{2\tau_1(1-\tau_1)}$. Substituting this bound and the expression for $x(\tau_1)$ above yields

$$\tilde{P}'(\tau_1) < x\left[\frac{(1 + \delta)^2(2 - \tau_1)(1 - 2\tau_1)}{2(1 - \tau_1)} + \delta\tau_1 + \frac{\delta\tau_1(1 - \tau_1)}{2 - \tau_1} - 2(1 + \delta)(1 - \tau_1)\right].$$

The expression in brackets is convex in δ, so its maximum on $[0, 1]$ is attained at an endpoint. At $\delta = 0$ it is $\frac{2\tau_1^2 - 3\tau_1 + 2}{2(\tau_1 - 1)} < 0$, and at $\delta = 1$ it is $\frac{\tau_1(2\tau_1^2 - 3\tau_1 - 1)}{(\tau_1 - 2)(\tau_1 - 1)} < 0$ for $0 < \tau_1 \leqslant \frac{1}{2}$. Thus $\tilde{P}'(\tau_1) < 0$.

It follows that a sufficiently small reduction in τ_1 that keeps $P = T$ makes both P and T strictly positive. It also lowers x. Since both changes strictly raise welfare, this contradicts optimality. Therefore $\tau_1 \leqslant \lambda$.

Step 3. I rule out $\tau_1 = \lambda$.

Suppose first that $\tau_1 = \lambda$ and $s = 0$. Then $x = \lambda(2 - \lambda)/A$ and

$$P = -\frac{1}{2}\lambda(2 - \lambda) + [1 + \delta(1 - \rho)]b_1\lambda(1 - \lambda) - \frac{\delta}{A}(1 - \rho^2)\lambda^2(1 - \lambda)(2 - \lambda).$$

Since $\tilde{b}_2 \leqslant \frac{1}{2}$, we have $b_1 \leqslant \frac{1}{2} + \rho\lambda(2 - \lambda)/A$. Hence

$$P \leqslant -\frac{1}{2}\lambda(2 - \lambda) + \frac{1}{2}[1 + \delta(1 - \rho)]\lambda(1 - \lambda) + \frac{1}{A}[\rho - \delta(1 - \rho)]\lambda^2(1 - \lambda)(2 - \lambda).$$

The right-hand side is affine in ρ. At $\rho = 0$ it is at most $-\lambda(2-\lambda)/2+\lambda(1-\lambda) = -\lambda^2/2 < 0$. At $\rho = 1$,

using $A > 1$, it is strictly smaller than $-\lambda/2 + \lambda^2(1-\lambda)(2-\lambda) < 0$, because $\lambda(1-\lambda)(2-\lambda) < 1/2$. Hence $s = 0$ is impossible.

Now suppose that $\tau_1 = \lambda$ and $s > 0$. We have $\frac{\partial W}{\partial x} = -s < 0$. Therefore P must bind. Otherwise, if $P > 0$, lowering x slightly raises welfare and raises T; for a sufficiently small change, P and s remain positive.

Next, T must also bind. If $T > 0$, reduce τ_1 slightly below λ, holding x fixed. Since the left derivative of P is strictly negative, this makes P strictly positive, while the first-order effect on welfare is zero. We can therefore also lower x by a sufficiently small amount proportional to the tax reduction. Since $\frac{\partial W}{\partial x} = -s < 0$, this makes the welfare change strictly positive. The constraint $P \geqslant 0$ remains satisfied if the proportional reduction in x is small enough, and $T > 0$ remains satisfied by continuity. This contradicts optimality. Hence $P = T = 0$.

As before, $P = T = 0$ implies $s = \tau_1 - \tau_1^2/2$. For $\tau_1 < \lambda$, combining this with the cutoff equation gives

$$x(\tau_1) = \frac{[1 + \delta(1-\rho)]\tau_1(2-\tau_1) + \delta\rho\lambda(2-\lambda)}{\delta A}.$$

In particular, $x(\lambda) = \frac{1+\delta}{\delta A}\lambda(2-\lambda)$ and $\frac{dx}{d\tau_1}\big|_{\tau_1=\lambda^-} = \frac{2[1+\delta(1-\rho)](1-\lambda)}{\delta A} > 0$.

Let $\tilde{P}(\tau_1)$ again denote the common value of P and T when $x = x(\tau_1)$. Differentiating T at $\tau_1 = \lambda^-$ gives

$$\begin{aligned}\tilde{P}'(\lambda^-) = {} & [1 + \delta(1-\rho)]b_1(1-2\lambda) + \delta(1-\rho)x\lambda \\ & - \left[\delta\lambda(1-\lambda) + \frac{2+\delta}{2}\lambda(2-\lambda)\right]\frac{2[1+\delta(1-\rho)](1-\lambda)}{\delta A}.\end{aligned}$$

Again this derivative is strictly negative. To see this, use $T = 0$ at $\tau_1 = \lambda$. Since $s = \lambda(2-\lambda)/2$,

$$[1 + \delta(1-\rho)]b_1\lambda(1-\lambda) < x\left[\delta\lambda(1-\lambda) + \frac{1}{2}\lambda(2-\lambda)\right].$$

Substituting this bound and the expression for $x(\lambda)$ above gives an upper bound that is affine in ρ. At $\rho = 0$, after dividing by the positive number x, the upper bound is $-\frac{4\delta(1-\lambda)^3 + (2-\lambda)(2\lambda^2-3\lambda+2)}{2(2-\lambda)(1-\lambda)} < 0$. At $\rho = 1$ it is

$$-\frac{-4\delta^2(2-\lambda)(1-\lambda)(\frac{1}{2}-\lambda) + \delta\lambda(4-3\lambda) - (\lambda-2)(2\lambda^2-3\lambda+2)}{2(1+\delta)(2-\lambda)(1-\lambda)}.$$

The numerator is concave in δ when $\lambda < 1/2$. At $\delta = 0$ it is $(2-\lambda)(2\lambda^2-3\lambda+2) > 0$, and at $\delta = 1$ it is $2\lambda(\lambda^2-5\lambda+5) > 0$. Hence the upper bound is negative at both $\rho = 0$ and $\rho = 1$, and therefore for every $\rho \in [0,1]$. Thus $\tilde{P}'(\lambda^-) < 0$.

A sufficiently small decrease in τ_1 keeping $P = T$ therefore makes both P and T positive. The

direct left derivative of welfare with respect to τ_1 is zero, while $\frac{\partial W}{\partial x} = -s < 0$ and $\frac{dx}{d\tau_1} > 0$. Hence $\frac{dW}{d\tau_1}\big|_{\tau_1=\lambda^-} < 0$, so this strictly raises welfare, a contradiction.

Step 4. I rule out $s = 0$ when $0 < \tau_1 < \lambda$.

Suppose that $0 < \tau_1 < \lambda$ and $s = 0$. Then $T = P + \tau_1 - \tau_1^2/2 > P$, so T is strictly slack whenever $P \geqslant 0$.

At $s = 0$, differentiating welfare and using the cutoff equation gives $\frac{dW}{ds} = \frac{2}{A}(1-\rho)(\lambda-\tau_1)(1-\tau_1) \geqslant 0$ and

$$\frac{dW}{d\tau_1} = (\lambda - \tau_1)[b_1 + \delta(1-\rho)b_2] + \frac{2\delta}{A}(1-\rho)^2(\lambda-\tau_1)(1-\tau_1)^2 > 0.$$

Also $\frac{dP}{ds} = \frac{1}{A}\mathrm{E}\left[A - 4\tau_2 + 3\tau_2^2\right] > 0$, because $\tau_2 \leqslant \lambda < 1/3$ implies $4\tau_2 - 3\tau_2^2 \leqslant 4\lambda - 3\lambda^2 < 1 < A$.

If $\rho < 1$, then $\frac{dW}{ds} > 0$. We can therefore increase s slightly, which raises both P and W; the constraint $T \geqslant 0$ remains satisfied because it is initially slack. This contradicts optimality.

If $\rho = 1$, the candidate is not feasible in the first place. In that case $b_2 = \tilde{b}_2$ and $P = -\tau_1 + \tau_1^2/2 + b_1\tau_1(1-\tau_1) < -\tau_1^2/2 < 0$, using $b_1 < 1$. Thus $s = 0$ is impossible whenever $0 < \tau_1 < \lambda$.

Step 5. I rule out $\tau_1 = s = 0$.

At $\tau_1 = s = 0$, we have $b_2 = \tilde{b}_2$ and therefore $P = T = 0$. Moreover, $\frac{dT}{d\tau_1} = b_1[1 + \delta(1-\rho)] - \frac{\delta}{A}\rho(1-\rho)\lambda(4-3\lambda)$ and $\frac{dT}{ds} = -\frac{\rho}{A}\lambda(4-3\lambda)$. Since $P = -\tau_1 + \tau_1^2/2 + s + T$, along the simple direction $\tau_1 = s = t$ we have $\frac{dP}{dt} = \frac{dT}{dt} = \frac{1+\delta(1-\rho)}{A}[Ab_1 - \rho\lambda(4-3\lambda)] > 0$, because $Ab_1 \geqslant 1$ and $\rho\lambda(4-3\lambda) < 1$ (since $\rho \leqslant 1$ and $\lambda < \frac{1}{3}$). Thus P and T both become strictly positive for small $t > 0$.

Finally, the formulas from Step 4 imply $\frac{dW}{d\tau_1} > 0$ at the origin and $\frac{dW}{ds} \geqslant 0$. Hence $\frac{dW}{dt} > 0$ along $\tau_1 = s = t$. This gives a feasible policy with strictly higher welfare, contradicting optimality of $(0, 0)$.

B.12. *Proof of Proposition A.1*

Let $x \equiv b_1 - b_2$. The proposer's gain relative to BAU can be written as

$$\Delta W = b_1\tau_1\left(\lambda - \frac{\tau_1}{2}\right) + \delta b_2\tau_2\left(\lambda - \frac{\tau_2}{2}\right) + \delta\lambda x - \frac{\delta A}{4}x^2.$$

First, the status quo is never an equilibrium.

Choose a sufficiently small $x > 0$ with $x < b_1 - \frac{1}{2}$, set $\tau_1 = 0$, and choose $s = \delta A x/2$. Then $b_2 = b_1 - x > \frac{1}{2}$. A polluting majority blocks any positive second-period tax, so $\tau_2 = 0$. A

noninvesting polluter gains $P^N = \delta\lambda x - \delta A x^2/2 > 0$ for sufficiently small x, so the proposal passes. The proposer gains $\Delta W = \delta\lambda x - \delta A x^2/4 > 0$. Hence BAU cannot be an equilibrium outcome.

Second, complete transition cannot occur.

Suppose that $b_2 = 0$. For a fixed $b_2 = 0$, welfare is maximized at $\tau_1 = \lambda$. Hence

$$\Delta W \leqslant \frac{\delta b_1}{4}\left(\frac{2\lambda^2}{\delta} + 4\lambda - 1 - \frac{2c}{\delta}\right) < 0$$

by Assumption 2. Since the first step exhibited a politically feasible policy with $\Delta W > 0$, we cannot have $b_2 = 0$ in equilibrium.

Third, I characterize the second-period policy.

Suppose first that $b_2 > \frac{1}{2}$. Ignoring constants, a polluting district's second-period payoff from a tax τ is $\frac{1}{2}(1-\tau)^2 + b_2\tau(1-\tau) - \lambda b_2(1-\tau)$. Its derivative is $b_2(1+\lambda) - 1 + (1-2b_2)\tau < 0$, since $b_2 \leqslant b_1$, $(1+\lambda)b_1 < 1$, and $b_2 > \frac{1}{2}$. Thus a polluting majority always prefers a lower tax. Since the proposer's ideal tax is λ, the continuation tax is $\tau_2 = \min\{\tau_1, \lambda\}$.

Suppose instead that $b_2 \leqslant \frac{1}{2}$, using the same convention as in the main model at equality. Green districts form a majority. Their second-period payoff has derivative $b_2(1+\lambda-2\tau)$, so their ideal tax is $(1+\lambda)/2$. It follows that the continuation tax is

$$\tau_2 = \begin{cases} \lambda, & \tau_1 \leqslant \lambda, \\ \tau_1, & \lambda < \tau_1 \leqslant (1+\lambda)/2, \\ 1+\lambda-\tau_1, & \tau_1 > (1+\lambda)/2. \end{cases}$$

Since BAU is not an equilibrium, there is some investment. Together with the previous step, this gives $0 < b_2 < b_1$. The marginal investing district is therefore indifferent, and subtracting the condition defining b_1 from its investment condition gives $s = \delta[Ax - (2\tau_2 - \tau_2^2)]/2$.

Fourth, consider an equilibrium with $b_2 > \frac{1}{2}$.

The noninvesting polluting districts themselves form a majority, so the proposal passes if and only if $P^N \geqslant 0$. Moreover, $P^N = G - (\tau_1 - \tau_1^2/2) - \delta(\tau_2 - \tau_2^2/2) \leqslant G$, so $P^N \geqslant 0$ implies $G \geqslant 0$. Thus, conditional on this continuation, the proposer maximizes W subject to $P^N \geqslant 0$.

I first show that $\tau_1 \leqslant \lambda$. If $\tau_1 > \lambda$, then $\tau_2 = \lambda$. A small reduction in τ_1 leaves investment and the continuation policy unchanged and strictly raises welfare. It also raises P^N, since $\partial P^N/\partial\tau_1 = -[1-(1+\lambda)b_1 + (2b_1-1)\tau_1] < 0$. Hence $\tau_1 > \lambda$ is impossible. We therefore have $\tau_2 = \tau_1$.

Suppose next that $\tau_1 = \lambda$ and $s > 0$. Holding the tax fixed and reducing x slightly reduces the required subsidy. At $\tau_1 = \tau_2 = \lambda$, $\partial\Delta W/\partial x = -s < 0$ and $\partial P^N/\partial x = -s - \delta A x/2 < 0$.

Thus reducing x strictly raises both welfare and the payoff of the winning coalition, contradicting optimality. Hence $\tau_1 = \lambda$ implies $s = 0$.

It remains to rule out $\tau_1 < \lambda$ and $s = 0$. In this case $\tau_2 = \tau_1$ and the investment condition implies $Ax = 2\tau_1 - \tau_1^2$. Holding τ_1 fixed, a marginal increase in the subsidy raises welfare at rate $2(1-\tau_1)(\lambda-\tau_1)/A > 0$. Thus, if P^N is slack, or if a marginal increase in the subsidy weakly relaxes P^N, there is an immediate profitable feasible deviation.

Consider the remaining case, in which $P^N = 0$ and increasing the subsidy tightens the constraint. Using (τ_1, x) as the policy variables, at $s = 0$ we have $\partial\Delta W/\partial x = \delta(1-\tau_1)(\lambda-\tau_1) > 0$ and $\partial\Delta W/\partial\tau_1 = (\lambda-\tau_1)(b_1+\delta b_2) > 0$. Moreover,

$$\frac{\partial P^N}{\partial x} = \delta\left[\lambda(1-\tau_1) - 2\tau_1 + \frac{3}{2}\tau_1^2\right],$$
$$\frac{\partial P^N}{\partial \tau_1} = (1+\delta)[(1+\lambda)b_1 - 1 + (1-2b_1)\tau_1] - \delta x(\lambda-\tau_1) < 0.$$

If $\partial P^N/\partial x \geqslant 0$, increasing x is itself a profitable feasible deviation. Suppose therefore that $\partial P^N/\partial x < 0$. We can then increase x and reduce τ_1 so as to keep $P^N = 0$.

Along this constraint, welfare increases with x if and only if

$$-\delta(1-\tau_1)\frac{\partial P^N}{\partial \tau_1} + (b_1+\delta b_2)\frac{\partial P^N}{\partial x} > 0.$$

Using $P^N = 0$ and $Ax = 2\tau_1 - \tau_1^2$, the left-hand side has the opposite sign of

$$[1-(1+\lambda)b_1]\left\{(\lambda-\tau_1)(4\tau_1^3 - 10\tau_1^2 + 8\tau_1 - 4) + 2\tau_1^2(\tau_1-2)\right\}$$
$$- 4(\lambda-\tau_1)^2(1-\tau_1)^3 + (\lambda-\tau_1)\tau_1^2(3\tau_1^2 - 8\tau_1 + 6) - \tau_1^3(1-\tau_1)(2-\tau_1).$$

The first line is strictly negative: $1-(1+\lambda)b_1 > 0$ by Assumption A.1, while both terms inside braces are negative for $0 < \tau_1 < \lambda < \frac{1}{2}$.

It remains to show that the second line is negative. Treat it as a quadratic in $\lambda - \tau_1$.

Suppose first that $\tau_1 \leqslant \frac{1}{3}$. Its discriminant is

$$\tau_1^3\left(25\tau_1^5 - 144\tau_1^4 + 324\tau_1^3 - 352\tau_1^2 + 180\tau_1 - 32\right).$$

The polynomial in parentheses is increasing on $[0, \frac{1}{3}]$. Indeed, its derivative is $125\tau_1^4 - 576\tau_1^3 + 972\tau_1^2 - 704\tau_1 + 180 \geqslant 125\tau_1^4 + 780\tau_1^2 - 704\tau_1 + 180 > 0$, where the first inequality uses $\tau_1 \leqslant \frac{1}{3}$, and the final quadratic is strictly positive because its discriminant is negative. At $\tau_1 = \frac{1}{3}$ the polynomial equals $-191/243 < 0$. Hence the discriminant is negative, and, since the leading coefficient of the

quadratic is negative, the second line is strictly negative.

Suppose next that $\frac{1}{3} < \tau_1 < \frac{1}{2}$. The vertex of the quadratic exceeds $\frac{1}{2} - \tau_1$ by

$$\frac{1 - 5(1-\tau_1)^4}{8(1-\tau_1)^3} > 0.$$

Since $0 < \lambda - \tau_1 < \frac{1}{2} - \tau_1$, the quadratic is increasing throughout the relevant interval. Its value is therefore smaller than its value at $\lambda - \tau_1 = \frac{1}{2} - \tau_1$, which is

$$-\frac{1}{2}\left(7\tau_1^4 - 26\tau_1^3 + 32\tau_1^2 - 14\tau_1 + 2\right) < 0.$$

The last inequality follows from

$$\begin{aligned} 7\tau_1^4 - 26\tau_1^3 + 32\tau_1^2 - 14\tau_1 + 2 = {} & 3(3\tau_1 - 1)^4 + 10(1 - 2\tau_1)(3\tau_1 - 1)^3 \\ & + 8(1 - 2\tau_1)^2(3\tau_1 - 1)^2 + (1 - 2\tau_1)^4 > 0. \end{aligned}$$

Thus the second line is strictly negative in both cases. Hence the entire expression is negative, so welfare strictly increases as we increase x along $P^N = 0$, contradicting optimality. Therefore $s = 0$ is impossible when $\tau_1 < \lambda$.

Therefore every non-first-best equilibrium with $b_2 > \frac{1}{2}$ satisfies $\tau_1 = \tau_2 < \lambda$ and $s > 0$.

Fifth, consider an equilibrium with $b_2 \leqslant \frac{1}{2}$.

The median polluting district invests, so its relevant constraint is $P^I \geqslant 0$. I first show that the initially green districts must also approve.

Suppose instead that $G < 0$. Then $P^N < G < 0$, so every noninvesting polluter rejects. Since the initially green districts also reject, the proposal can pass only with a majority consisting entirely of investing polluters. This requires $x > \frac{1}{2}$. The least productive member of such a majority is $i = b_1 - \frac{1}{2}$. Its gain relative to BAU is $G + s - (\tau_1 - \tau_1^2/2) - \delta A/4$. If it supports the proposal while $G < 0$, then necessarily $s > \delta A/4$.

Reduce x slightly, keeping the taxes fixed and adjusting s to preserve the investment cutoff. Every green-majority continuation has $\tau_2 \geqslant \lambda$, and $\partial \Delta W/\partial x = \delta(\lambda - \tau_2)(1 - \tau_2) - s < 0$. Thus reducing x strictly raises welfare. The payoff of the pivotal member of the polluter-only coalition has derivative $\delta(\lambda - \tau_2)(1 - \tau_2) - s + \delta A(1 - x)/2 < 0$, because $x > \frac{1}{2}$ and $s > \delta A/4$. Thus the same deviation also raises the payoff of the pivotal supporter, contradicting optimality. Therefore $G \geqslant 0$.

Hence, conditional on a green-majority continuation, political feasibility is exactly $P^I \geqslant 0$ and $G \geqslant 0$, and the proposer maximizes W subject to these constraints.

Sixth, I show that $\tau_1 \leqslant \lambda$.

Suppose first that $\tau_1 > (1+\lambda)/2$. Then $\tau_2 = 1+\lambda-\tau_1$. Let $\hat{\tau} \equiv 1+\lambda-\tau_1$. Replace the first-period tax by $\hat{\tau}$, leaving s unchanged. The continuation tax remains $\hat{\tau}$, so investment, b_2, subsidy expenditure, and the second-period allocation are unchanged.

The green districts' first-period payoff depends on the tax through $\tau_1(1+\lambda-\tau_1)$, which is the same at τ_1 and $\hat{\tau}$. Hence G is unchanged. In contrast, P^I strictly increases because $\hat{\tau} < \tau_1$ reduces the median polluter's first-period tax loss. Welfare also strictly increases because $b_1\tau_1(\lambda-\tau_1/2)$ is strictly decreasing for $\tau_1 > \lambda$. This is a contradiction.

It remains to consider $\lambda < \tau_1 \leqslant (1+\lambda)/2$, where $\tau_2 = \tau_1$. Hold x fixed and vary τ_1, adjusting s to preserve the investment cutoff. We have $d\Delta W/d\tau_1 = (b_1+\delta b_2)(\lambda-\tau_1) < 0$, while

$$\frac{dP^I}{d\tau_1} = (1+\delta)[b_1(1+\lambda-2\tau_1)+\tau_1-1]+\delta x(\tau_1-\lambda) < 0.$$

The last expression is linear in τ_1: at $\tau_1 = \lambda$ it is $(1+\delta)(1-\lambda)(b_1-1) < 0$, and at $\tau_1 = (1+\lambda)/2$ it is $(1-\lambda)[-(1+\delta)+\delta x]/2 < 0$.

A reduction in τ_1 could tighten the green constraint, so it remains to show that G cannot bind. If $G = 0$, substituting the binding constraint into welfare gives

$$\Delta W = \frac{1}{4}\left\{-2(1+\delta)b_1\tau_1(1-\lambda)+\delta x\left[2\lambda(1-\tau_1)+\tau_1^2\right]\right\}.$$

Since $x < b_1$, this is strictly smaller than the same expression with x replaced by b_1. The expression in braces is strictly decreasing in $\tau_1 \geqslant \lambda$, and at $\tau_1 = \lambda$ it equals $\lambda[-2+(2+\delta)\lambda] < 0$. Hence $G = 0$ implies $\Delta W < 0$, contradicting the first step.

Therefore $G > 0$. A sufficiently small reduction in τ_1, holding x fixed and adjusting s to preserve the investment cutoff, remains politically feasible and strictly raises welfare. Hence $\tau_1 > \lambda$ is impossible.

Seventh, $b_2 = \frac{1}{2}$ cannot occur in equilibrium.

We have just shown that $\tau_1 \leqslant \lambda$, so a green-majority continuation has $\tau_2 = \lambda$. If $b_2 = \frac{1}{2}$, then $P^I = P^N$ and $x = b_1 - \frac{1}{2}$. Since $s \geqslant 0$, the investment cutoff implies $Ax \geqslant \lambda(2-\lambda)$. Substituting this bound into P^N gives

$$P^I = P^N \leqslant \frac{\delta\lambda}{2}(\lambda+2b_1-2)+\tau_1[(1+\lambda)b_1-1]-\tau_1^2\left(b_1-\frac{1}{2}\right) < 0.$$

Indeed, $(1+\lambda)b_1 < 1$, and it also implies $\lambda+2b_1-2 < \lambda+2/(1+\lambda)-2 < 0$. Thus the median polluting district rejects, a contradiction. Therefore every green-majority equilibrium has $0 < b_2 < \frac{1}{2}$.

Eighth, I characterize the instrument mix when $0 < b_2 < \frac{1}{2}$.

We now have $\tau_2 = \lambda$. Suppose first that $\tau_1 = \lambda$ and $s > 0$. The argument in the previous step showing that $G = 0$ implies $\Delta W < 0$ also applies at $\tau_1 = \lambda$, so $G > 0$. Holding τ_1 fixed, $\partial \Delta W / \partial x = -s < 0$, so reducing x raises welfare.

If this reduction tightens P^I, reduce τ_1 slightly as well. From the left at $\tau_1 = \lambda$, $\partial P^I / \partial \tau_1 = -(1 - b_1)(1 - \lambda) < 0$, so a reduction in τ_1 relaxes P^I. The direct first-order welfare effect of changing τ_1 is zero at $\tau_1 = \lambda$, whereas reducing x has a strictly positive first-order welfare effect. Since G is strictly slack, this gives a profitable feasible deviation. Hence $\tau_1 = \lambda$ implies $s = 0$.

Finally suppose that $\tau_1 < \lambda$ and $s = 0$. Then $G = b_1 \tau_1 (1 + \lambda - \tau_1) + \delta \lambda b_1 > 0$, so the green constraint is slack. At this point $\partial \Delta W / \partial \tau_1 = b_1(\lambda - \tau_1) > 0$ and $\partial \Delta W / \partial x = 0$. Moreover, $\partial P^I / \partial \tau_1 = b_1(1 + \lambda - 2\tau_1) + \tau_1 - 1 < 0$, while $\partial P^I / \partial x = \delta(A - 2\lambda + \lambda^2)/2 > 0$.

If P^I is slack, a small increase in τ_1 is a profitable feasible deviation. If P^I binds, increase τ_1 and x simultaneously so as to keep $P^I = 0$. The increase in x has zero first-order welfare cost at $s = 0$, while the increase in τ_1 has a strictly positive first-order benefit. Since G is slack, this is again a profitable feasible deviation. Hence $\tau_1 < \lambda$ requires $s > 0$.

We have therefore shown that every non-first-best green-majority equilibrium has $0 < b_2 < \frac{1}{2}$, $\tau_1 < \lambda$, $\tau_2 = \lambda$, and $s > 0$.

Finally, whenever $\tau_1 = \lambda$ we have shown that $s = 0$. The continuation tax is then also λ, and Proposition 1 implies that the resulting allocation is the first best. Combining the preceding steps gives the three cases in the statement.

B.13. *Proof of Proposition A.2*

Suppose that the status quo is replaced and $\sigma > 0$. The gain from accepting the proposal for an investing polluting district $i \in [b_2, b_1)$ is $P_i = \frac{1}{2}(1 - \tau_1)^2 + s - c + (1 + \sigma)^2 \frac{\delta A}{2} i + T - \frac{1+\delta}{2}$, while the gain for an initially green district $i \in [b_1, 1]$ is $G_i = \frac{\delta A}{2}\left[(1 + \sigma)^2 - 1\right] i + T$. Both are increasing in i.

As in the proof of Proposition 2, every accepted proposal that changes the status quo has $0 < b_2 < \frac{1}{2}$ and $\tau_1 \leqslant \frac{1}{2}$. Indeed, if $b_2 \geqslant \frac{1}{2}$, the median district does not invest. Its gain from accepting is

$$P_{1/2} = -\tau_1 \left[1 - b_1 + \left(b_1 - \frac{1}{2}\right)\tau_1\right] - \delta \tau_2 \left[1 - b_2 + \left(b_2 - \frac{1}{2}\right)\tau_2\right]$$
$$- s(b_1 - b_2) - \frac{\delta A}{2} \sigma (1 + \sigma)(1 - b_2^2) < 0$$

unless $\tau_1 = \tau_2 = s = \sigma = 0$. Hence the median district must invest, and $b_2 < \frac{1}{2}$. The cases $b_2 = 0$ and $\tau_1 > \frac{1}{2}$ are ruled out exactly as in the proof of Proposition 2. Since $b_2 < \frac{1}{2}$, green districts form a majority in period 2, and therefore $\tau_2 = \max\{\tau_1, \lambda\}$.

The proof has three steps. I first rule out a majority composed entirely of initially polluting districts. I then rule out a mixed coalition. Finally, I show that if the pivotal polluter and all initially green districts support the proposal, reducing the production subsidy is profitable.

First, I rule out a majority composed entirely of initially polluting districts.

Suppose that no initially green district supports the proposal. Since support among investing polluters is increasing in i, passage requires district $b_1 - \frac{1}{2}$ to support it.

Consider a policy with the same τ_1, τ_2, and b_2, but with $\sigma = 0$. Increase the investment subsidy so that b_2 remains unchanged. The investment condition implies that the required subsidy is $s' = s + \frac{\delta A}{2}\left[(1+\sigma)^2 - 1\right]b_2 > s$. This is therefore a feasible policy in the baseline model. By the proof of Proposition 2, district $b_1 - \frac{1}{2}$ strictly rejects it.

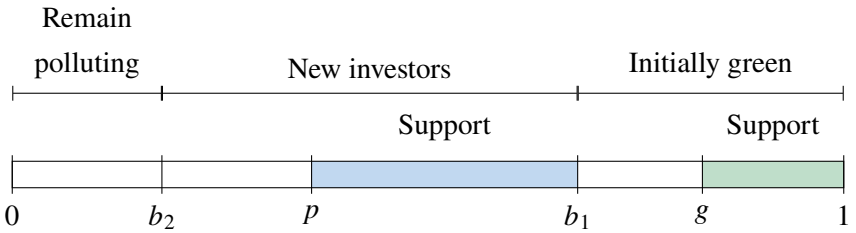
The difference between district $b_1 - \frac{1}{2}$'s payoff under the original policy and under this baseline policy is

$$\frac{\delta A}{4}\sigma\left[-2b_2^2 - 2b_2(1-b_1)(2+\sigma) + (1+\sigma)(2b_1-3) + (2b_1-1)\right].$$

The expression in brackets is strictly negative. The first two terms are nonpositive, while $(1+\sigma)(2b_1-3) + (2b_1-1) \leqslant 4(b_1-1) < 0$. Hence district $b_1 - \frac{1}{2}$ obtains an even lower payoff under the original policy, a contradiction. Thus a winning coalition cannot consist entirely of initially polluting districts.

Second, I rule out a mixed coalition.

Suppose that not all initially green districts support the proposal. Since support is increasing in i within each group, let p be the least productive investing polluter that supports and let $g \in (b_1, 1)$ be the least productive initially green district that supports. The supporting coalition is therefore $[p, b_1) \cup [g, 1]$.



Supporting districts have measure $(b_1 - p) + (1 - g)$.
Holding b_1 fixed, a fall in $p + g$ expands the coalition.

The investment condition gives $s = \frac{\delta A}{2}\left[b_1 - (1+\sigma)^2 b_2\right] - \frac{\delta}{2}(2\tau_2 - \tau_2^2)$. Substituting this into

P_i gives $P_i = T - (\tau_1 - \frac{1}{2}\tau_1^2) - \delta(\tau_2 - \frac{1}{2}\tau_2^2) + \frac{\delta A}{2}(1+\sigma)^2(i - b_2)$. Hence

$$p = b_2 + \frac{2\Big[\tau_1 - \frac{1}{2}\tau_1^2 + \delta\Big(\tau_2 - \frac{1}{2}\tau_2^2\Big) - T\Big]}{\delta A(1+\sigma)^2}, \qquad g = -\frac{2T}{\delta A\big[(1+\sigma)^2 - 1\big]}.$$

Hold τ_1 and σ fixed and increase b_2, adjusting s so that the investment condition continues to hold. Differentiating the transfer gives

$$\frac{\partial T}{\partial b_2} = \frac{\delta A}{2}\left[b_1\Big((1+\sigma)^2 + 1\Big) - 2b_2(1+\sigma) - \frac{\tau_2^2}{A}\right].$$

Therefore

$$\frac{\partial(p+g)}{\partial b_2} = 1 - \left[b_1\Big((1+\sigma)^2 + 1\Big) - 2b_2(1+\sigma) - \frac{\tau_2^2}{A}\right]\left(\frac{1}{(1+\sigma)^2} + \frac{1}{(1+\sigma)^2 - 1}\right).$$

I claim that this derivative is strictly negative. Since $s \geqslant 0$, the investment condition implies $A(1+\sigma)^2 b_2 \leqslant Ab_1 - (2\tau_2 - \tau_2^2)$. It follows that

$$b_1\Big((1+\sigma)^2 + 1\Big) - 2b_2(1+\sigma) - \frac{\tau_2^2}{A} > \frac{(1+\sigma)^2\big[(1+\sigma)^2 - 1\big]}{2(1+\sigma)^2 - 1}.$$

Here is an explicit bound. The left-hand side is at least $b_1[(1+\sigma)^2 + 1 - 2/(1+\sigma)] + [2(2\tau_2 - \tau_2^2)/(1+\sigma) - \tau_2^2]/A$. If $0 < \sigma \leqslant 1$, the second term in square brackets is nonnegative. Since $b_1 > 1/2$, subtracting the claimed lower bound leaves strictly more than $\sigma^2(3\sigma + 2)/[2(1+\sigma)(2(1+\sigma)^2 - 1)] > 0$. If $\sigma \geqslant 1$, use $1/A \leqslant b_1$ and $\tau_2^2 \leqslant 1/4$ to bound the left-hand side below by $b_1((1+\sigma)^2 + 3/4 - 2/(1+\sigma))$. Subtracting the claimed lower bound now leaves strictly more than $(10(1+\sigma)^3 - 16(1+\sigma)^2 - 3(1+\sigma) + 8)/[8(1+\sigma)(2(1+\sigma)^2 - 1)] > 0$. The numerator is positive at $\sigma = 1$ and increasing for $\sigma \geqslant 1$. Hence $\frac{\partial(p+g)}{\partial b_2} < 0$: increasing b_2 strictly increases the measure of districts that support the proposal.

The same deviation raises the proposer's payoff. A mixed coalition requires $s > 0$. Indeed,

$$G_{b_1} - P_{1/2} = \tau_1 - \frac{1}{2}\tau_1^2 - s + \frac{\delta A}{2}(1+\sigma)^2\left(b_1 - \frac{1}{2}\right).$$

If $s = 0$, then $P_{1/2} \geqslant 0$ implies $G_{b_1} > 0$, contradicting the assumption that not all initially green districts support.

Holding τ_1 and σ fixed and adjusting s to preserve the investment condition,

$$\frac{\partial W}{\partial b_2} = s + \delta(\tau_2 - \lambda)(1 - \tau_2) + \delta A b_2 \sigma(1 + \sigma) > 0.$$

Thus a small increase in b_2 both raises welfare and expands the winning coalition. A mixed-coalition equilibrium with $\sigma > 0$ is therefore impossible.

Third, I rule out $\sigma > 0$ when all initially green districts support.

Hold τ_1, τ_2, and b_2 fixed and reduce σ, adjusting s so that the investment cutoff remains b_2. The investment condition gives $\frac{ds}{d\sigma} = -\delta A b_2(1 + \sigma)$. Along this change,

$$\frac{dW}{d\sigma} = -\frac{\delta A}{2}\sigma(1 - b_2^2) < 0,$$

while the payoff of the pivotal polluter satisfies

$$\frac{dP_{1/2}}{d\sigma} = -\frac{\delta A}{2}\left[\sigma + b_2^2 + 2b_2(1 - b_1)(1 + \sigma)\right] < 0.$$

Hence reducing σ strictly raises welfare and relaxes the pivotal polluter's political constraint.

If all initially green districts continue to support after a sufficiently small reduction in σ, this is immediately a profitable feasible deviation. If instead some of the least productive initially green districts stop supporting, the perturbed policy enters the mixed-coalition case considered above. In that case $s > 0$, so a sufficiently small increase in b_2 is feasible; by the previous step it restores political support and raises welfare further.

Thus every candidate equilibrium with $\sigma > 0$ admits a profitable feasible deviation. Therefore every equilibrium has $\sigma = 0$.

B.14. *Proof of Proposition A.3*

Suppose that s is not available and the status quo is replaced. Let P denote the gain of the pivotal polluting district and G the gain of district b_1 among the initially green districts. Since

$$G - P = \tau_1 - \frac{1}{2}\tau_1^2 + \frac{\delta A}{2}(1 + \sigma)^2\left(b_1 - \frac{1}{2}\right) > 0,$$

$P \geqslant 0$ implies $G > 0$. Thus the pivotal polluter's constraint is the relevant political constraint.

The proof has two steps. I first show that $\tau_1 < \lambda$, and hence $\tau_2 = \lambda$. I then characterize when the production subsidy is positive.

First, I show that $\tau_1 < \lambda$.

By the preceding proof, every accepted proposal that changes the status quo has $\tau_1 \leqslant \frac{1}{2}$ and $\tau_2 = \max\{\tau_1, \lambda\}$.

Suppose first that $\tau_1 > \lambda$, so $\tau_2 = \tau_1$. Since $s = 0$, the investment condition gives

$$b_2 = \frac{Ab_1 - 2\tau_1 + \tau_1^2}{A(1+\sigma)^2}.$$

Holding σ fixed and taking the induced change in b_2 into account,

$$\frac{dW}{d\tau_1} = -(b_1 + \delta b_2)(\tau_1 - \lambda) - \frac{2\delta(1-\tau_1)}{A(1+\sigma)^2}[(\tau_1 - \lambda)(1-\tau_1) + Ab_2\sigma(1+\sigma)] < 0.$$

Thus a small reduction in τ_1 strictly raises welfare. Direct differentiation of the pivotal polluter's payoff shows that $\frac{dP}{d\tau_1} < 0$, so the same reduction also relaxes the political constraint. Hence $\tau_1 > \lambda$ is impossible.

Now suppose $\tau_1 = \lambda$. If $\sigma = 0$, the policy is $\tau_1 = \tau_2 = \lambda$, which is ruled out by the same argument as in the proof of Proposition 3.

Suppose instead that $\sigma > 0$. From the left, reducing τ_1 leaves $\tau_2 = \lambda$, and

$$\frac{\partial P}{\partial \tau_1} = -[1 - b_1 + (2b_1 - 1)\tau_1] < 0.$$

Thus lowering τ_1 relaxes the political constraint. At $\tau_1 = \lambda$, the direct welfare effect of changing τ_1 is zero, whereas

$$\frac{dW}{d\sigma} = -\frac{\delta A}{2}\sigma(1 + 3b_2^2) < 0.$$

Hence a small reduction in σ raises welfare to first order. If it also relaxes P, reduce σ alone. If it tightens P, accompany it with a sufficiently small reduction in τ_1 so that the political constraint continues to hold. The latter adjustment has only a second-order welfare cost at $\tau_1 = \lambda$. In either case welfare strictly rises. Therefore $\tau_1 = \lambda$ is impossible.

We conclude that $\tau_1 < \lambda$, and therefore $\tau_2 = \lambda$.

Second, I characterize when $\sigma > 0$.

Since $\tau_2 = \lambda$ and $s = 0$, the investment condition gives

$$b_2 = \frac{Ab_1 - 2\lambda + \lambda^2}{A(1+\sigma)^2}.$$

For fixed σ, b_2 does not depend on τ_1. Hence $\frac{\partial W}{\partial \tau_1} = b_1(\lambda - \tau_1) > 0$, while $\frac{\partial P}{\partial \tau_1} = -[1 - b_1 + (2b_1 - 1)\tau_1] < 0$. The political constraint must therefore bind: otherwise the proposer could increase τ_1 slightly toward λ.

Differentiating welfare with respect to σ, taking into account the induced change in b_2, gives

$$\frac{\partial W}{\partial \sigma} = -\frac{\delta A}{2}\sigma(1 + 3b_2^2).$$

Thus a production subsidy has zero first-order welfare cost at $\sigma = 0$, but strictly lowers welfare at the margin whenever $\sigma > 0$.

At $\sigma = 0$, the investment condition gives $Ab_2 = Ab_1 - 2\lambda + \lambda^2$. Differentiating the pivotal polluter's political constraint and substituting this expression gives

$$\left.\frac{\partial P}{\partial \sigma}\right|_{\sigma=0} = \frac{\delta b_2}{2}[Ab_2 - 4\lambda(1-\lambda)] = \frac{\delta b_2}{2}\left[Ab_1 - 6\lambda + 5\lambda^2\right].$$

Since $b_2 > 0$, a small production subsidy relaxes the political constraint at $\sigma = 0$ if and only if $Ab_1 > 6\lambda - 5\lambda^2$.

It remains to show that this local condition is sufficient. For arbitrary σ,

$$\frac{\partial P}{\partial \sigma} = -\frac{\delta}{2(1+\sigma)^4}\left[A(1+\sigma)^4\sigma + \frac{(Ab_1 - 2\lambda + \lambda^2)^2}{A}(2\sigma - 1) + 4\frac{Ab_1 - 2\lambda + \lambda^2}{A}\lambda(1-\lambda)(1+\sigma)\right].$$

The expression in brackets is strictly increasing in $\sigma \geqslant 0$. At $\sigma = 0$, its sign is the opposite of $Ab_1 - 6\lambda + 5\lambda^2$.

Suppose first that $Ab_1 \leqslant 6\lambda - 5\lambda^2$. Then $\frac{\partial P}{\partial \sigma} \leqslant 0$ at $\sigma = 0$ and, since the expression in brackets is increasing, $\frac{\partial P}{\partial \sigma} < 0$ for every $\sigma > 0$. A production subsidy therefore never relaxes the political constraint, while it strictly lowers welfare whenever $\sigma > 0$. Hence $\sigma = 0$.

Suppose instead that $Ab_1 > 6\lambda - 5\lambda^2$. Then $\left.\frac{\partial P}{\partial \sigma}\right|_{\sigma=0} > 0$. If an equilibrium had $\sigma = 0$, the political constraint would bind. A small increase in σ would then create political slack at first order while having zero first-order welfare cost. The proposer could use that slack to increase τ_1 slightly toward λ, which raises welfare at first order. This contradicts optimality.

Therefore $\sigma > 0$ if and only if $Ab_1 > 6\lambda - 5\lambda^2$.

B.15. *Proof of Proposition A.4*

Write $x \equiv b_1 - b_2$. Recall that

$$V(b,\tau) = \frac{\sigma^2}{1-\sigma}\left(b(1+\tau)^{1-\frac{1}{\sigma}} + \int_b^1 H(i)\,di\right) + b\tau(1+\tau)^{-\frac{1}{\sigma}}.$$

The gain of an initially green district relative to the no-policy status quo is

$$G = V(b_1, \tau_1) + \delta V(b_2, \tau_2) - \delta\sigma(s + \sigma f)x - (1 + \delta)V(b_1, 0).$$

Whenever $b_2 < \frac{1}{2}$, the median district invests. Its gain relative to the no-policy status quo can be written as $P = P_1 + \delta P_2$, where

$$P_1 = V(b_1, \tau_1) - V(b_1, 0) + \sigma\left[(1 + \tau_1)^{1-\frac{1}{\sigma}} - 1\right]$$

is its first-period gain, and

$$P_2 = V(b_2, \tau_2) - V(b_1, 0) - \sigma(s + \sigma f)x + \sigma\left[-(1 - \sigma)f + s + H\left(\frac{1}{2}\right) - 1\right]$$

is its second-period gain, including the investment subsidy and the fiscal consequences of investment. Thus (PC_G) and (PC_P) are $G \geqslant 0$ and $P \geqslant 0$, respectively.

I will repeatedly use two cutoff conditions. The initial cutoff satisfies $H(b_1) = 1 + (1 - \sigma)f$. If $0 < b_2 < b_1$, district b_2 is indifferent between remaining polluting and investing, so

$$H(b_2) - (1 - \sigma)f + s = (1 + \tau_2)^{1-\frac{1}{\sigma}}. \tag{MI}$$

I refer to (MI) as the marginal-investor condition.

Step 1. I show that policy change requires $b_2 < \frac{1}{2}$.

Consider first the effect of a carbon tax on a district that remains polluting. Holding the share b of polluting districts fixed, its one-period gain from a tax τ is

$$V(b, \tau) - V(b, 0) + \sigma\left[(1 + \tau)^{1-\frac{1}{\sigma}} - 1\right].$$

Differentiating and using the expression for $\frac{\partial V}{\partial \tau}$ gives

$$\frac{\partial V}{\partial \tau} + \sigma\left(1 - \frac{1}{\sigma}\right)(1 + \tau)^{-\frac{1}{\sigma}} = -\frac{(1 + \tau)^{-1-\frac{1}{\sigma}}}{\sigma}[\sigma(1 - \sigma)(1 - b)(1 + \tau) + b\tau] < 0.$$

Since the gain is zero at $\tau = 0$, a polluting district weakly loses from every positive carbon tax.

Suppose now that $b_2 \geqslant \frac{1}{2}$, so the median district remains polluting in both periods. Its total gain relative to the no-policy status quo is

$$P_1 + \delta\left\{V(b_2, \tau_2) - V(b_2, 0) + \sigma\left[(1 + \tau_2)^{1-\frac{1}{\sigma}} - 1\right]\right\}$$

$$+\delta\left\{\frac{\sigma^2}{1-\sigma}\int_{b_2}^{b_1}[H(i)-H(b_1)]\,di-\sigma sx\right\}.$$

The first two terms are nonpositive by the calculation above. The last term is also nonpositive because H is increasing. Equality throughout requires $\tau_1=\tau_2=0$ and no subsidized investment, which gives the no-policy status quo. Hence the median district strictly rejects every nontrivial proposal with $b_2\geqslant\frac{1}{2}$. Since support among initially polluting districts is increasing in i, such a proposal cannot obtain a majority. Therefore $b_2<\frac{1}{2}$.

Step 2. I show that policy change requires $b_2>0$.

Suppose instead that $b_2=0$. First consider first-period welfare, holding b_1 fixed. Relative to the no-policy status quo, the gain from a tax τ_1 is

$$b_1\left\{\frac{\sigma}{1-\sigma}\left[(1+\tau_1)^{1-\frac{1}{\sigma}}-1\right]+(\tau_1-\lambda)(1+\tau_1)^{-\frac{1}{\sigma}}+\lambda\right\}.$$

Its derivative is

$$\frac{b_1(\lambda-\tau_1)}{\sigma}(1+\tau_1)^{-1-\frac{1}{\sigma}},$$

so it is maximized at $\tau_1=\lambda$. At that tax the gain is

$$\lambda b_1+\frac{\sigma b_1}{1-\sigma}\left[(1+\lambda)^{1-\frac{1}{\sigma}}-1\right]<\lambda b_1,$$

because $1-\frac{1}{\sigma}<0$.

If every initially polluting district invests, the gain from the second-period allocation and the investments, relative to the no-policy status quo, is

$$\delta\lambda b_1-\frac{\delta\sigma}{1-\sigma}\int_0^{b_1}[H(b_1)-H(i)]\,di.$$

The first term is the maximum possible environmental gain from eliminating second-period emissions. The second is the excess resource cost of switching all initially polluting districts to the green technology. Therefore the total welfare gain from any proposal with $b_2=0$ is strictly smaller than

$$(1+\delta)\lambda b_1-\frac{\delta\sigma}{1-\sigma}\int_0^{b_1}[H(b_1)-H(i)]\,di\leqslant(1+\delta)\lambda b_1-\frac{\delta\sigma m}{2(1-\sigma)}b_1^2<0,$$

where the last inequality follows from H2. The proposer therefore prefers the no-policy status quo. Together with Step 1, this proves $0<b_2<\frac{1}{2}$.

Step 3. I derive the second-period policy.

Since $b_2 < \frac{1}{2}$, green districts form a second-period majority. Holding b_2 fixed,

$$\frac{\partial V}{\partial \tau} = b_2 \frac{(1+\tau)^{-1-\frac{1}{\sigma}}}{\sigma}[\sigma(1-\sigma)(1+\tau) - \tau].$$

Thus their ideal tax is $\overline{\tau} = \frac{\sigma(1-\sigma)}{1-\sigma(1-\sigma)}$. The proposer's second-period payoff satisfies

$$\frac{\partial W}{\partial \tau} = \frac{b_2(\lambda - \tau)}{\sigma}(1+\tau)^{-1-\frac{1}{\sigma}},$$

so the proposer's ideal tax is $\lambda < \overline{\tau}$.

I first show that $\tau_1 \leqslant \overline{\tau}$. Suppose instead that $\tau_1 > \overline{\tau}$, and let $\hat{\tau} < \overline{\tau}$ be the tax chosen in the second period against the status quo τ_1. Replace the first-period tax τ_1 by $\hat{\tau}$. The second-period policy remains $\hat{\tau}$, so investment and b_2 are unchanged. Green districts weakly prefer the deviation, every polluting district strictly prefers the lower first-period tax, and the proposer strictly prefers it because $\lambda \leqslant \hat{\tau} < \tau_1$. This contradicts equilibrium. Hence $\tau_1 \leqslant \overline{\tau}$.

If $\tau_1 \leqslant \lambda$, green districts approve λ and the proposer implements it. If $\lambda < \tau_1 \leqslant \overline{\tau}$, green districts reject every reduction from τ_1 toward λ, while the proposer does not want to raise the tax. Therefore $\tau_2 = \max\{\tau_1, \lambda\}$.

Step 4. I show that every winning coalition contains the initially green districts.

Suppose instead that initially green districts reject the proposal. Since support among initially polluting districts is increasing in i, a majority consisting only of polluting districts requires district $b_1 - \frac{1}{2}$ to approve.

Suppose first that $b_1 - \frac{1}{2} \leqslant b_2$, so this district does not invest. Its gain is exactly the expression derived in Step 1 for a district that remains polluting. Every term in that expression is nonpositive, and at least one is strictly negative for a nontrivial proposal. Hence it rejects.

Suppose instead that $b_1 - \frac{1}{2} > b_2$. Then the district invests and $x = b_1 - b_2 > \frac{1}{2}$. Its first-period gain is $P_1 \leqslant 0$. Its second-period gain is

$$V(b_2, \tau_2) - V(b_1, 0) - \sigma(s + \sigma f)x + \sigma\left\{H\left(b_1 - \frac{1}{2}\right) - 1 - (1-\sigma)f + s\right\}.$$

Use the marginal-investor condition $H(b_2) - (1-\sigma)f + s = (1+\tau_2)^{1-\frac{1}{\sigma}}$ to eliminate s, and use $H(b_1) = 1 + (1-\sigma)f$. After expanding V, the terms that depend on τ_2 are

$$\left\{\frac{\sigma b_2}{1-\sigma} + \sigma(1-b_1)\right\}\left[(1+\tau_2)^{1-\frac{1}{\sigma}} - 1\right] + b_2\tau_2(1+\tau_2)^{-\frac{1}{\sigma}}.$$

These are nonpositive. Indeed,

$$\frac{\sigma}{1-\sigma}\left[1-(1+\tau_2)^{1-\frac{1}{\sigma}}\right]=\int_0^{\tau_2}(1+z)^{-\frac{1}{\sigma}}\,dz\geqslant\tau_2(1+\tau_2)^{-\frac{1}{\sigma}}.$$

After removing these tax terms, the remaining second-period gain, divided by σ, is

$$H\left(b_1-\frac{1}{2}\right)-H(b_2)+\frac{\sigma\int_{b_2}^{b_1}[H(i)-H(b_2)]\,di-x[H(b_1)-H(b_2)]}{1-\sigma}.$$

H1 implies that this is at most

$$M\left(x-\frac{1}{2}\right)+\frac{\frac{\sigma M}{2}x^2-mx^2}{1-\sigma}\leqslant 2m\left(x-\frac{1}{2}\right)-mx^2=-m(1-x)^2<0.$$

Thus district $b_1-\frac{1}{2}$ strictly rejects. A winning coalition cannot therefore consist only of initially polluting districts.

Hence every winning coalition contains the initially green districts. Since their measure is $1-b_1<\frac{1}{2}$, passage also requires the support of the median polluting district. A proposal therefore passes iff $G\geqslant 0$ and $P\geqslant 0$, and the proposer chooses τ_1, s to maximize W subject to (PC_G) and (PC_P).

Step 5. I show that the first-best policy is not politically feasible.

Consider $\tau_1=\tau_2=\lambda$ and $s=0$. Since $\lambda>0$, Step 1 implies immediately that $P_1<0$.

The marginal-investor condition and the initial-cutoff condition give

$$H(b_2)-H(b_1)=(1+\lambda)^{1-\frac{1}{\sigma}}-1.$$

Substituting this equality into P_2 gives

$$\begin{aligned}P_2=\frac{\sigma^2}{1-\sigma}&\left\{-b_2\left[1-(1+\lambda)^{1-\frac{1}{\sigma}}\right]+\int_{b_2}^{b_1}[H(i)-H(b_1)]\,di\right\}\\&+b_2\lambda(1+\lambda)^{-\frac{1}{\sigma}}+\sigma\left[H\left(\frac{1}{2}\right)-H(b_1)\right].\end{aligned}$$

Since $H'(i)\geqslant m$,

$$P_2\leqslant-\frac{\sigma^2 m}{1-\sigma}\left\{b_2x+\frac{x^2}{2}\right\}+b_2\lambda(1+\lambda)^{-\frac{1}{\sigma}}-\sigma m\left(b_1-\frac{1}{2}\right).$$

In particular,

$$P_2 < b_2\lambda(1+\lambda)^{-\frac{1}{\sigma}} - \sigma m\left(b_1 - \frac{1}{2}\right) < 0,$$

where the last inequality follows from H2. Hence $P = P_1 + \delta P_2 < 0$, and the first-best policy is not politically feasible.

Step 6. I rule out the only problematic case with $\tau_1 \geqslant \lambda$: both political constraints binding.

Suppose $\lambda \leqslant \tau_1 \leqslant \overline{\tau}$. By Step 3, $\tau_2 = \tau_1$. Suppose, toward a contradiction, that $P = G = 0$. Subtracting the two constraints gives

$$(1+\tau_1)^{1-\frac{1}{\sigma}} - 1 + \delta\left\{-(1-\sigma)f + s + H\left(\frac{1}{2}\right) - 1\right\} = 0.$$

Using the marginal-investor condition to replace $-(1-\sigma)f + s$ gives

$$H\left(\frac{1}{2}\right) - H(b_2) = \frac{1+\delta}{\delta}\left[1 - (1+\tau_1)^{1-\frac{1}{\sigma}}\right]. \tag{1}$$

Now substitute the marginal-investor condition and (1) into G. Using

$$\frac{\sigma}{1-\sigma}\left[1 - (1+\tau_1)^{1-\frac{1}{\sigma}}\right] \geqslant \tau_1(1+\tau_1)^{-\frac{1}{\sigma}},$$

we obtain

$$G \leqslant \delta\sigma\left\{b_2\left[H\left(\frac{1}{2}\right) - H(b_2)\right] - x\left[H(b_1) - H\left(\frac{1}{2}\right)\right] - \frac{\sigma}{1-\sigma}\int_{b_2}^{b_1}[H(b_1) - H(i)]\,di\right\}.$$

Hence

$$G < \delta\sigma\left\{b_2\left[H\left(\frac{1}{2}\right) - H(b_2)\right] - x\left[H(b_1) - H\left(\frac{1}{2}\right)\right]\right\}.$$

But $x > b_1 - \frac{1}{2}$, $b_2 < \frac{1}{2}$, and $H(b_1) - H(\frac{1}{2}) \geqslant m(b_1 - \frac{1}{2})$. Using (1) and $\tau_1 \leqslant \overline{\tau}$,

$$x\left[H(b_1) - H\left(\frac{1}{2}\right)\right] - b_2\left[H\left(\frac{1}{2}\right) - H(b_2)\right] > m\left(b_1 - \frac{1}{2}\right)^2 - \frac{1+\delta}{2\delta}\left[1 - (1+\overline{\tau})^{-\frac{1-\sigma}{\sigma}}\right] > 0$$

by H2. Therefore $G < 0$, a contradiction. Thus the two political constraints cannot bind simultaneously when $\tau_1 \geqslant \lambda$.

Step 7. I show that $\tau_1 < \lambda$.

Suppose first that $\tau_1 > \lambda$. Then $\tau_2 = \tau_1$. Consider changes in τ_1 that hold b_2 fixed, adjusting s

so that the marginal-investor condition continues to hold. Along such changes,

$$\frac{dW}{d\tau_1} = \frac{(b_1 + \delta b_2)(\lambda - \tau_1)}{\sigma}(1+\tau_1)^{-1-\frac{1}{\sigma}} < 0,$$

while

$$\frac{dP}{d\tau_1} = -\frac{(1+\tau_1)^{-1-\frac{1}{\sigma}}}{\sigma}\{\sigma(1-\sigma)(1+\delta)(1-b_1)(1+\tau_1) + \tau_1(b_1+\delta b_2)\} < 0.$$

Thus a small reduction in τ_1, holding b_2 fixed, raises both welfare and the pivotal polluter's surplus.

It is also useful to vary b_2. Holding τ_1 fixed and adjusting s according to the marginal-investor condition gives $\frac{ds}{db_2} = -H'(b_2)$. Direct differentiation then gives

$$\frac{dW}{db_2} = \delta\left\{\frac{\sigma s}{1-\sigma} + (\tau_1 - \lambda)(1+\tau_1)^{-\frac{1}{\sigma}}\right\} > 0$$

and

$$\frac{dG}{db_2} = \delta\left\{\tau_1(1+\tau_1)^{-\frac{1}{\sigma}} + \frac{\sigma s}{1-\sigma} + \sigma H'(b_2)x\right\} > 0.$$

I first claim that $P = 0$. If $P > 0$ and $s > 0$, a small increase in b_2 reduces the subsidy, raises welfare, and relaxes G, while P remains satisfied by continuity. If instead $P > 0$ and $s = 0$, then

$$P - G = \sigma\left[(1+\tau_1)^{1-\frac{1}{\sigma}} - 1\right] + \delta\sigma\left[H\left(\frac{1}{2}\right) - H(b_1)\right] < 0,$$

so $G > P > 0$. A small reduction in τ_1, holding b_2 fixed, then raises welfare and P while preserving $G \geqslant 0$. Both cases contradict optimality. Hence $P = 0$.

If $G > 0$, the same small reduction in τ_1 raises welfare and relaxes P, while G remains satisfied. Hence $G = 0$. But Step 6 rules out $P = G = 0$. Therefore $\tau_1 > \lambda$ is impossible.

Now suppose $\tau_1 = \lambda$. Then $\tau_2 = \lambda$. Step 5 implies $s > 0$: if $s = 0$, the marginal-investor condition pins down exactly the first-best investment cutoff, so the policy would be the infeasible first best.

If $P > 0$, a small increase in b_2 raises welfare and G, while P remains satisfied. Hence $P = 0$. Step 6 then implies $G > 0$.

Increase b_2 slightly, reducing s according to the marginal-investor condition. At $\tau_1 = \tau_2 = \lambda$,

$$\frac{dW}{db_2} = \frac{\delta\sigma s}{1-\sigma} > 0.$$

If this change weakly raises P, it is already a profitable feasible deviation. If it lowers P, accompany it with a small reduction in τ_1. Since τ_2 then remains equal to λ, $\frac{\partial P}{\partial \tau_1} < 0$, so lowering τ_1 restores

$P = 0$. Moreover, $\frac{\partial W}{\partial \tau_1} = 0$ at $\tau_1 = \lambda$, whereas the increase in b_2 has a strictly positive first-order welfare effect. Since $G > 0$, the combined deviation is feasible for sufficiently small changes and strictly raises welfare. This contradiction rules out $\tau_1 = \lambda$.

Therefore $\tau_1 < \lambda$.

Step 8. I show that $s > 0$.

Since $\tau_1 < \lambda$, Step 3 gives $\tau_2 = \lambda$. Suppose $s = 0$. Then

$$P - G = \sigma\left[(1+\tau_1)^{1-\frac{1}{\sigma}} - 1\right] + \delta\sigma\left[H\left(\frac{1}{2}\right) - H(b_1)\right] < 0.$$

Thus every feasible policy with $s = 0$ has $G > P \geqslant 0$, so the green constraint is strictly slack.

Moreover,

$$\frac{\partial W}{\partial \tau_1} = \frac{b_1(\lambda - \tau_1)}{\sigma}(1+\tau_1)^{-1-\frac{1}{\sigma}} > 0, \qquad \frac{\partial P}{\partial \tau_1} < 0.$$

Hence $P = 0$: otherwise a small increase in τ_1 would raise welfare and remain politically feasible.

Finally, differentiate the marginal-investor condition with respect to s. Since $\tau_2 = \lambda$ is fixed, $\frac{db_2}{ds} = -1/H'(b_2)$. Taking this induced change in b_2 into account gives

$$\left.\frac{dP}{ds}\right|_{s=0} = \delta\left\{\sigma(1 - b_1 + b_2) - \frac{\lambda(1+\lambda)^{-\frac{1}{\sigma}}}{H'(b_2)}\right\} > 0.$$

Indeed, $H'(b_2) \geqslant m$ and $1 - b_1 + b_2 > 1 - b_1 \geqslant \min\{b_1 - \frac{1}{2}, 1 - b_1\}$, so the inequality follows from H2. Thus a small subsidy relaxes the binding pivotal-polluter constraint.

Taking the induced change in b_2 into account, the direct welfare effect of the subsidy is

$$\frac{dW}{ds} = -\frac{\delta\sigma s}{(1-\sigma)H'(b_2)},$$

which is zero at $s = 0$. After increasing s slightly, the proposer can therefore increase τ_1 slightly so as to keep $P = 0$. The green constraint is strictly slack, so this deviation remains feasible. The subsidy has zero first-order welfare cost at $s = 0$, while the increase in τ_1 has a strictly positive first-order welfare benefit because $\tau_1 < \lambda$. Hence the deviation strictly raises welfare, a contradiction.

Therefore $s > 0$, completing the proof.